TRACK W (NONLINEAR WAVE): BLUEPRINT

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Abstract. A proof plan for track W: two-frequency quasi-periodic solutions of the cubic Dirichlet wave equation $u_{tt} = u_{xx} - mu - u^3$ on $[0, \pi]$ for every fixed $m > 0$, excited in the modes $J = \{1, 2\}$ (owner decision D30). Part 0 fixes every normalization once and proves milestone W0: the cubic attachment, the quartic coefficients, the partial Birkhoff normal-form action coefficients, nondegeneracy for every $m > 0$, and the Duffing consistency check. The proposed headline W-NLW2 closes the document. Nothing here is frozen or approved. The milestone plan is in `PLAN.md`.

Contents

1. Conventions and normalizations 2
1.1. Sources and locators 2
1.2. The equation 2
1.3. Sequence spaces 3
1.4. Coordinates, Hamiltonian and equations of motion 3
1.5. Complex coordinates, actions, angles 4
1.6. Comparison with the printed conventions of the sources 4
2. Milestone W0: cubic attachment and quartic normalization 5
2.1. Reconstruction 5
2.2. The quartic coefficients 6
2.3. The cubic gradient 7
2.4. PDE $\leftrightarrow$ coefficient system 8
2.5. Quartic nonresonance 9
2.6. Partial quartic Birkhoff normal form: the action coefficients 10
2.7. Nondegeneracy for every $m > 0$ 11
2.8. Duffing consistency check 12
2.9. Ledger: proved, cited, source gaps 12
3. Milestone W1: weighted sequence calculus 12
3.1. Realization in Lean and transport maps 13
3.2. External facts 13
3.3. Weighted spaces, embeddings, multipliers 14
3.4. Two-sided spaces, Young and the algebra 17
3.5. The cubic field 18
3.6. The sine system and analytic reconstruction 21
3.7. Analytic maps between weighted spaces 23
3.8. Support for the KAM step 25
3.9. Ledger 26
4. Milestone W3: the analytic partial Birkhoff normal form 27
4.1. External facts 27
4.2. Coordinates, symplectic form, Hamiltonian fields 28
4.3. The quartic Hamiltonian in two-sided coordinates 30
4.4. Quantitative small divisors 31
4.5. The generating Hamiltonian 33

4.6. The Lie series of the flow 34
4.7. The normal form 36
4.8. Interface to milestones W4–W6 38
4.9. Constants and their dependence on m 39
4.10. Summary of status 39
5. The proposed headline W-NLW2 39
5.1. Definitions 39
5.2. Statement 40

1. Conventions and normalizations

This part fixes every normalization of track W *once*. Later parts, the proposed headline (§5) and all future Lean statements use exactly these conventions; any departure must be declared as a deviation. Section 1.6 compares them with the printed conventions of the wave source, which are not internally consistent (§1.6, and `docs/tracks/W/errata-check-W0.md`).

1.1. Sources and locators.

- **P96-W**: J. Pöschel, *Quasi-periodic solutions for a nonlinear wave equation*, Comment. Math. Helv. **71** (1996), 269–296 (ETH E-Periodica scan). Printed page = PDF page +267.
- **P96-K**: J. Pöschel, *A KAM-theorem for some nonlinear partial differential equations*, Ann. Scuola Norm. Sup. Pisa (4) **23** (1996), 119–148 (Numdam scan). Printed page = PDF page +117.
- **KP96**: S. Kuksin, J. Pöschel, *Invariant Cantor manifolds of quasi-periodic oscillations for a nonlinear Schrödinger equation*, Ann. of Math. **143** (1996), 149–179; *author version 2.1*, cited by its own page numbers (not the journal's).

Versions and hashes are those of `docs/explorations/pde-kam-and-leapfrogging.md`, §1. Every formula of a scanned source relied on below was read on a page image, not only on the OCR layer.

1.2. The equation.

Standing assumption 1.1 (Mass)**.** $m > 0$ is fixed. All constants may depend on m unless stated otherwise; no uniformity as $m \to 0$ or $m \to \infty$ is claimed anywhere.

Definition 1.2 (Cubic Dirichlet wave equation)**.** A *classical solution* on an open interval $I \subset \mathbb{R}$ is a function $u : I \times [0,\pi] \to \mathbb{R}$ such that $u, u_t, u_x, u_{tt}, u_{xx}$ exist and are continuous on $I \times [0,\pi]$ (one-sided in x at the endpoints) and

$$u_{tt} = u_{xx} - mu - u^3 \quad \text{on } I \times [0,\pi], \qquad u(t,0) = u(t,\pi) = 0 \quad (t \in I). \tag{1}$$

Source: P96-W (1) and p. 272 with $f(u) = u^3$; the source writes $u_{tt} = u_{xx} - mu - f(u)$ with f odd, analytic, $f(u) = au^3 + O(u^5)$, $a \neq 0$; we take $a = 1$ and no higher terms (declared specialization).

Definition 1.3 (Dirichlet basis and linear frequencies)**.** For $j \geq 1$ let $\phi_j(x) = \sqrt{2/\pi}\,\sin(jx)$ and $\lambda_j = \sqrt{j^2 + m} > 0$. Then $\int_0^\pi \phi_i \phi_j \, dx = \delta_{ij}$, $-\phi_j'' + m\phi_j = \lambda_j^2 \phi_j$ and $\phi_j(0) = \phi_j(\pi) = 0$.

Source: P96-W p. 277, identical.

We record the elementary facts about λ used repeatedly: $j < \lambda_j < j + \sqrt{m}$, $j/\lambda_j < 1$, the map $j \mapsto \lambda_j$ is strictly increasing, the function $f(s) = \sqrt{s^2 + m}$ is strictly convex on $\mathbb{R}$ with $0 < f'(s) = s/f(s) < 1$ for $s > 0$, f' is strictly increasing, and $\lambda_{j+1} - \lambda_j = (2j+1)/(\lambda_{j+1} + \lambda_j) < 1$.

1.3. **Sequence spaces.**

Definition 1.4 (Weighted sequence spaces)**.** For $a \geq 0$ and $p \in \mathbb{R}$, $\ell^{a,p}$ is the space of real sequences $w = (w_j)_{j\geq 1}$ with

$$\|w\|_{a,p}^2 = \sum_{j\geq 1} |w_j|^2 \, j^{2p} e^{2aj} < \infty,$$

and $\ell^{a,p}_{\mathbb{C}}$ its complex analogue. The phase space is $E_{a,p} = \ell^{a,p} \times \ell^{a,p}$ with $\|(q,p)\|_E = \max(\|q\|_{a,p}, \|p\|_{a,p})$. The symbol p in $\ell^{a,p}$ is the regularity exponent; the second component of $E_{a,p}$ is the momentum, also written p; context separates them.

Source: P96-W p. 277 (the source's s is our p).

Convention 1.5 (Headline weights)**.** The headline and milestone W0 use $a = 1$ and $p = 1$: phase space $E_{1,1}$, nonlinear field in $\ell^{1,2}$. The source allows any $a > 0$, $s > 0$; this specialization removes two headline parameters. Lower exponents $\ell^{1,p}$, $p \leq 1$, appear only as target spaces of the unbounded linear part.

Definition 1.6 (Two-sided spaces)**.** For $a \geq 0$, $s \in \mathbb{R}$, $\ell^{a,s}_b$ is the space of complex sequences $(c_l)_{l\in\mathbb{Z}}$ with

$$\|c\|_{a,s}^2 = \sum_{l\in\mathbb{Z}} \langle l\rangle^{2s} e^{2a|l|} |c_l|^2 < \infty, \qquad \langle l\rangle = \max(1, |l|).$$

At $l = 0$ the weight is 1, so this is exactly the norm $|c_0|^2 + \sum_{l\neq 0} |c_l|^2 |l|^{2s} e^{2|l|a}$ of P96-W p. 279. The convolution is $(c * d)_l = \sum_{i\in\mathbb{Z}} c_i d_{l-i}$.

1.4. **Coordinates, Hamiltonian and equations of motion.**

Definition 1.7 (Reconstruction)**.** For $(q,p) \in E_{a,p}$ with $a > 0$ put

$$U_m(q)(x) = \sum_{j\geq 1} \frac{q_j}{\sqrt{\lambda_j}} \phi_j(x), \qquad V_m(p)(x) = \sum_{j\geq 1} \sqrt{\lambda_j}\, p_j \phi_j(x).$$

The intended dictionary is $u = U_m(q)$, $u_t = V_m(p)$. Equivalently, if $u = \sum x_j \phi_j$ and $u_t = \sum y_j \phi_j$, then $q_j = \sqrt{\lambda_j}\, x_j$ and $p_j = y_j/\sqrt{\lambda_j}$; this is a symplectic rescaling, $dq_j \wedge dp_j = dx_j \wedge dy_j$.

Source: P96-W p. 277 ($u = \mathcal{S}q$, $v = \mathcal{S}'p$), identical.

Lemma 2.1 shows that for $a > 0$ these series converge absolutely and uniformly on $[0,\pi]$ together with all their x-derivatives, for every $p \in \mathbb{R}$.

Definition 1.8 (Hamiltonian)**.** For $(q,p) \in E_{1,1}$,

$$H_m(q,p) = \Lambda(q,p) + G(q), \qquad \Lambda(q,p) = \frac{1}{2}\sum_{j\geq 1} \lambda_j \big(q_j^2 + p_j^2\big), \qquad G(q) = \frac{1}{4}\int_0^\pi U_m(q)(x)^4 \, dx.$$

Λ is finite on $E_{a,p}$ for $p \geq \frac{1}{2}$; in particular on $E_{1,1}$.

Source: P96-W (8) p. 280 and (9) p. 280 with $g(u) = u^4/4$; the energy $\frac{1}{2}\langle v, v\rangle + \frac{1}{2}\langle Au, u\rangle + \int g(u)$ of p. 277 in the coordinates of Definition 1.7.

Definition 1.9 (Cubic gradient)**.** For $q \in \ell^{1,1}$ let $G_q(q)$ be the sequence

$$G_q(q)_l = \frac{1}{\sqrt{\lambda_l}} \int_0^\pi U_m(q)(x)^3 \phi_l(x)\, dx \qquad (l \geq 1).$$

By Lemma 2.3, $G_q(q)_l = \partial G/\partial q_l$; by Theorem 2.5 it lies in $\ell^{1,2}$.

Definition 1.10 (Symplectic form and equations of motion)**.** The symplectic form is $\sum_j dq_j \wedge dp_j$ and the Hamiltonian equations are $\dot q_j = \partial H/\partial p_j$, $\dot p_j = -\partial H/\partial q_j$, i.e.

$$\dot q_j = \lambda_j p_j, \qquad \dot p_j = -\lambda_j q_j - G_q(q)_j \qquad (j \geq 1). \tag{2}$$

The linear part $q \mapsto (\lambda_j q_j)$ maps $\ell^{a,p+1}$ boundedly to $\ell^{a,p}$ (norm $\leq \sqrt{1+m}$) but not $\ell^{a,p}$ to itself; the system is read in $\ell^{1,0}$ for $(q,p) \in E_{1,1}$, or coordinatewise. The nonlinear part G_q is smoothing of order one (Theorem 2.5).

Source: P96-W (6) p. 277 and p. 280 ($X_G = \sum(\partial_{p_j} G\, \partial_{q_j} - \partial_{q_j} G\, \partial_{p_j})$), identical.

1.5. **Complex coordinates, actions, angles.**

Definition 1.11 (Complex coordinates)**.** $z_j = (q_j + ip_j)/\sqrt{2}$, $\bar z_j = (q_j - ip_j)/\sqrt{2}$, so $q_j = \sqrt{2}\,\mathrm{Re}\, z_j$, $p_j = \sqrt{2}\,\mathrm{Im}\, z_j$, $|z_j|^2 = (q_j^2 + p_j^2)/2$, and

$$\sum_j dq_j \wedge dp_j = i \sum_j dz_j \wedge d\bar z_j, \qquad \Lambda = \sum_j \lambda_j |z_j|^2.$$

Under the linear flow $\dot q_j = \lambda_j p_j$, $\dot p_j = -\lambda_j q_j$ one has $\dot z_j = -i\lambda_j z_j$.

Source: P96-W p. 281 (definition of z, form $i\sum dz \wedge d\bar z$, $\Lambda = \sum \lambda_j |z_j|^2$), identical; the recovery formula and the rotation sign printed later in P96-W differ, see §1.6.

Definition 1.12 (Actions and radian angles)**.** The (oscillator) action of mode j is $I_j = (q_j^2 + p_j^2)/2 = |z_j|^2$. On $\{I_j > 0\}$ the radian angle φ_j is defined by

$$q_j = \sqrt{2I_j}\cos\varphi_j, \qquad p_j = -\sqrt{2I_j}\sin\varphi_j, \qquad z_j = \sqrt{I_j}\, e^{-i\varphi_j}.$$

Then $dq_j \wedge dp_j = d\varphi_j \wedge dI_j$ (Jacobian determinant 1), $\Lambda = \sum \lambda_j I_j$, and the linear flow is $\dot\varphi_j = \lambda_j$, $\dot I_j = 0$. For tangential modes $j \in J = \{1,2\}$ we write $I = (I_1, I_2)$; for normal modes $j \geq 3$ we write $Z_j = I_j$.

Convention 1.13 (Period-one torus angles and frequencies)**.** Torus parameters $\theta \in \mathbb{R}^2$ have period one: $\varphi_j = 2\pi\theta_j$. A torus with radian frequency vector ω (period 2π) has campaign frequency $\nu = \omega/(2\pi)$. The pair (θ, Y) with $Y = 2\pi I$ is canonical, $d\varphi \wedge dI = d\theta \wedge dY$. Small divisors in campaign units are $2\pi\, k \cdot \nu + l \cdot \Omega$. The linear torus of action $I \in (0,\infty)^2$ is

$$K_I^0(\theta) = (Q^0, P^0), \qquad Q_j^0 = \sqrt{2I_j}\cos(2\pi\theta_j), \quad P_j^0 = -\sqrt{2I_j}\sin(2\pi\theta_j) \quad (j = 1,2),$$

and $Q_j^0 = P_j^0 = 0$ for $j \geq 3$, which is invariant under the linear flow with $\nu = (\lambda_1, \lambda_2)/(2\pi)$.

1.6. **Comparison with the printed conventions of the sources.**

The following table maps each normalization to P96-W (and, where relevant, KP96). "Same" means checked identical on the page image. Details and derivations are in `docs/tracks/W/errata-check-W0.md`.

Object	This blueprint	P96-W (printed)
basis, λ_j	$\phi_j = \sqrt{2/\pi}\sin jx$, $\lambda_j = \sqrt{j^2+m}$	same (p. 277)
$\ell^{a,p}$	$\sum \lvert w_j\rvert^2 j^{2p} e^{2aj}$	same, exponent called s (p. 277)
two-sided $\ell^{a,s}_b$	weight $\langle l\rangle^{2s} e^{2a\lvert l\rvert}$	same (p. 279)
u, u_t	$U_m(q)$, $V_m(p)$	same, $\mathcal{S}q$, $\mathcal{S}'p$ (p. 277)
H	$\frac12\sum\lambda_j(q_j^2+p_j^2)+\frac14\int U^4$	same, (8)–(9) p. 280
quartic coefficient	$g_{ijkl}=\int_0^\pi \phi_i\phi_j\phi_k\phi_l$ (unweighted)	$G_{ijkl}=g_{ijkl}/\sqrt{\lambda_i\lambda_j\lambda_k\lambda_l}$, (10) p. 281; (11) correct
symplectic form	$\sum dq\wedge dp = i\sum dz\wedge d\bar z$	same (pp. 278, 281)
z	$(q+ip)/\sqrt2$	same (p. 281)
recovery of q,p	$q=\sqrt2\,\mathrm{Re}\,z$, $p=\sqrt2\,\mathrm{Im}\,z$	$q=\mathrm{Re}\,z$, $p=\mathrm{Im}\,z$ (p. 290): **disagrees**
action	$I_j=\lvert z_j\rvert^2=(q_j^2+p_j^2)/2$	$I_j=q_j^2+p_j^2$ (p. 276, informal); $I=\frac12\lvert z\rvert^2$ (p. 287); $\Lambda=\langle\alpha,I\rangle$ with $\alpha=\lambda$ (p. 290) requires $I=\lvert z\rvert^2$: **inconsistent**
quartic in z	$\frac{1}{16}\sum g_{ijkl}\prod(z+\bar z)/\sqrt{\lambda\lambda\lambda\lambda}$	$\frac14\sum G_{ijkl}\prod(z+\bar z)$, then $\sum' G_{ijkl}w^4$ (p. 282): **two factor-4 slips**
normal form	$\frac12\langle AI,I\rangle+\langle BI,Z\rangle$, $A_{ii}=\frac{9}{8\pi\lambda_i^2}$, $A_{ij}=B_{ji}=\frac{3}{2\pi\lambda_i\lambda_j}$	$\bar G_{ij}=\frac{6}{\pi}\frac{4-\delta_{ij}}{\lambda_i\lambda_j}$ (pp. 282, 284), $A=B=\bar G$ (p. 290): 16× ours
rotation sign	$\dot z_j=-i\lambda_j z_j$, φ increases	$\dot z=i\omega z$, flow $e^{i\omega t}$ (pp. 287, 289–290): time-reversed
torus angles	period one, $\nu=\omega/2\pi$	period 2π

Remark 1.14 (Why the disagreements are harmless for existence but fatal for constants). Every nondegeneracy condition used by the source (P96-W (A_1)–(A_3), p. 288) is invariant under multiplying (A,B) by a common positive constant and under rescaling I, and H is even in p, so the time reversal $(q,p)(t)\mapsto(q,-p)(-t)$ maps solutions to solutions. Hence the qualitative theorem is unaffected. The printed constants, frequency shifts, and the amplitude–action dictionary are not, and must not be copied.

2. Milestone W0: cubic attachment and quartic normalization

Throughout, Standing assumption 1.1 ($m>0$) and the conventions of §1 are in force. The results of this part are: (a) the coefficient system (2) is equivalent to the PDE (Proposition 2.6); (b) $G_q:\ell^{1,1}\to\ell^{1,2}$ is a bounded cubic polynomial with an explicit constant (Theorem 2.5); (c) closed form of the quartic coefficients (Lemma 2.2); (d) the action coefficients A,B of the partial quartic Birkhoff normal form for $J=\{1,2\}$ (Theorem 2.10); (e) the nondegeneracy conditions of the source for every $m>0$ (Theorem 2.13); (f) the Duffing consistency check (Proposition 2.15). All proofs are complete except where a step is explicitly cited to a pinned source and labelled as such; those citations belong to later milestones (W1, W3) and are listed at the end (§2.9).

2.1. **Reconstruction.**

Lemma 2.1 (Reconstruction is bounded into C^k)**.** *Let $a>0$, $p\in\mathbb{R}$, $k\in\mathbb{N}$. For $c\in\ell^{a,p}_{\mathbb{C}}$,*

$$\sum_{j\ge1}\lvert c_j\rvert\, j^k \le M_{a,p,k}\|c\|_{a,p},\qquad M_{a,p,k}^2=\sum_{j\ge1} j^{2k-2p}e^{-2aj}<\infty.$$

Consequently the series $\sum_j c_j\phi_j$ and all its termwise x-derivatives of order $\le k$ converge absolutely and uniformly on $[0,\pi]$; $c\mapsto\sum_j c_j\phi_j$ is a bounded linear map $\ell^{a,p}\to C^k([0,\pi])$ of norm $\le\sqrt{2/\pi}\,M_{a,p,k}$, its value vanishes at $x=0,\pi$, and $\int_0^\pi(\sum_j c_j\phi_j)\phi_l\,dx=c_l$. In particular U_m, V_m and $q\mapsto U_m(\lambda q)$ are bounded linear maps $\ell^{a,p}\to C^k([0,\pi])$ for every $p\in\mathbb{R}$, because $\lambda_j^{\pm1/2}\le$

$(1+m)^{1/4} j^{1/2}$ and multiplication by (λ_j^r) maps $\ell^{a,p}$ into $\ell^{a,p-|r|}$ for $|r| \le 2$, $r \ge 0$, with norm $\le (1+m)^{r/2}$.

Proof. Cauchy–Schwarz: $\sum |c_j| j^k = \sum (|c_j| j^p e^{aj})(j^{k-p} e^{-aj}) \le \|c\|_{a,p} M_{a,p,k}$, and $M_{a,p,k} < \infty$ because $a > 0$. Since $|\partial_x^r \phi_j| \le \sqrt{2/\pi}\, j^r$, the Weierstrass M-test gives uniform convergence of the derivative series of order $r \le k$, hence termwise differentiation. Boundary values and the coefficient formula follow termwise from $\phi_j(0) = \phi_j(\pi) = 0$ and orthonormality. The bound on λ_j^r is $j^2 + m \le (1+m) j^2$. □

Source: P96-W Lemma 1, p. 278, first two sentences (asserted there without the estimate).

2.2. **The quartic coefficients.**

Lemma 2.2 (Closed form, (c))**.** *For $i, j, k, l \ge 1$ let $g_{ijkl} = \int_0^\pi \phi_i \phi_j \phi_k \phi_l \, dx$. Then*

$$g_{ijkl} = \frac{1}{2\pi}\Big([i+l=j+k] + [i+k=j+l] + [i+j=k+l] \\ - [i=j+k+l] - [j=i+k+l] - [k=i+j+l] - [l=i+j+k] \Big),$$

where $[\cdot]$ is the Iverson bracket. In particular g is symmetric, $g_{ijkl} = 0$ unless $\pm i \pm j \pm k \pm l = 0$ for some choice of signs, $g_{iiii} = \frac{3}{2\pi}$, and $g_{iijj} = \frac{1}{\pi}$ for $i \ne j$. Equivalently, for $l \ge 1$ and $i, j, k \ge 1$,

$$g_{ijkl} = \frac{1}{2\pi} \sum_{\varepsilon \in \{\pm 1\}^3} s(\varepsilon)\, [\, l = \varepsilon_1 i + \varepsilon_2 j + \varepsilon_3 k \,], \qquad s(\varepsilon) = (-1)^{1 + \#\{t : \varepsilon_t = -1\}}, \tag{3}$$

so that $|g_{ijkl}| \le \frac{1}{2\pi} \sum_\varepsilon [\, l = \varepsilon_1 i + \varepsilon_2 j + \varepsilon_3 k \,]$.

Proof. **Step S2.1: Product to sum.** $\sin\alpha \sin\beta = \frac{1}{2}(\cos(\alpha-\beta) - \cos(\alpha+\beta))$ gives

$$\phi_i \phi_j \phi_k \phi_l = \frac{1}{\pi^2} \big(\cos(i-j)x - \cos(i+j)x\big)\big(\cos(k-l)x - \cos(k+l)x\big).$$

Step S2.2: Cosine integrals. For integers A, B: $\int_0^\pi \cos Ax \cos Bx \, dx = \frac{\pi}{2}\big([A=B] + [A=-B]\big)$ (this includes $A = B = 0$, value π). Hence

$$g_{ijkl} = \frac{1}{2\pi}\Big([i-j=k-l] + [i-j=l-k] - [i-j=k+l] - [i-j=-k-l] \\ - [i+j=k-l] - [i+j=l-k] + [i+j=k+l] + [i+j=-k-l] \Big).$$

The last bracket vanishes for positive indices; rewriting the others gives the stated form. The special values are read off: for (i,i,i,i) the three positive brackets hold and no negative one; for (i,i,j,j), $i \ne j$, exactly $[i+l=j+k]$ and $[i+k=j+l]$ hold (a negative bracket would force one index to be the sum of the other three, e.g. $i = i + 2j$, impossible).

Step S2.3: Signed form. The seven relations, solved for l, are $l = -i+j+k$, $l = i-j+k$, $l = i+j-k$ (sign $+$, one minus), $l = i-j-k$, $l = -i+j-k$, $l = -i-j+k$ (sign $-$, two minuses), $l = i+j+k$ (sign $-$, no minus); the eighth pattern $l = -i-j-k$ is impossible for $l \ge 1$ and may be added with either sign. This is (3). □

Source: P96-W (10)–(11), p. 281 (only the selection rule and G_{iijj} are stated, "by an elementary calculation – see [6]"); KP96 p. 13 (computation of G_{ijij}). The weighted coefficient of P96-W is $G_{ijkl} = g_{ijkl}/\sqrt{\lambda_i \lambda_j \lambda_k \lambda_l}$; its (11) agrees with the lemma.

2.3. **The cubic gradient.**

Lemma 2.3 (G and its gradient as absolutely convergent sums)**.** *Let $a > 0$, $p \geq 0$, $q \in \ell^{a,p}$ and write $\hat{q}_j = q_j/\sqrt{\lambda_j}$. Then*

$$G(q) = \frac{1}{4} \sum_{i,j,k,l \geq 1} g_{ijkl} \hat{q}_i \hat{q}_j \hat{q}_k \hat{q}_l, \qquad G_q(q)_l = \frac{1}{\sqrt{\lambda_l}} \sum_{i,j,k \geq 1} g_{ijkl} \hat{q}_i \hat{q}_j \hat{q}_k,$$

both sums converging absolutely; G is Gâteaux differentiable along every coordinate with $\partial G/\partial q_l = G_q(q)_l$; and $\int_0^\pi U_m(q)^3 \phi_l \, dx = \sqrt{\lambda_l}\, G_q(q)_l$.

Proof. By Lemma 2.1 with $k = 0$, $S = \sum_j |\hat{q}_j| < \infty$ and $U_m(q) = \sum \hat{q}_j \phi_j$ uniformly. Since $|\phi_j| \leq \sqrt{2/\pi}$, the quadruple series $\sum \hat{q}_i \hat{q}_j \hat{q}_k \hat{q}_l \phi_i \phi_j \phi_k \phi_l$ is dominated by $(2/\pi)^2 S^4$ uniformly in x; integrate termwise (dominated convergence for series) to obtain both formulas and their absolute convergence. Differentiating $\frac{1}{4} \int U_m(q + t e_l)^4$ at $t = 0$ under the integral (the integrand is a polynomial in t with coefficients continuous in x) gives $\int U_m(q)^3 \phi_l / \sqrt{\lambda_l}$, which is $G_q(q)_l$ by Definition 1.9 and the triple-series formula. □

Lemma 2.4 (Weighted convolution algebra)**.** *Let $a \geq 0$ and $s \geq 1$. For $c, d \in \ell_b^{a,s}$,*

$$\|c * d\|_{a,s} \leq K_s \|c\|_{a,s} \|d\|_{a,s}, \qquad K_s = 2^s \Big(1 + 2\zeta(2s)\Big)^{1/2}.$$

In particular $K_{3/2}^2 = 8(1 + 2\zeta(3))$. The convolution sum converges absolutely for each l.

Proof. Let $w_l = \langle l \rangle^s e^{a|l|}$, $C_l = w_l |c_l|$, $D_l = w_l |d_l|$.

Step S2.4: Weight inequality. For $i, l \in \mathbb{Z}$: $e^{a|l|} \leq e^{a|i|} e^{a|l-i|}$ and $\langle l \rangle \leq \langle i \rangle + \langle l - i \rangle$ (if $|l| \leq 1$ the left side is 1; otherwise $|l| \leq |i| + |l - i|$). By convexity of $t \mapsto t^s$, $(\alpha + \beta)^s \leq 2^{s-1}(\alpha^s + \beta^s)$, so $w_l \leq 2^{s-1}(w_i e^{a|l-i|} + e^{a|i|} w_{l-i}) \leq 2^{s-1}(w_i w_{l-i}/\langle l - i \rangle^s + w_i w_{l-i}/\langle i \rangle^s)$.

Step S2.5: Young and Cauchy–Schwarz. Hence $w_l |(c * d)_l| \leq 2^{s-1} \sum_i \big(C_i \, |d_{l-i}| e^{a|l-i|} + |c_i| e^{a|i|} D_{l-i} \big) = 2^{s-1} \big((C * D')_l + (C' * D)_l \big)$ with $C_i' = C_i/\langle i \rangle^s$, $D_i' = D_i/\langle i \rangle^s$. Young's inequality $\|X * Y\|_{\ell^2} \leq \|X\|_{\ell^2} \|Y\|_{\ell^1}$ and $\|D'\|_{\ell^1} \leq \|D\|_{\ell^2} (\sum_i \langle i \rangle^{-2s})^{1/2} = \|D\|_{\ell^2} (1 + 2\zeta(2s))^{1/2}$ give the claim; the same bounds give absolute convergence of each convolution sum. □

Source: P96-W Lemma 2, p. 279 and Appendix A, pp. 294–295; KP96 Lemma 2 p. 12 and Appendix A, pp. 31–32 (constant "depending only on s", not made explicit there). This lemma is the W1 node `bpW:w1:algebra`; it is proved here because (b) needs it.

Theorem 2.5 (Cubic gradient, (b))**.** *Let $a \geq 0$, $p \geq \frac{1}{2}$, and define the symmetric trilinear map T on $(\ell_{\mathbb{C}}^{a,p})^3$ by*

$$T(q, q', q'')_l = \frac{1}{\sqrt{\lambda_l}} \sum_{i,j,k \geq 1} g_{ijkl} \, \hat{q}_i \hat{q}_j' \hat{q}_k'' \qquad (\hat{q}_j = q_j/\sqrt{\lambda_j}).$$

The sums converge absolutely, T is a bounded trilinear map into $\ell_{\mathbb{C}}^{a,p+1}$, real on real arguments, and

$$\|T(q, q', q'')\|_{a,p+1} \leq C_* \|q\|_{a,p} \|q'\|_{a,p} \|q''\|_{a,p}, \qquad C_* = \frac{K_{p+1/2}^2}{\pi}.$$

For $a > 0$, $G_q(q) = T(q, q, q)$ (Lemma 2.3). Consequently, for $a = p = 1$, $G_q : \ell^{1,1} \to \ell^{1,2}$ is real analytic (indeed a continuous cubic polynomial), and

$$\|G_q(q)\|_{1,2} \leq C_m \|q\|_{1,1}^3, \qquad C_m = \frac{8(1 + 2\zeta(3))}{\pi} \approx 8.67.$$

The constant is independent of $m > 0$ (we only use $\lambda_j \geq j$).

Proof. **Step S2.6: Even majorants.** Given $q \in \ell^{a,p}_{\mathbb{C}}$ define the even two-sided sequence $x_i = |q_{|i|}|/\sqrt{\lambda_{|i|}}$ for $i \neq 0$, $x_0 = 0$; similarly x', x''. Using $\lambda_i \geq i$,

$$\|x\|^2_{a,p+1/2} = 2\sum_{i\geq 1} \frac{i}{\lambda_i}\, i^{2p} e^{2ai} |q_i|^2 \leq 2\|q\|^2_{a,p}.$$

Step S2.7: Selection rule bound. By (3), for $l \geq 1$,

$$\sum_{i,j,k\geq 1} |g_{ijkl}|\, |\hat{q}_i \hat{q}'_j \hat{q}''_k| \leq \frac{1}{2\pi} \sum_{i,j,k\geq 1} \sum_{\varepsilon} [\, l = \varepsilon_1 i + \varepsilon_2 j + \varepsilon_3 k\,]\, x_i x'_j x''_k \leq \frac{1}{2\pi}(x * x' * x'')_l,$$

since $(i,j,k,\varepsilon) \mapsto (\varepsilon_1 i, \varepsilon_2 j, \varepsilon_3 k)$ is injective into $(\mathbb{Z}\setminus 0)^3$ and x is even. Thus the sum defining T converges absolutely.

Step S2.8: Smoothing. With $X = x * x' * x''$ (even, by evenness of the factors) and $l/\lambda_l \leq 1$,

$$\|T\|^2_{a,p+1} \leq \sum_{l\geq 1} \frac{l^{2p+2} e^{2al}}{\lambda_l} \frac{X_l^2}{4\pi^2} \leq \frac{1}{4\pi^2} \sum_{l\geq 1} l^{2p+1} e^{2al} X_l^2 \leq \frac{1}{8\pi^2} \|X\|^2_{a,p+1/2}.$$

By Lemma 2.4 with $s = p + \frac{1}{2} \geq 1$, $\|X\|_{a,p+1/2} \leq K_s^2 \|x\|\|x'\|\|x''\| \leq K_s^2 2^{3/2} \|q\|\|q'\|\|q''\|$. Hence $\|T\|_{a,p+1} \leq (2\sqrt{2}\pi)^{-1} 2^{3/2} K_s^2 \prod \|\cdot\| = (K_s^2/\pi) \prod \|\cdot\|$.

Step S2.9: Analyticity. Trilinearity and symmetry are clear from the formula and the symmetry of g. A bounded symmetric trilinear map T between complex Banach spaces defines the continuous homogeneous cubic polynomial $q \mapsto T(q,q,q)$, which is analytic on the whole space, with the finite power series $T(q_0+h)^3$ expanded by multilinearity (in Lean: a `ContinuousMultilinearMap` is analytic). It maps real sequences to real sequences. For $a > 0$ the identification with G_q is Lemma 2.3. □

Source: P96-W Lemma 3, pp. 279–280 (with $O(\cdot)$ instead of a constant) and (13)–(14), p. 284; KP96 p. 16. The source deduces analyticity from coordinatewise analyticity and local boundedness [P96-W ref. 9, Appendix A]; the polynomial structure makes that principle unnecessary here.
SOURCE_GAP: P96-W p. 280 asserts $\|\mathcal{S}q\|_{a,s+1/2} \leq \|q\|_{a,s}$ in the two-sided norm. The correct bound is $\leq \sqrt{2}\|q\|_{a,s}$ (Step S2.6); the operator norm is exactly $\sqrt{2}$ for every fixed $m > 0$ (take $q = e_j$, $j \to \infty$), so the printed bound fails for every $m > 0$ (e.g. $q = e_1$, $a = s = 0$, $m = 1$). Constants only; recorded as new item N-1 in the errata check.

2.4. PDE ↔ coefficient system.

Proposition 2.6 (Equivalence, (a)). *Let $I \subset \mathbb{R}$ be an open interval and $q, p : I \to \ell^{1,1}$.*

(i) (Coefficients ⇒ PDE.) Suppose q, p are continuous into $\ell^{1,1}$, continuously differentiable into $\ell^{1,0}$, and satisfy (2) as an identity in $\ell^{1,0}$. Then $u(t,x) = U_m(q(t))(x)$ is a classical solution of (1) on I (Definition 1.2) with $u_t(t,\cdot) = V_m(p(t))$.

(ii) (PDE ⇒ coefficients.) Suppose $u(t,x) = U_m(q(t))(x)$ is a classical solution of (1) on I and $u_t(t,\cdot) = V_m(p(t))$ for every t. Then every q_j, p_j is C^1 on I and (2) holds coordinatewise.

In both directions the cubic term is realized exactly: for $q \in \ell^{1,1}$,

$$U_m(q)^3 = \sum_{l\geq 1} \sqrt{\lambda_l}\, G_q(q)_l\, \phi_l \quad \text{uniformly on } [0,\pi], \qquad -\partial_x^2 U_m(q) + m U_m(q) = U_m(\lambda^2 q), \tag{4}$$

where $(\lambda^2 q)_j = \lambda_j^2 q_j$ and the right sides converge in C^0 (first) and C^0 (second, as $\lambda^2 q \in \ell^{1,-1}$).

Proof. **Step S2.10: Force identity.** Let $f = U_m(q)^3 \in C([0,\pi])$. By Lemma 2.3, $\int f\phi_l = \sqrt{\lambda_l} G_q(q)_l$; by Theorem 2.5 and Lemma 2.1 (the sequence $(\sqrt{\lambda_l} G_q(q)_l)$ lies in $\ell^{1,1}$), $S = \sum_l \sqrt{\lambda_l} G_q(q)_l \phi_l$ converges uniformly to a continuous S with the same sine coefficients as

f. Since $(\phi_l)_{l\ge1}$ is a complete orthonormal system of $L^2(0,\pi)$, $f = S$ almost everywhere, hence everywhere by continuity. The second identity is termwise differentiation (Lemma 2.1 with $k = 2$) and $-\phi_j'' + m\phi_j = \lambda_j^2\phi_j$.

Step S2.11: Time regularity. In (i), $\dot q = \lambda p$ is continuous into $\ell^{1,0}$, and since p is C^1 into $\ell^{1,0}$ and multiplication by λ is bounded $\ell^{1,0} \to \ell^{1,-1}$, $\dot q$ is C^1 into $\ell^{1,-1}$ with $\ddot q = \lambda\dot p = -\lambda^2 q - \lambda G_q(q)$. As $U_m : \ell^{1,-1} \to C^2([0,\pi])$ is bounded linear, $t \mapsto u(t,\cdot)$ is C^2 into $C^2([0,\pi])$; therefore $u, u_t, u_{tt}, u_x, u_{xx}$ exist and are jointly continuous, $u_t = U_m(\lambda p) = V_m(p)$ and $u_{tt} = U_m(\ddot q)$.

Step S2.12: Identification. By (4), $U_m(\lambda^2 q) = -u_{xx} + mu$ and $U_m(\lambda G_q(q)) = \sum_l \sqrt{\lambda_l} G_q(q)_l \phi_l = u^3$; hence $u_{tt} = u_{xx} - mu - u^3$. The boundary conditions hold by Lemma 2.1.

Step S2.13: Converse. In (ii), $\int_0^\pi u(t,x)\phi_l(x)\,dx = q_l(t)/\sqrt{\lambda_l}$ and $\int_0^\pi u_t\phi_l = \sqrt{\lambda_l}p_l$. Differentiation under the integral sign (u_t, u_{tt} continuous on compact sets) gives $\dot q_l = \sqrt{\lambda_l}\int u_t\phi_l = \lambda_l p_l$ and $\sqrt{\lambda_l}\dot p_l = \int u_{tt}\phi_l$. Two integrations by parts, using $u(t,0) = u(t,\pi) = 0 = \phi_l(0) = \phi_l(\pi)$, give $\int u_{xx}\phi_l = -l^2\int u\phi_l$, so $\int u_{tt}\phi_l = -\lambda_l^2 q_l/\sqrt{\lambda_l} - \sqrt{\lambda_l}G_q(q)_l$ by Lemma 2.3. Divide by $\sqrt{\lambda_l}$. □

Source: P96-W Lemma 1, p. 278, and (7), p. 278.
Expansion/strengthening of the source proof: P96-W Lemma 1 (p. 278) argues via convergence in complex neighbourhoods, termwise time differentiation and completeness, for real-analytic curves. The proof above expands these steps and proves a stronger C^1-curve version (i); this is not a source gap. (Separately, the second line of the source's u_{tt} display drops a minus sign on $\lambda_j^2 q_j$: errata item N-4; the proof above uses the correct sign.)
completeness of (ϕ_l) in $L^2(0,\pi)$ is a standard external fact (Fourier basis of L^2 on the circle applied to odd extensions). The all-time real-analytic version used by the headline is W2.

2.5. **Quartic nonresonance.**

Lemma 2.7 (Qualitative nonresonance of all quartic terms)**.** *Let $m > 0$, let $a, b, c, d \ge 1$ with $g_{abcd} \ne 0$, and let $\sigma \in \{\pm1\}^4$. If $\sigma_1\lambda_a + \sigma_2\lambda_b + \sigma_3\lambda_c + \sigma_4\lambda_d = 0$, then the indices with $\sigma = +1$ and those with $\sigma = -1$ form equal multisets (in particular there are two of each).*

Proof. By Lemma 2.2, $g_{abcd} \ne 0$ implies a signed relation $\pm a \pm b \pm c \pm d = 0$; in particular each index is at most the sum of the other three.

Step S2.14: Four or three equal signs. All signs equal is impossible ($\lambda > 0$). If three signs agree, say $\lambda_a + \lambda_b + \lambda_c = \lambda_d$, then $d \le a + b + c$, so $\lambda_d \le \lambda_{a+b+c}$, while $\lambda_x + \lambda_y > \lambda_{x+y}$ for $x, y \ge 1$ (square both sides: $2\sqrt{(x^2+m)(y^2+m)} > 2xy - m$); hence $\lambda_a + \lambda_b + \lambda_c > \lambda_{a+b+c} \ge \lambda_d$.

Step S2.15: Two against two. Let $\lambda_a + \lambda_b = \lambda_c + \lambda_d$ with $\{a,b\} \ne \{c,d\}$ as multisets, and let $f(s) = \sqrt{s^2+m}$. Up to renaming within and between the two sides, the signed relation is one of: (1) $a+b = c+d$: for fixed sum, $f(x)+f(y)$ is strictly increasing in $|x-y|$ by strict convexity of f, so equality forces $|a-b| = |c-d|$ and $\{a,b\} = \{c,d\}$; (2) $a-b = d-c =: t$ (that is, $a+c = b+d$): then $f(b+t)+f(b) = f(c)+f(c+t)$ and $y \mapsto f(y)+f(y+t)$ is strictly increasing on $\{y \ge 1,\ y+t \ge 1\}$, so $b = c$, $a = d$; (3) one index is the sum of the other three, and after renaming it is $a = b+c+d$ (the side containing the largest index is called the left side). Then $\lambda_a - \lambda_c = \int_c^{b+c+d} f'$ and $\lambda_d - \lambda_b = \int_b^d f'$ (nonpositive if $d \le b$). Since $f' > 0$ is increasing, $\int_c^{c+b+d} f' > \int_{c+2b}^{c+b+d} f' \ge \int_b^d f'$ when $d > b$ (an interval of the same length $d-b$ shifted right by $c+b > 0$). So $\lambda_a + \lambda_b > \lambda_c + \lambda_d$. Every signed relation among four positive integers with two signs on each side is of type (1) or (2) after renaming, and every relation with one sign different is of type (3); this exhausts the cases. □

Remark 2.8. The lemma is qualitative and covers *all* indices. It does not give a uniform lower bound: divisors such as $\lambda_{j+1} - \lambda_j - \lambda_{k+1} + \lambda_k$ tend to 0, which is why P96-W (p. 276) says no complete quartic normal form is available. The quantitative bound for quartic terms with at least

one index in $\{1,\dots,n\}$, $|\lambda'_i+\lambda'_j+\lambda'_k+\lambda'_l| \geq cm/\sqrt{n^2+m}^3$, is P96-W Lemma 4, p. 283 (proof pp. 285–286); it belongs to W3 and is cited, not reproved, here.

2.6. **Partial quartic Birkhoff normal form: the action coefficients.** Write $\hat\lambda = (\lambda_j)$ and, in the complex coordinates of Definition 1.11, $q_j = (z_j+\bar z_j)/\sqrt{2}$. Then

$$G = \frac{1}{4}\sum_{i,j,k,l} \frac{g_{ijkl}}{\sqrt{\lambda_i\lambda_j\lambda_k\lambda_l}} q_iq_jq_kq_l = \frac{1}{16}\sum_{i,j,k,l} \frac{g_{ijkl}}{\sqrt{\lambda_i\lambda_j\lambda_k\lambda_l}} (z_i+\bar z_i)(z_j+\bar z_j)(z_k+\bar z_k)(z_l+\bar z_l). \tag{5}$$

Expanding gives a sum of monomials $z^\alpha\bar z^\beta$ with $|\alpha|+|\beta|=4$. Along the linear flow ($\dot z_j = -i\lambda_j z_j$) such a monomial evolves by $e^{-i\langle\alpha-\beta,\hat\lambda\rangle t}$; it is *resonant* if $\langle\alpha-\beta,\hat\lambda\rangle = 0$. Let $J=\{1,2\}$, $N=\{j\geq 3\}$, and split $G = G^{\tan}+G^{\nor}$, where $G^{\nor}$ collects the monomials all of whose indices lie in N.

Definition 2.9 (Partial quartic normal form)**.** $\bar G$ is the sum of the resonant monomials of $G^{\tan}$ (with their coefficients), and the partial quartic normal form is $\Lambda+\bar G+G^{\nor}$.

Theorem 2.10 (Action coefficients, (d))**.** *For $J=\{1,2\}$ and every $m>0$, with $I=(I_1,I_2)$, $Z=(I_j)_{j\geq 3}$ and $I_j=|z_j|^2$,*

$$\bar G = \frac{1}{2}\langle AI,I\rangle + \langle BI,Z\rangle, \qquad A_{ii}=\frac{9}{8\pi\lambda_i^2}, \quad A_{12}=A_{21}=\frac{3}{2\pi\lambda_1\lambda_2}, \quad B_{ji}=\frac{3}{2\pi\lambda_j\lambda_i}\ (j\geq 3,\ i\in J).$$

Equivalently $A=\frac{3}{8\pi}D(4\mathbf{1}\mathbf{1}^{\mathsf T}-\mathrm{Id})D$ with $D=\mathrm{diag}(\lambda_1^{-1},\lambda_2^{-1})$, and $(BI)_j=\frac{3}{2\pi\lambda_j}\big(\frac{I_1}{\lambda_1}+\frac{I_2}{\lambda_2}\big)$. More generally, the full angle average of G is

$$\langle G\rangle = \frac{3}{16\pi}\Big(4\Big(\sum_{j\geq 1}\frac{I_j}{\lambda_j}\Big)^2 - \sum_{j\geq 1}\frac{I_j^2}{\lambda_j^2}\Big). \tag{6}$$

Proof. **Step S2.16: Resonant equals averaged.** By Lemma 2.7, a monomial $z^\alpha\bar z^\beta$ of (5) with nonzero coefficient is resonant iff $\alpha=\beta$, i.e. it is $|z_a|^2|z_b|^2 = I_aI_b$. With $z_j=\sqrt{I_j}e^{-i\varphi_j}$ the average over independent angles of $z^\alpha\bar z^\beta$ is $\delta_{\alpha\beta}I^\alpha$. Hence $\bar G$ is the part of $\langle G\rangle$ (average over all angles, computed on finitely supported q and extended by the absolutely convergent formula) consisting of the terms that contain at least one tangential action.

Step S2.17: Averages of monomials. With $q_j=\sqrt{2I_j}\cos\varphi_j$: $\langle q_iq_jq_kq_l\rangle = 0$ unless the indices pair up; $\langle q_i^4\rangle = 4I_i^2\cdot\frac{3}{8}=\frac{3}{2}I_i^2$ and $\langle q_i^2q_j^2\rangle = I_iI_j$ for $i\neq j$. Among ordered quadruples, (i,i,i,i) occurs once and the multiset $\{i,i,j,j\}$ $(i<j)$ six times.

Step S2.18: Summation. Therefore, with $y_j=I_j/\lambda_j$ and Lemma 2.2,

$$\langle G\rangle = \frac{1}{4}\Big(\sum_i \frac{3}{2\pi}\cdot\frac{3}{2}y_i^2 + \sum_{i<j} 6\cdot\frac{1}{\pi}\,y_iy_j\Big) = \frac{9}{16\pi}\sum_i y_i^2 + \frac{3}{2\pi}\sum_{i<j} y_iy_j,$$

which is (6). Matching with $\frac{1}{2}\sum_{i,j}A_{ij}I_iI_j = \frac{1}{2}\sum_i A_{ii}I_i^2 + \sum_{i<j}A_{ij}I_iI_j$ on J and with $\sum_{j\geq 3,i\in J}B_{ji}I_iZ_j$ on the mixed terms gives the coefficients. □

Proposition 2.11 (Formal partial Birkhoff transformation)**.** *Let F be the quartic polynomial whose coefficient on each nonresonant monomial $z^\alpha\bar z^\beta$ of $G^{\tan}$ with coefficient $c_{\alpha\beta}$ is $c_{\alpha\beta}/(-i\langle\alpha-\beta,\hat\lambda\rangle)$, and zero otherwise. Then F is real ($\overline{F}=F$ under $z\leftrightarrow\bar z$), $X_\Lambda F = G^{\tan}-\bar G$ where $X_\Lambda F=\frac{d}{dt}F$ along the linear flow, and formally*

$$H\circ\Phi_F^1 = \Lambda+\bar G+G^{\nor}+(\textit{terms of degree } \geq 6),$$

where Φ_F^1 is the time-one map of X_F. The quartic part of the normal form is independent of the choice of F among solutions of $X_\Lambda F = G^{\tan}-\bar G$.

Proof. The divisors are nonzero by Lemma 2.7; $X_\Lambda(z^\alpha \bar z^\beta) = -i\langle \alpha-\beta, \hat\lambda\rangle z^\alpha \bar z^\beta$ gives the homological identity. Reality: the coefficient of $\bar z^\alpha z^\beta$ is $\bar c_{\alpha\beta}$ (since G is real) and the divisor changes sign and is conjugated. The Lie series $H\circ\Phi_F^1 = H+\{H,F\}+\frac12\{\{H,F\},F\}+\dots$ with $\{H,F\} = X_F H = -X_H F$ has quartic part $G - X_\Lambda F = G^{\mathrm{nor}} + \bar G$. Any two solutions F differ by a resonant quartic F_0, and $X_\Lambda F_0 = 0$. □

Source: P96-W Main Proposition, pp. 281–282; proof pp. 282–284, (12) and the Taylor formula on p. 283. The analytic realization of Φ_F^1 on a neighbourhood of 0 in $E_{a,p}$, the smoothing of X_F, and the bound $O(\|z\|^6)$ on the remainder are W3 (they use P96-W Lemma 4 and (13)–(14)). The *coefficients* printed there, $\bar G_{ij} = \frac{6}{\pi}\frac{4-\delta_{ij}}{\lambda_i\lambda_j}$, are 16 times those of Theorem 2.10; see Remark 2.12.

Remark 2.12 (Comparison with the printed coefficients). P96-W p. 282 expands G as $\frac14\sum G_{ijkl}(z_i+\bar z_i)\cdots(z_l+\bar z_l)$; by (5) the prefactor is $\frac{1}{16}$. The next line rewrites this as $\sum' G_{ijkl}w_iw_jw_kw_l$ *without* any prefactor, and the multiplicity count of p. 284 ($\bar G_{ij} = 24G_{iijj}$, $\bar G_{ii} = 12G_{iiii}$) is correct for that line. Two factors of 4 are therefore lost; with the prefactor $\frac{1}{16}$ the same count gives exactly A and B above. The factor is common to A and B, so the nondegeneracy arguments of P96-W Lemmas 5–6 are unaffected (Remark 1.14).

2.7. **Nondegeneracy for every $m>0$.** The Cantor Manifold Theorem used by P96-W (p. 288, assumption A) requires, with $\omega(I) = \lambda_J + AI$ and $\Omega(I) = \lambda_N + BI$, ($\mathrm{A}_1$) $\det A \neq 0$; (A_2) $\langle l,\lambda_N\rangle \neq 0$; ($\mathrm{A}_3$) $\langle k,\omega(I)\rangle + \langle l,\Omega(I)\rangle \not\equiv 0$ as an affine function of I, for all $(k,l)\in\mathbb{Z}^2\times\mathbb{Z}^N$ with $1\le|l|\le 2$ ($|l| = \sum|l_j|$).

Theorem 2.13 (Nondegeneracy, (e)). *For $J=\{1,2\}$ and every $m>0$:*

(i) $\det A = -\dfrac{63}{64\pi^2\lambda_1^2\lambda_2^2} = -\dfrac{63}{64\pi^2(1+m)(4+m)} \neq 0$;
(ii) $\langle l,\lambda_N\rangle \neq 0$ *for* $1\le|l|\le 2$;
(iii) *(A_3) holds.*

Proof. **Step S2.19: Determinant.** $\det A = (\frac{3}{8\pi})^2\lambda_1^{-2}\lambda_2^{-2}\det\left(\begin{smallmatrix}3&4\\4&3\end{smallmatrix}\right) = -\frac{63}{64\pi^2}\lambda_1^{-2}\lambda_2^{-2}$. (For general n: $\det(4\mathbf{1}\mathbf{1}^{\mathsf T} - \mathrm{Id}) = (-1)^{n-1}(4n-1)$.)

Step S2.20: Pure normal. $\langle l,\lambda_N\rangle$ is $\pm\lambda_a$, $\pm2\lambda_a$, $\pm(\lambda_a+\lambda_b)$ or $\pm(\lambda_a-\lambda_b)$ with $a\neq b$ in N; all are nonzero since $\lambda>0$ is strictly increasing.

Step S2.21: Reduction of ($\mathbf{A}_3$). Let $b = (\lambda_1^{-1},\lambda_2^{-1})$ and $s = \sum_{j\ge3} l_j/\lambda_j$. Then $A = \frac{3}{8\pi}(4bb^{\mathsf T} - D^2)$ and $B^{\mathsf T} l = \frac{3}{2\pi} s\, b$, so the linear part of the affine map is $\frac{3}{8\pi}\big(4(\langle b,k\rangle + s)b - D^2k\big)$. It vanishes iff $k_i/\lambda_i = c$ for $i=1,2$ with $c = 4(\langle b,k\rangle + s)$; then $\langle b,k\rangle = 2c$ and $s = -\frac74 c$. The constant part is $c(\lambda_1^2+\lambda_2^2) + \langle l,\lambda_N\rangle$.

Step S2.22: The case $|l|=1$. $l = \sigma e_\nu$, $\nu\ge3$: $s = \sigma/\lambda_\nu$, $c = -4\sigma/(7\lambda_\nu)$, and the constant part vanishes iff $7\lambda_\nu^2 = 4(\lambda_1^2+\lambda_2^2)$. Then $k_i^2 = c^2\lambda_i^2 = \frac{4\lambda_i^2}{7(\lambda_1^2+\lambda_2^2)} \in (0,\frac47)$, not the square of an integer. So (A_3) holds.

Step S2.23: The case $|l|=2$. $|s|\le 2/\lambda_3 < \frac23$, so $|c| < \frac{8}{21}$, and $k_2-k_1 = c(\lambda_2-\lambda_1)$ with $0<\lambda_2-\lambda_1<1$; integrality forces $k_1=k_2$, hence $c=0$, hence $s=0$. But $s\neq0$ for $|l|=2$: $s=0$ would need $l = e_a - e_b$ with $\lambda_a=\lambda_b$, $a\neq b$. So the linear part does not vanish. □

Source: P96-W assumption A p. 288; Lemma 5 pp. 291–292 ((A_1), with the printed matrix; the printed $(1-4n)\prod\lambda_j^{-2}$ agrees after scaling); Lemma 6 pp. 292–293 ((A_3) for admissible J). Our proof of (iii) is the specialization of Lemma 6 to $J=\{1,2\}$; the second branch of the source's $|l|=2$ argument ($|\langle w,l\rangle| \ge (4n-1)/(4n-4)$) cannot occur for $J=\{1,2\}$ since $|\langle w,l\rangle| \le 2/\lambda_3 < 1$.

Remark 2.14 (Spectral asymptotics with the corrected B). $\Omega_j(I) = \lambda_j + \frac{3}{2\pi\lambda_j}\langle b, I\rangle = j + \frac{m_I}{j} + O(j^{-3})$ with $m_I = \frac{m}{2} + \frac{3}{2\pi}\langle b, I\rangle$, so $(\Omega_i - \Omega_j)/(i-j) = 1 + O(j^{-2})$ uniformly for bounded I: the exponent $\kappa = 2$ of P96-W p. 291 is unchanged by the correction (only $v = \frac{24}{\pi}b$ becomes $\frac{3}{2\pi}b$).

2.8. **Duffing consistency check.**

Proposition 2.15 (One-mode frequency shift, (f))**.** *Restrict (1) to $u = x(t)\phi_j(x)$ with only mode j excited and drop the forcing of other modes (which is $O(x^3)$ and does not enter the first-order frequency shift). The coefficient equation is the Duffing equation $\ddot{x} + \lambda_j^2 x + \varepsilon x^3 = 0$ with $\varepsilon = g_{jjjj} = \frac{3}{2\pi}$, whose Lindstedt–Poincaré frequency at amplitude a is $\lambda_j + \frac{3\varepsilon a^2}{8\lambda_j} + O(a^4)$. With $q_j = \sqrt{\lambda_j}x$ and $I_j = \frac{1}{2}\lambda_j a^2$ this is*

$$\lambda_j + \frac{9}{16\pi\lambda_j}a^2 = \lambda_j + \frac{9}{8\pi\lambda_j^2}I_j = \lambda_j + A_{jj}I_j,$$

the frequency $\partial_{I_j}(\lambda_j I_j + \frac{1}{2}A_{jj}I_j^2)$ of the normal form. The printed P96-W value $\bar{G}_{jj} = \frac{18}{\pi\lambda_j^2}$ would predict a shift 16 *times larger.*

Proof. $\int(x\phi_j)^3\phi_j = x^3 g_{jjjj}$ gives the Duffing coefficient. The Lindstedt shift $3\varepsilon a^2/(8\omega_0)$ is classical (e.g. by averaging εx^3 against cos over a period: $\frac{2}{T}\int \varepsilon a^3\cos^4 = \frac{3}{4}\varepsilon a^3$). Along a solution $x = a\cos\varphi$, $q_j = \sqrt{\lambda_j}a\cos\varphi$ and $p_j = \dot{q}_j/\lambda_j$, so to leading order $I_j = \frac{1}{2}(q_j^2 + p_j^2) = \frac{1}{2}\lambda_j a^2$. Substitute. □

Remark 2.16. The relation between I_j and the amplitude holds to leading order only; the agreement checked is of the coefficient of the first-order shift, which is what the normal form determines. `scripts/wave/w0_checks.py` also integrates the Duffing equation numerically for $m \in \{0.1, 1, 10\}$ and confirms $|\omega_{\text{num}} - (\lambda_1 + A_{11}I_1)| = O(I_1^2)$.

2.9. **Ledger: proved, cited, source gaps.**

- *Proved here*: Lemmas 2.1, 2.2, 2.3, 2.4, 2.7; Theorems 2.5, 2.10, 2.13; Propositions 2.6, 2.11 (formal level), 2.15.
- *External standard facts*: completeness of (ϕ_l) in $L^2(0,\pi)$; analyticity of continuous multilinear maps; Young's inequality; $\zeta(3) < \infty$; the classical Lindstedt shift.
- *Cited, for later milestones*: quantitative divisor bound (P96-W Lemma 4, W3); analytic realization of the Birkhoff map and its remainder (P96-W Main Proposition, W3).
- *Expansions of source proofs*: P96-W Lemma 1 (expanded and strengthened to C^1 curves).
- *Source gaps / normalization defects*: P96-W p. 280 constant (N-1); P96-W p. 278 sign typo (N-4); KP96 p. 13 factor 2 (N-5); P96-W p. 282/284 factor 16 (E-PDE-W-1); P96-W pp. 287, 290 action/recovery/sign conventions (E-PDE-W-2).

3. Milestone W1: weighted sequence calculus

Throughout, Standing assumption 1.1 ($m > 0$) and the conventions of §1 are in force, and $\mathbb{K}$ denotes $\mathbb{R}$ or $\mathbb{C}$. This part supplies the sequence-space analysis used by every later milestone: the weighted spaces $\ell^{a,p}$ as transports of ℓ^2 (the realization of the frozen `KAM.Wave.WaveSpace`), embeddings and diagonal multipliers, the two-sided spaces and the convolution algebra, the cubic gradient as a continuous trilinear map with its analyticity, smoothing and gradient structure, the Hamiltonian field, completeness of the sine system, analyticity of the reconstruction, and the Banach-space analytic toolkit (complexification, Cauchy estimate, uniform limits, holomorphic flows, Hilbert-valued Fourier series, the Hilbert-matrix operator bound) that the Birkhoff normal form (W3), the KAM step and iteration of P96-K in the setting $d = 1$, $\delta = -1$, $\bar{p} - p = 1$ (W4–W6) and the assembly (W2, W8) consume. Every result is proved here or is an external fact from the list §3.2.

Route. Every weighted space is the image of one Hilbert space $\ell^2(\mathbb{N}_+;\mathbb{K})$ under an explicit diagonal isometry, so that each map between weighted spaces becomes a map on ℓ^2, and a diagonal map becomes a bounded multiplier. The cubic gradient is a trilinear form in the coefficients whose coefficients obey the selection rule of Lemma 2.2; majorizing by an even two-sided sequence turns it into a triple convolution, which the algebra property controls with a gain of one power of j. Multilinearity then gives analyticity for free, over $\mathbb{R}$ and over $\mathbb{C}$, with no appeal to the principle "coordinatewise analytic and locally bounded implies analytic" used by the source. The analytic toolkit of §3.7 works with holomorphic maps on open subsets of complex Banach spaces, defined as complex Fréchet differentiable maps; it never needs the (absent in Mathlib) theorem that such maps are analytic on infinite-dimensional domains, because all analyticity conclusions of the headline have finite-dimensional domains.

3.1. **Realization in Lean and transport maps.**

Definition 3.1 (Weights, transport and coefficient maps)**.** For $a,p\in\mathbb{R}$ and $j\in\mathbb{N}_+$ let $\omega_{a,p}(j)=j^p e^{aj}>0$ (frozen `KAM.Wave.ellWeight a p j`). Write $\ell^2_{\mathbb{K}}=\ell^2(\mathbb{N}_+;\mathbb{K})$ (Lean `lp (fun _ : PNat => K) 2`) and, for $a\ge 0$, let $\ell^{a,p}_{\mathbb{K}}$ be the space of Definition 1.4 ($\ell^{a,p}=\ell^{a,p}_{\mathbb{R}}$). Define

$$\tau_{a,p}:\ell^{a,p}_{\mathbb{K}}\to\ell^2_{\mathbb{K}},\quad (\tau_{a,p}w)_j=\omega_{a,p}(j)\,w_j,\qquad \kappa_{a,p}:\ell^2_{\mathbb{K}}\to\ell^{a,p}_{\mathbb{K}},\quad (\kappa_{a,p}y)_j=\frac{y_j}{\omega_{a,p}(j)}.$$

A vector $y\in\ell^2_{\mathbb{K}}$ *represents* $w=\kappa_{a,p}y\in\ell^{a,p}_{\mathbb{K}}$; we call y the *weighted coordinates* of w at weight (a,p). A map A from $\ell^{a,p}_{\mathbb{K}}$ to $\ell^{a',p'}_{\mathbb{K}}$ is represented by $\hat A=\tau_{a',p'}\circ A\circ\kappa_{a,p}$ on $\ell^2_{\mathbb{K}}$.

Source: ours (the frozen encoding, `docs/tracks/W/W-STATEMENTS.md`, design summary). Lean guard: the division in $\kappa_{a,p}$ is by $\omega_{a,p}(j)>0$; no case split.
Realization table. Every space of this part is realized as an ℓ^2 transport; no new normed type is introduced. The frozen objects are the case $\mathbb{K}=\mathbb{R}$, $a=p=1$.

Space	Lean carrier	Transport / coefficient map
$\ell^{a,p}_{\mathbb{K}}$	`lp (fun _ : PNat => K) 2`	$\tau_{a,p}$, $\kappa_{a,p}$; for $\mathbb{K}=\mathbb{R}$, $a=p=1$: $\kappa_{1,1}=$ frozen `coeff`
$E_{a,p}$ (Def. 1.4)	`lp ... 2 × lp ... 2` (max norm)	$\tau_{a,p}\times\tau_{a,p}$; $E_{1,1}=$ frozen `WaveSpace`
$E^{\mathbb{C}}_{a,p}$ (complexification)	`lp (fun _ : PNat => Complex) 2 × lp ... 2`	same weights; ι, re of Def. 3.22
$\ell^{a,s}_b$ (Def. 1.6)	`lp (fun _ : Int => Complex) 2`	weight $\langle l\rangle^s e^{a\lvert l\rvert}$ (Def. 3.9)
ℓ^p_∞ (Def. 3.7)	`lp (fun _ : PNat => Real) ∞`	weight j^p
$\mathcal{P}^{a,p}$ (Def. 3.30)	`(Fin 2 -> Complex) × (Fin 2 -> Complex) × lp ... 2 × lp ... 2`	weights $\omega_{a,p}$ on the last two factors

In Lean every weighted statement below is stated on the carrier, with κ applied where the mathematics speaks of coefficients. The non-frozen helper `KAM.Wave.coeffW a p y j := y j / ellWeight a p j` (over $\mathbb{K}$) satisfies `coeffW 1 1 = coeff` on real sequences by unfolding; it is a distinct name, not a substitute for a frozen declaration. Statements uniform in $\mathbb{K}$ are written once over `[RCLike K]`.

3.2. **External facts.**

The proofs of this part use the following external (Mathlib-level) facts, besides those of track K's Part 1 (`bp:p1:ext-*`, cited by label in the text).

(W1-E1) *ℓ^2 spaces.* $\ell^2(\iota;\mathbb{K})$ is a Hilbert space for every index type ι (`lp.instInnerProductSpace`, completeness `lp.instCompleteSpace`), with $\langle f,g\rangle=\sum_i \overline{f_i}g_i$ (`lp.inner_eq_tsum`); $|f_i|\le\|f\|$ (`lp.norm_apply_le_norm`); membership and norm bounds follow from bounds on finite partial sums (`mem$\ell$p_gen_iff`, `lp.norm_le_of_forall_sum_le`); $f=\sum_i f_i\,e_i$ with $e_i=$ `lp.single 2 i 1` (`lp.hasSum_single`).

(W1-E2) *Series.* $\sum_{k\geq 1} k^{-2} = \pi^2/6$ (`hasSum_zeta_two`); $\sum_{k\geq 1} k^{-\sigma} < \infty$ for $\sigma > 1$ (`Real.summable_one_div_nat_rpow`); Tonelli for nonnegative double series and Fubini for absolutely summable ones (`tsum_prod'`, `Summable.tsum_comm`).

(W1-E3) *Analytic maps.* A continuous multilinear map is analytic (indeed a continuous polynomial) everywhere (`ContinuousMultilinearMap.analyticAt`, `.cpolynomialAt`); a continuous linear or bilinear map is analytic (`ContinuousLinearMap.analyticAt`, `.analyticAt_bilinear`); compositions and pairs of analytic maps are analytic (`AnalyticAt.comp`, `AnalyticAt.comp2`, `AnalyticAt.prod`); $\mathbb{C}$-analytic maps are $\mathbb{R}$-analytic (`AnalyticAt.restrictScalars`); analytic maps are Fréchet differentiable (`AnalyticAt.differentiableAt`).

(W1-E4) *One complex variable, Banach values.* For a complete complex normed space F: the Cauchy estimate $\|\varphi'(c)\| \leq R^{-1} \sup_{|\zeta-c|=R} \|\varphi(\zeta)\|$ for φ continuous on the closed disc and holomorphic inside (`Complex.norm_deriv_le_of_forall_mem_sphere_norm_le`); the Cauchy integral formula (`bp:p1:ext-cauchy`, valid for complete F); holomorphic maps on open subsets of $\mathbb{C}$ are analytic (`DifferentiableOn.analyticOnNhd`); a locally uniform limit of holomorphic maps on an open subset of $\mathbb{C}$ is holomorphic (`TendstoLocallyUniformlyOn.differentiableOn`).

(W1-E5) *Uniform limits and derivatives.* If $f_k \to f$ pointwise on an open set U of a normed space over $\mathbb{K}$, each f_k is differentiable on U and $Df_k \to g$ uniformly on U, then f is differentiable on U with $Df = g$ (`hasFDerivAt_of_tendstoUniformlyOn`). Mean value inequality on convex sets (`bp:p1:ext-mvt`).

(W1-E6) *Integrals.* Bochner integrals of continuous maps of a compact interval or box into a complete space, the estimate $\|\int F\| \leq \int \|F\|$, differentiation under the integral sign with a dominating bound on the derivative (`hasFDerivAt_integral_of_dominated_of_fderiv_le`), dominated convergence (`bp:p1:ext-integral`).

(W1-E7) *ODE.* Picard–Lindelöf existence and uniqueness for Lipschitz fields in a Banach space (`IsPicardLindelof`, `ODE_solution_unique`); Gronwall's inequality (`norm_le_gronwallBound_of_norm_deriv_right_le`).

(W1-E8) *Fourier analysis.* The characters $e_n(x) = e^{inx}$, $n \in \mathbb{Z}$, form a Hilbert basis of L^2 of the circle $\mathbb{R}/2\pi\mathbb{Z}$ with normalized Haar measure (`fourierBasis` on `AddCircle (2 * Real.pi)`), and the characters $e^{2\pi i k\cdot x}$, $k \in \mathbb{Z}^n$, one of $L^2(\mathbb{R}^n/\mathbb{Z}^n)$ (`UnitAddTorus.mFourierBasis`); Parseval (`tsum_sq_fourierCoeff`); continuous periodic functions with absolutely summable coefficients are the sums of their Fourier series (`bp:p1:ext-fourier`).

3.3. **Weighted spaces, embeddings, multipliers.**

Lemma 3.2 (The weighted spaces)**.** *Let $a \geq 0$, $p \in \mathbb{R}$.*

(i) $\tau_{a,p}$ and $\kappa_{a,p}$ are mutually inverse $\mathbb{K}$-linear bijections and $\|\tau_{a,p} w\|_{\ell^2} = \|w\|_{a,p}$. Hence $\ell^{a,p}_{\mathbb{K}}$, with the inner product $\langle w, w'\rangle_{a,p} = \sum_j \omega_{a,p}(j)^2\, \overline{w_j} w'_j$, is a Hilbert space isometric to $\ell^2_{\mathbb{K}}$; it is complete and separable.

(ii) Coordinate evaluation is continuous: $|(\kappa_{a,p} y)_j| \leq \|y\|/\omega_{a,p}(j)$, and $y \mapsto (\kappa_{a,p} y)_j$ is the continuous linear functional $\omega_{a,p}(j)^{-1}\, \mathrm{ev}_j$.

(iii) Finite truncations converge: with $P_N y = \sum_{j\leq N} y_j e_j$, $\|y - P_N y\| \to 0$ as $N \to \infty$.

(iv) For $\mathbb{K} = \mathbb{R}$ and $a = p = 1$, $\kappa_{1,1}$ is the frozen `coeff`, and the frozen `WaveSpace` $= \ell^2_{\mathbb{R}} \times \ell^2_{\mathbb{R}}$ with its product (max) norm is isometric to $E_{1,1}$ with $\|(q,p)\|_E = \max(\|q\|_{1,1}, \|p\|_{1,1})$ via $\kappa_{1,1} \times \kappa_{1,1}$.

Proof. **Step S3.1: Isometry.** $\|\tau_{a,p}w\|^2 = \sum_j \omega_{a,p}(j)^2|w_j|^2 = \|w\|_{a,p}^2$ by Definition 1.4; so $\tau_{a,p}$ maps $\ell_{\mathbb{K}}^{a,p}$ into $\ell_{\mathbb{K}}^2$, $\kappa_{a,p}$ maps back, and both compositions are the identity coordinatewise. Transport the inner product and completeness of (W1-E1).

Step S3.2: Separability. Finitely supported sequences with entries in $\mathbb{Q}$ (resp. $\mathbb{Q}+i\mathbb{Q}$) form a countable set; it is dense by (iii) and density of $\mathbb{Q}$ in each of finitely many coordinates. (Separability is not consumed by later milestones.)

Step S3.3: Coordinates and truncation. (ii) is (W1-E1) ($|y_j| \le \|y\|$) divided by $\omega_{a,p}(j)$. (iii): $\|y - P_N y\|^2 = \sum_{j>N}|y_j|^2$ is the tail of a convergent series.

Step S3.4: Frozen case. `coeff y j` is $y_j/\omega_{1,1}(j)$ by its frozen definition, and Mathlib's norm on a product is the max of the norms. □

Source: P96-W p. 277 and P96-K p. 120 ("$\ell^{a,p}$ is the Hilbert space ..."), asserted without proof; the transport is ours.

Lemma 3.3 (Derivatives of coordinates)**.** *Let $c : \mathbb{R} \to \ell_{\mathbb{K}}^2$ be differentiable at t, and let (a,p), (a',p') be weights with $a \ge a'$, $p \ge p'$ and ι the inclusion of Lemma 3.5, represented by $\hat{\iota}$. For $v \in \ell_{\mathbb{K}}^2$ the following are equivalent: (a) for every j, $t \mapsto (\kappa_{a,p}c(t))_j$ has derivative $(\kappa_{a',p'}v)_j$ at t; (b) $\hat{\iota}\,c'(t) = v$. In particular, if $K : \mathbb{R}^2 \to$ `WaveSpace` is differentiable and X^w is the field of Proposition 3.18(v), then the frozen invariance conclusion (4) of `KAM.Wave.nlw2` at (I,θ) is equivalent to $\hat{\iota}\big(DK(\theta)\nu\big) = X^w(K(\theta))$ with $\hat{\iota}$ the representative of $E_{1,1} \subset E_{1,0}$.*

Proof. **Step S3.5: Coordinates of the derivative.** By Lemma 3.2(ii), $c \mapsto (\kappa_{a,p}c)_j$ is a continuous linear functional, so the coordinate curve has derivative $(\kappa_{a,p}c'(t))_j = (\kappa_{a',p'}\hat{\iota}c'(t))_j$ (the inclusion does not change coefficients, Lemma 3.5).

Step S3.6: Comparison. Hence (a) says $(\kappa_{a',p'}\hat{\iota}c'(t))_j = (\kappa_{a',p'}v)_j$ for all j, which is (b) because $\kappa_{a',p'}$ is injective. For the frozen statement apply this to $c(t) = K(\theta + t\nu)$ componentwise at $t = 0$, $c'(0) = DK(\theta)\nu$, with $(a,p) = (1,1)$, $(a',p') = (1,0)$, and compare with Proposition 3.18(v). □

Source: ours.

Lemma 3.4 (Diagonal multipliers)**.** *Let $\mu : \mathbb{N}_+ \to \mathbb{K}$ with $M = \sup_j|\mu_j| < \infty$. Then $D_\mu y = (\mu_j y_j)_j$ is a bounded linear operator on $\ell_{\mathbb{K}}^2$ with $\|D_\mu\| = M$. Consequently a diagonal map $w \mapsto (m_j w_j)$ is bounded from $\ell_{\mathbb{K}}^{a,p}$ to $\ell_{\mathbb{K}}^{a',p'}$ iff $\sup_j|m_j|\omega_{a',p'}(j)/\omega_{a,p}(j) < \infty$; its representative is D_μ with $\mu_j = m_j\,\omega_{a',p'}(j)/\omega_{a,p}(j)$, and its norm is that supremum.*

Proof. $\sum_{j\le N}|\mu_j y_j|^2 \le M^2\|y\|^2$ for every N, so $D_\mu y \in \ell^2$ with $\|D_\mu y\| \le M\|y\|$ ((W1-E1)); $\|D_\mu e_j\| = |\mu_j|$ gives equality of the norm. The representative is computed coordinatewise: $\tau_{a',p'}(m\,\kappa_{a,p}y)_j = \omega_{a',p'}(j)m_j y_j/\omega_{a,p}(j)$. □

Source: standard.

Lemma 3.5 (Embeddings)**.** *Let $a \ge a' \ge 0$ and $p, p' \in \mathbb{R}$.*

(i) If $p \ge p'$, then $\ell_{\mathbb{K}}^{a,p} \subset \ell_{\mathbb{K}}^{a',p'}$ with $\|w\|_{a',p'} \le \|w\|_{a,p}$; the representative of the inclusion is D_μ with $\mu_j = j^{p'-p}e^{(a'-a)j} \in (0,1]$.

(ii) If $a > a'$ and $\sigma = p' - p > 0$, then $\|w\|_{a',p'} \le \big(\sigma/(e(a-a'))\big)^\sigma\|w\|_{a,p}$.

(iii) The inclusions do not change coefficients: $\kappa_{a',p'}D_\mu = \kappa_{a,p}$. They are injective and compose: the inclusion $(a,p) \to (a'',p'')$ is the composite through (a',p').

Proof. (i) and (ii) are Lemma 3.4 with $m_j = 1$: $\mu_j = \omega_{a',p'}(j)/\omega_{a,p}(j)$. In (i) both factors of μ_j are at most 1. In (ii), $t^\sigma e^{-(a-a')t}$ is maximal on $(0,\infty)$ at $t = \sigma/(a-a')$, with value $(\sigma/(e(a-a')))^\sigma$. (iii) is the identity $\mu_j/\omega_{a',p'}(j) = 1/\omega_{a,p}(j)$; injectivity because $\mu_j \ne 0$; composition because the μ's multiply. □

Source: standard (implicit in P96-W pp. 277–280 and P96-K p. 121).

Lemma 3.6 (Frequency multipliers and the linear flow)**.** *Let $a \geq 0$, $p \in \mathbb{R}$, $r \in \mathbb{R}$, and let λ^r denote the multiplier $w \mapsto (\lambda_j^r w_j)$.*

(i) λ^r is bounded $\ell^{a,p+r}_{\mathbb{K}} \to \ell^{a,p}_{\mathbb{K}}$ with norm $\sup_j(\lambda_j/j)^r$, which is $(1+m)^{r/2}$ for $r \geq 0$ and at most 1 for $r < 0$; its representative is D_μ with $\mu_j = (\lambda_j/j)^r$. It is an isomorphism $\ell^{a,p+r} \to \ell^{a,p}$ with inverse λ^{-r}.

(ii) For $r > 0$, λ^r is not bounded from $\ell^{a,p}$ to itself: $\|\lambda^r e_j\|_{a,p}/\|e_j\|_{a,p} = \lambda_j^r \to \infty$.

(iii) (Complex rotations.) For $t \in \mathbb{R}$, $R_t : z \mapsto (e^{-i\lambda_j t} z_j)$ is a linear isometry of $\ell^{a,p}_{\mathbb{C}}$, $R_{t+s} = R_t R_s$, and $t \mapsto R_t z$ is continuous.

(iv) (Real linear flow.) For $(q,p) \in E_{a,p}$ put $L_t(q,p) = \big(\cos(\lambda t)q + \sin(\lambda t)p,\ -\sin(\lambda t)q + \cos(\lambda t)p\big)$ (coordinatewise). Then L_t preserves $\|q\|^2_{a,p} + \|p\|^2_{a,p}$, $\|L_t(q,p)\|_E \leq \sqrt{2}\,\|(q,p)\|_E$, $L_{t+s} = L_t L_s$, and every coordinate solves $\dot q_j = \lambda_j p_j$, $\dot p_j = -\lambda_j q_j$. In the complex coordinates of Definition 1.11, L_t is R_t.

Proof. **Step S3.7: Ratio.** By Lemma 3.4, $\mu_j = \lambda_j^r \omega_{a,p}(j)/\omega_{a,p+r}(j) = (\lambda_j/j)^r$. Since $j^2 < \lambda_j^2 = j^2 + m \leq (1+m)j^2$, $1 < \lambda_j/j \leq \sqrt{1+m}$ with equality at $j = 1$; so the supremum is $(1+m)^{r/2}$ for $r \geq 0$ (attained at $j = 1$) and ≤ 1 for $r < 0$. The inverse statement is the same computation with $-r$.

Step S3.8: Unboundedness and rotations. (ii) is immediate. (iii): $|e^{-i\lambda_j t}| = 1$, so R_t is D_μ with $|\mu_j| = 1$ (an isometry), and the group law is coordinatewise. Continuity: $\|R_t z - R_s z\|^2 = \sum_j |e^{-i\lambda_j t} - e^{-i\lambda_j s}|^2 \omega_{a,p}(j)^2 |z_j|^2 \to 0$ as $t \to s$ by dominated convergence for series (terms $\leq 4\omega^2|z_j|^2$).

Step S3.9: Real flow. Coordinatewise $(q_j(t), p_j(t))$ is a rotation of (q_j, p_j), so $q_j(t)^2 + p_j(t)^2 = q_j^2 + p_j^2$; multiplying by $\omega_{a,p}(j)^2$ and summing gives the first claim, and then $\max(\|q(t)\|, \|p(t)\|)^2 \leq \|q\|^2 + \|p\|^2 \leq 2\|(q,p)\|_E^2$. The derivative and the group law are coordinatewise trigonometry; with $z_j = (q_j + ip_j)/\sqrt{2}$ one gets $z_j(t) = e^{-i\lambda_j t} z_j$. □

Source: P96-W p. 277 and Definition 1.10 (the multiplier bound is stated there without the norm); the factor $\sqrt{2}$ in (iv) is the price of the max norm of the frozen phase space and is not in the source, which uses $\ell^{a,s} \times \ell^{a,s}$ without specifying the product norm.

Definition 3.7 (Sup-weighted sequences)**.** For $p \in \mathbb{R}$, ℓ^p_∞ is the space of real sequences $(w_j)_{j \geq 1}$ with $|w|_p = \sup_j |w_j| j^p < \infty$, realized as `lp (fun _ : PNat => Real) ∞` by $w \mapsto (j^p w_j)$. For a map $\tilde\Omega : \Pi \to \ell^p_\infty$ on a set $\Pi \subset \mathbb{R}^2$, $|\tilde\Omega|^{\mathcal{L}}_{p,\Pi} = \sup_{\xi \neq \zeta} |\tilde\Omega(\xi) - \tilde\Omega(\zeta)|_p / |\xi - \zeta|$ (max norm on $\mathbb{R}^2$).

Source: P96-K p. 120 (Assumption B) and p. 121 (Lipschitz semi-norms).

Lemma 3.8 (Spectral asymptotics of the wave normal frequencies)**.** *Let $b = (\lambda_1^{-1}, \lambda_2^{-1})$ and, for $\xi \in \mathbb{R}^2$ and $j \geq 3$, $\Omega_j(\xi) = \lambda_j + \frac{3}{2\pi\lambda_j}\langle b, \xi\rangle$ (the normal frequencies of Theorem 2.10, $\Omega = \lambda_N + B\xi$). Put $\bar\Omega_j = j$ and $\tilde\Omega_j = \Omega_j - \bar\Omega_j$ ($j \geq 3$). Then*

$$|\tilde\Omega(\xi)|_1 \leq \frac{m}{2} + \frac{3}{2\pi}|\langle b, \xi\rangle|, \qquad |\tilde\Omega(\xi) - \tilde\Omega(\zeta)|_1 \leq \frac{3}{2\pi}\Big(\frac{1}{\lambda_1} + \frac{1}{\lambda_2}\Big)|\xi - \zeta|.$$

So Assumption B of P96-K holds with $d = 1$, $\delta = -1$ (which satisfies $\delta < d - 1$), and, with $\bar p = p + 1$ for the perturbation field, the convention $p - \bar p \leq \delta$ of P96-K p. 121 holds with equality. Moreover $|\Omega_i(\xi) - \Omega_j(\xi) - (i - j)| \leq \big(m + \frac{3}{\pi}|\langle b, \xi\rangle|\big)/\min(i,j)$ for $i, j \geq 3$.

Proof. $\lambda_j - j = m/(\lambda_j + j) \in (0, m/(2j)]$, and $j \cdot \frac{3}{2\pi\lambda_j} \leq \frac{3}{2\pi}$ since $\lambda_j > j$. Multiply by j and take the supremum; the Lipschitz bound is the second term with $|\langle b, \xi - \zeta\rangle| \leq (\lambda_1^{-1} + \lambda_2^{-1})|\xi - \zeta|$. The last bound is $|\tilde\Omega_i| + |\tilde\Omega_j| \leq 2|\tilde\Omega|_1/\min(i,j)$. □

Source: ours; checks P96-K Assumption B (p. 120) for the wave normal form of P96-W p. 290 with the corrected B of Theorem 2.10. Normal modes are indexed by $j \geq 3$ here; P96-K indexes them from 1, and the shift $j \mapsto j-2$ changes $\bar{\Omega}_j$ only by the constant 2, a lower-order term allowed by Assumption B. In Lean, normal-mode sequences are `PNat`-indexed with the entries $j = 1, 2$ set to zero.

3.4. **Two-sided spaces, Young and the algebra.**

Definition 3.9 (Two-sided transport)**.** For $a \geq 0$, $s \in \mathbb{R}$ let $\omega^b_{a,s}(l) = \langle l\rangle^s e^{a|l|}$ ($l \in \mathbb{Z}$, $\langle l\rangle = \max(1,|l|)$). The space $\ell^{a,s}_b$ of Definition 1.6 is realized as `lp (fun _ : Int => Complex) 2` through $c \mapsto (\omega^b_{a,s}(l)c_l)_l$, an isometry as in Lemma 3.2. For c with $\sum_i |c_i||d_{l-i}| < \infty$ for all l, $(c*d)_l = \sum_{i\in\mathbb{Z}} c_i d_{l-i}$ (Lean: a `tsum` guarded by the summability proved before each use).

Source: P96-W p. 279 and §6 p. 294.

Lemma 3.10 (Passage between one- and two-sided sequences)**.** *Let $a \geq 0$, $p \in \mathbb{R}$.*

(i) (Even and odd extensions.) For $q \in \ell^{a,p}_{\mathbb{C}}$ define $(E^{\pm}q)_l = (\pm 1)^{[l<0]} q_{|l|}/\sqrt{\lambda_{|l|}}$ for $l \neq 0$ and $(E^{\pm}q)_0 = 0$. Then $E^{\pm} : \ell^{a,p}_{\mathbb{C}} \to \ell^{a,p+1/2}_b$ is linear with $\|E^{\pm}q\|_{a,p+1/2} \leq \sqrt{2}\|q\|_{a,p}$, and the same holds for the majorant $l \mapsto |(E^+q)_l|$. The even sequence $x_l = |q_{|l|}|/\sqrt{|l|}$ ($l \neq 0$), $x_0 = 0$, satisfies $\|x\|^2_{a,p+1/2} = 2\|q\|^2_{a,p}$.

(ii) (Restriction with gain.) For $X \in \ell^{a,p+1/2}_b$, the sequence $(X_l/\sqrt{\lambda_l})_{l\geq 1}$ lies in $\ell^{a,p+1}_{\mathbb{C}}$ with norm at most $\big(\sum_{l\geq 1} l^{2p+1}e^{2al}|X_l|^2\big)^{1/2} \leq \|X\|_{a,p+1/2}$, and at most $\|X\|_{a,p+1/2}/\sqrt{2}$ if $|X|$ is even.

(iii) (Pairing.) $J(z,\bar{z})_j = z_j$, $J(z,\bar{z})_{-j} = \bar{z}_j$ ($j \geq 1$), $J(z,\bar{z})_0 = 0$ defines a linear isometry from $\ell^{a,p}_{\mathbb{C}} \times \ell^{a,p}_{\mathbb{C}}$ with the norm $(\|z\|^2 + \|\bar{z}\|^2)^{1/2}$ onto $\{w \in \ell^{a,p}_b : w_0 = 0\}$; in particular $\|J(z,\bar{z})\|_{a,p} \leq \sqrt{2}\max(\|z\|_{a,p}, \|\bar{z}\|_{a,p})$. (Here $\bar{z}$ is an independent variable, as in P96-W p. 282.)

Proof. **Step S3.10: Extensions.** $\|E^{\pm}q\|^2 = 2\sum_{j\geq 1} j^{2p+1}e^{2aj}|q_j|^2/\lambda_j \leq 2\sum_j j^{2p}e^{2aj}|q_j|^2$ since $j/\lambda_j < 1$; the majorant has the same norm; with $\sqrt{|l|}$ the factor j/λ_j becomes 1. This is Step S2.6.

Step S3.11: Restriction. $\sum_{l\geq 1} l^{2p+2}e^{2al}|X_l|^2/\lambda_l \leq \sum_{l\geq 1} l^{2p+1}e^{2al}|X_l|^2$ since $l/\lambda_l < 1$; for even $|X|$ the sum over $l \geq 1$ is at most half of the full sum. This is Step S2.8.

Step S3.12: Pairing. $\|J(z,\bar{z})\|^2 = \sum_{j\geq 1} \omega_{a,p}(j)^2(|z_j|^2 + |\bar{z}_j|^2)$, since $\langle \pm j\rangle = j$ for $j \geq 1$. □

Source: P96-W p. 280 (proof of Lemma 3, with the corrected constant $\sqrt{2}$, errata item N-1, see Theorem 2.5) and p. 282 (the coordinates w).

Lemma 3.11 (Young and ℓ^1 control)**.** *(i) For $X \in \ell^2(\mathbb{Z})$ and $Y \in \ell^1(\mathbb{Z})$, each sum $\sum_i |X_i||Y_{l-i}|$ converges, and $\|X*Y\|_{\ell^2} \leq \|X\|_{\ell^2}\|Y\|_{\ell^1}$.*

(ii) For $s > \frac{1}{2}$, $\sum_{l\in\mathbb{Z}}\langle l\rangle^{-2s} = 1 + 2\zeta(2s) < \infty$, and for $a \geq 0$ and $Y \in \ell^{a,s}_b$, $\|Y\|_{\ell^1} \leq (1 + 2\zeta(2s))^{1/2}\|Y\|_{0,s} \leq (1+2\zeta(2s))^{1/2}\|Y\|_{a,s}$.

(iii) For nonnegative sequences X, Y, Z with $X, Y, Z \in \ell^{0,s}_b$, $s > \frac{1}{2}$, convolution is commutative and associative, and monotone: $0 \leq X \leq X'$ pointwise implies $X*Y \leq X'*Y$. For complex sequences, $|c*d| \leq |c|*|d|$ pointwise.

Proof. **Step S3.13: Weighted Cauchy–Schwarz.**

$$\Big(\sum_i |X_i||Y_{l-i}|\Big)^2 \leq \Big(\sum_i |X_i|^2|Y_{l-i}|\Big)\Big(\sum_i |Y_{l-i}|\Big) = \|Y\|_{\ell^1}\sum_i |X_i|^2|Y_{l-i}|,$$

finite since $|X_i|^2 \leq \|X\|^2$.

Step S3.14: Summation. Summing over l and exchanging the nonnegative double sum ((W1-E2)) gives $\sum_l |(X*Y)_l|^2 \leq \|Y\|_{\ell^1}\sum_i |X_i|^2 \sum_l |Y_{l-i}| = \|Y\|^2_{\ell^1}\|X\|^2_{\ell^2}$.

Step S3.15: ℓ^1 control. $\sum_l \langle l\rangle^{-2s} = 1 + 2\sum_{l\geq 1} l^{-2s}$ ($\langle 0\rangle = 1$), finite by (W1-E2). Then $\sum_l |Y_l| = \sum_l \langle l\rangle^{-s}\cdot\langle l\rangle^s |Y_l| \leq (1+2\zeta(2s))^{1/2}\|Y\|_{0,s}$, and $\|Y\|_{0,s}\leq \|Y\|_{a,s}$ since $e^{a|l|}\geq 1$.

Step S3.16: Algebraic rules. By (ii) all sequences involved are in $\ell^1\cap\ell^2$, so the double series defining $(X*Y)*Z$ and $X*(Y*Z)$ converge absolutely and may be rearranged (Fubini for series); commutativity is the substitution $i\mapsto l-i$; monotonicity and $|c*d|\leq |c|*|d|$ are termwise. □

Source: standard; used in P96-W §6 p. 294 and KP96 App. A pp. 31–32.

Lemma 3.12 (Convolution algebra)**.** *For $a\geq 0$, $s\geq 1$ and $c,d\in \ell^{a,s}_b$: $\|c*d\|_{a,s}\leq K_s\|c\|_{a,s}\|d\|_{a,s}$ with $K_s = 2^s(1+2\zeta(2s))^{1/2}$, and for three factors $\|c*d*e\|_{a,s}\leq K_s^2\|c\|\|d\|\|e\|$.*

Proof. This is Lemma 2.4, whose Step S2.5 uses Lemma 3.11(i)–(ii); the three-factor form applies it twice, with associativity from Lemma 3.11(iii). In Lean the lemma is proved once, in the W1 module, and W0 imports it. □

Source: P96-W Lemma 2, p. 279, and §6, pp. 294–295 (the proof there is labelled "Appendix A" on p. 279 but printed as §6); KP96 Lemma 2 p. 12.

Remark 3.13 (The algebra for $\frac12 < s < 1$). P96-W §6 proves the algebra for all $s>\frac12$ by a different weight split: $\langle l\rangle/(\langle l-i\rangle\langle i\rangle)\leq \langle l-i\rangle^{-1}+\langle i\rangle^{-1}$ and $(\alpha+\beta)^{2s}\leq 2^{2s-1}(\alpha^{2s}+\beta^{2s})$ give the same constant K_s. No later milestone needs $s<1$ (the cubic estimates use $s=p+\frac12\geq 1$), so this extension is not part of the plan.

3.5. **The cubic field.** The next lemma isolates the only property of the quartic coefficients that the estimates use: the selection rule of Lemma 2.2 together with the decay $(ijkl)^{-1/2}$. It serves both the cubic gradient (here) and the Birkhoff generating field of milestone W3, whose coefficients are those of G divided by the divisors of P96-W Lemma 4.

Lemma 3.14 (Selection-rule trilinear maps)**.** *Let $a\geq 0$, $p\geq\frac12$, $s=p+\frac12$, $M\geq 0$.*

(a) (One-sided.) Let $c:\mathbb{N}_+^4\to\mathbb{C}$ satisfy $|c_{ijkl}|\leq M(ijkl)^{-1/2}\sum_{\varepsilon\in\{\pm1\}^3}[\,l=\varepsilon_1 i+\varepsilon_2 j+\varepsilon_3 k\,]$. For $q,q',q''\in\ell^{a,p}_{\mathbb{C}}$ the sums $T_c(q,q',q'')_l=\sum_{i,j,k\geq 1} c_{ijkl}q_iq'_jq''_k$ converge absolutely, and

$$\|T_c(q,q',q'')\|_{a,p+1}\leq 2MK_s^2\,\|q\|_{a,p}\|q'\|_{a,p}\|q''\|_{a,p}.$$

(b) (Two-sided.) Let $c:(\mathbb{Z}\setminus 0)^4\to\mathbb{C}$ satisfy $|c_{ijkl}|\leq M|ijkl|^{-1/2}\sum_{\varepsilon\in\{\pm1\}^3}[\,l=\varepsilon_1 i+\varepsilon_2 j+\varepsilon_3 k\,]$. For $w,w',w''\in\ell^{a,p}_b$ the sums $T_c(w,w',w'')_l=\sum_{i,j,k\neq 0} c_{ijkl}w_iw'_jw''_k$ ($l\neq 0$) converge absolutely; with $T_c(\cdot)_0=0$,

$$\|T_c(w,w',w'')\|_{a,p+1}\leq 64MK_s^2\,\|w\|_{a,p}\|w'\|_{a,p}\|w''\|_{a,p}.$$

In both cases T_c is trilinear; it is symmetric if c is symmetric in its first three indices.

Proof. **Step S3.17: One-sided majorant.** In (a) let $x_i=|q_{|i|}|/\sqrt{|i|}$ ($i\neq 0$), $x_0=0$, even; similarly x',x''. By Lemma 3.10(i), $\|x\|_{a,s}=\sqrt{2}\|q\|_{a,p}$.

Step S3.18: One-sided selection. $\sum_{i,j,k}|c_{ijkl}q_iq'_jq''_k|\leq Ml^{-1/2}\sum_{i,j,k\geq1}\sum_{\varepsilon}[l=\varepsilon_1 i+\varepsilon_2 j+\varepsilon_3 k]\,x_ix'_jx''_k\leq Ml^{-1/2}X_l$, $X=x*x'*x''$, because $(i,j,k,\varepsilon)\mapsto(\varepsilon_1 i,\varepsilon_2 j,\varepsilon_3 k)$ is injective into $(\mathbb{Z}\setminus 0)^3$ and the x's are even (as in Step S2.7). X is finite and even by Lemmas 3.11 and 3.12.

Step S3.19: One-sided smoothing. $\|T_c\|^2_{a,p+1}\leq M^2\sum_{l\geq1} l^{2p+1}e^{2al}X_l^2\leq\frac{M^2}{2}\|X\|^2_{a,s}$ (evenness), and $\|X\|_{a,s}\leq K_s^2\|x\|\|x'\|\|x''\|=2^{3/2}K_s^2\prod\|q\|$. Hence the bound $2MK_s^2\prod\|q\|$.

Step S3.20: Two-sided majorant. In (b) let $\tilde{x}_i=(|w_i|+|w_{-i}|)/\sqrt{|i|}$ ($i\neq 0$), $\tilde{x}_0=0$, even, and similarly $\tilde{x}',\tilde{x}''$. Then $\|\tilde{x}\|^2_{a,s}=\sum_{i\neq0}|i|^{2p}e^{2a|i|}(|w_i|+|w_{-i}|)^2\leq 4\|w\|^2_{a,p}$.

Step S3.21: Two-sided selection. For fixed ε, the substitution $i' = \varepsilon_1 i$, $j' = \varepsilon_2 j$, $k' = \varepsilon_3 k$ is a bijection of $(\mathbb{Z}\setminus 0)^3$, and $|w_i|/\sqrt{|i|} \le \tilde{x}_{i'}$. Hence each of the eight ε-terms is at most $(\tilde{x} * \tilde{x}' * \tilde{x}'')_l =: \tilde{X}_l$, and $|T_c(w, w', w'')_l| \le 8M|l|^{-1/2}\tilde{X}_l$.

Step S3.22: Two-sided smoothing. $\|T_c\|^2_{a,p+1} \le 64M^2 \sum_{l\neq 0} |l|^{2p+1} e^{2a|l|} \tilde{X}_l^2 \le 64M^2 \|\tilde{X}\|^2_{a,s}$ and $\|\tilde{X}\|_{a,s} \le K_s^2 \cdot 8 \prod \|w\|$, so $\|T_c\| \le 64MK_s^2 \prod \|w\|$. Trilinearity and symmetry are termwise, the sums being absolutely convergent. □

Source: P96-W pp. 279–280 (Lemma 3) and p. 284 ((13)–(14), estimate of $\partial F/\partial w_l$, constants not made explicit); KP96 p. 16. The explicit constants are ours.

Theorem 3.15 (The cubic map as a continuous trilinear map)**.** *Let $a \ge 0$, $p \ge \frac{1}{2}$, $C_* = K^2_{p+1/2}/\pi$, and $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$.*

(i) There is a unique continuous symmetric $\mathbb{K}$-trilinear map $\mathcal{T}^{a,p}_{\mathbb{K}} : (\ell^2_{\mathbb{K}})^3 \to \ell^2_{\mathbb{K}}$ (Lean `ContinuousMultilinearMap K (fun _ : Fin 3 => lp ... 2) (lp ... 2)`) with $\kappa_{a,p+1}\mathcal{T}^{a,p}_{\mathbb{K}}(y, y', y'') = T(\kappa_{a,p}y, \kappa_{a,p}y', \kappa_{a,p}y'')$, where T is the trilinear map of Theorem 2.5; and $\|\mathcal{T}^{a,p}_{\mathbb{K}}\| \le C_*$.

(ii) (Real structure.) With ι and conjugation of Definition 3.22, $\mathcal{T}^{a,p}_{\mathbb{C}}(\iota y, \iota y', \iota y'') = \iota\, \mathcal{T}^{a,p}_{\mathbb{R}}(y, y', y'')$ and $\overline{\mathcal{T}^{a,p}_{\mathbb{C}}(z, z', z'')} = \mathcal{T}^{a,p}_{\mathbb{C}}(\bar{z}, \bar{z}', \bar{z}'')$.

(iii) (Compatibility.) For $a \ge a' \ge 0$, $p \ge p' \ge \frac{1}{2}$, with $\hat{\imath}$, $\hat{\imath}'$ the representatives of $\ell^{a,p} \subset \ell^{a',p'}$ and $\ell^{a,p+1} \subset \ell^{a',p'+1}$: $\mathcal{T}^{a',p'}(\hat{\imath}y, \hat{\imath}y', \hat{\imath}y'') = \hat{\imath}'\mathcal{T}^{a,p}(y, y', y'')$.

(iv) (Frozen identification.) For $y \in \ell^2_{\mathbb{R}}$ and $l \ge 1$,

$$\texttt{cubicGrad m hm y l} = \frac{\mathcal{T}^{1,1}_{\mathbb{R}}(y, y, y)_l}{\omega_{1,2}(l)}, \qquad \text{i.e.} \qquad G_q(\kappa_{1,1}y) = \kappa_{1,2}\, \mathcal{T}^{1,1}_{\mathbb{R}}(y, y, y).$$

For $a > 0$ and $p \ge \frac{1}{2}$ likewise $G_q(\kappa_{a,p}y) = \kappa_{a,p+1}\mathcal{T}^{a,p}_{\mathbb{R}}(y, y, y)$.

Proof. **Step S3.23: Coordinates.** For $y, y', y'' \in \ell^2_{\mathbb{K}}$ put $q = \kappa_{a,p}y$ etc. By Theorem 2.5 (equivalently Lemma 3.14(a) with $c_{ijkl} = g_{ijkl}(\lambda_i\lambda_j\lambda_k\lambda_l)^{-1/2}$ and $M = 1/(2\pi)$, using $\lambda \ge j$ and the bound after (3)), each sum $T(q, q', q'')_l$ is an absolutely convergent `tsum` over $\mathbb{N}_+^3$ and $T(q, q', q'') \in \ell^{a,p+1}_{\mathbb{K}}$ with norm $\le C_* \prod \|q\|_{a,p} = C_* \prod \|y\|$. Define $\mathcal{T}(y, y', y'') = \tau_{a,p+1}T(q, q', q'')$.

Step S3.24: Continuity. Trilinearity and symmetry hold termwise (symmetry of g, Lemma 2.2); the bound makes the multilinear map continuous with norm $\le C_*$ (Lean: `MultilinearMap.mkContinuous`). Uniqueness: a map with the stated coordinates is determined, since $\kappa_{a,p+1}$ is injective.

Step S3.25: Real structure and compatibility. The coefficients $g_{ijkl}(\lambda_i\lambda_j\lambda_k\lambda_l)^{-1/2}$ are real, so the coordinate sums commute with ι and with conjugation (absolute convergence). (iii) holds because the inclusions do not change coefficients (Lemma 3.5(iii)), so both sides have coefficients $T(q, q', q'')$.

Step S3.26: Frozen identification. By the frozen definitions, `reconU m hm y` is $U_m(\kappa_{1,1}y)$ (Definition 1.7; the `tsum` is absolutely convergent by Lemma 2.1), the interval integral over $[0, \pi]$ is the integral of Definition 1.9, and $\sqrt{\lambda_l} > 0$. So `cubicGrad m hm y l` $= G_q(\kappa_{1,1}y)_l$, which is $T(q, q, q)_l$ by Lemma 2.3 ($a = 1 > 0$). The general case is the same with $a > 0$. □

Source: P96-W Lemma 3, pp. 279–280; Theorem 2.5 for the bound. The Lean packaging is ours.

Corollary 3.16 (Analyticity and Lipschitz bounds of the cubic gradient)**.** *Let $a \ge 0$, $p \ge \frac{1}{2}$, $\mathcal{T} = \mathcal{T}^{a,p}_{\mathbb{K}}$, C_* as above, and $\mathcal{N}(y) = \mathcal{T}(y, y, y)$.*

(i) $\mathcal{N} : \ell^2_{\mathbb{K}} \to \ell^2_{\mathbb{K}}$ is $\mathbb{K}$-analytic on all of $\ell^2_{\mathbb{K}}$ (Lean `AnalyticOnNhd K N Set.univ`), with $\|\mathcal{N}(y)\| \le C_*\|y\|^3$.

(ii) $D\mathcal{N}(y)h = 3\mathcal{T}(y,y,h)$, $D^2\mathcal{N}(y)(h,k) = 6\mathcal{T}(y,h,k)$, $D^3\mathcal{N}(y) = 6\mathcal{T}$; *so* $\|D\mathcal{N}(y)\| \le 3C_*\|y\|^2$ *and* $\|D^2\mathcal{N}(y)\| \le 6C_*\|y\|$.
(iii) $\|\mathcal{N}(y) - \mathcal{N}(y')\| \le 3C_*\|y - y'\| \max(\|y\|, \|y'\|)^2$.
(iv) (Frozen form.) The map $\mathcal{G}^w : \ell^2_{\mathbb{R}} \to \ell^2_{\mathbb{R}}$, $\mathcal{G}^w(y)_l = \omega_{1,2}(l)\cdot$ `cubicGrad m hm y l`, *is* $\mathcal{N}$ *for* $(a,p) = (1,1)$, $\mathbb{K} = \mathbb{R}$; *it is real analytic everywhere and* $\|\mathcal{G}^w(y)\| \le C_m\|y\|^3$, $C_m = 8(1+2\zeta(3))/\pi$. *The coefficient sequence* $G_q(\kappa_{1,1}y)$ *has weighted coordinates* $\mathcal{G}^w(y)$ *at weight* $(1,2)$, $(l^{-1}\mathcal{G}^w(y)_l)$ *at weight* $(1,1)$ *and* $(l^{-2}\mathcal{G}^w(y)_l)$ *at weight* $(1,0)$.

Proof. **Step S3.27: Analyticity.** $\mathcal{N} = \mathcal{T}\circ\Delta$ with $\Delta(y) = (y,y,y)$ continuous linear; (W1-E3).

Step S3.28: Derivatives. Expanding $\mathcal{T}(y+h, y+h, y+h)$ by trilinearity, the linear term is $\mathcal{T}(h,y,y) + \mathcal{T}(y,h,y) + \mathcal{T}(y,y,h) = 3\mathcal{T}(y,y,h)$ by symmetry and the rest is $O(\|h\|^2)$ with constant $\le 3C_*(\|y\| + \|h\|)\|h\|^2$; the higher derivatives are the same computation (Lean: `ContinuousMultilinearMap.hasFDerivAt` composed with Δ).

Step S3.29: Lipschitz. $\mathcal{N}(y) - \mathcal{N}(y') = \mathcal{T}(y-y', y, y) + \mathcal{T}(y', y-y', y) + \mathcal{T}(y', y', y-y')$, so the difference is at most $C_*\|y-y'\|(\|y\|^2 + \|y\|\|y'\| + \|y'\|^2) \le 3C_*\|y-y'\|\max^2$.

Step S3.30: Frozen form. Theorem 3.15(iv) and $\omega_{1,1}(l)/\omega_{1,2}(l) = l^{-1}$, $\omega_{1,0}(l)/\omega_{1,2}(l) = l^{-2}$; $C_* = K_{3/2}^2/\pi = C_m$ (Theorem 2.5). □

Source: P96-W Lemma 3, p. 279 (analyticity near 0, with $O(\|q\|^3)$); the global polynomial form and the constants are ours.

Proposition 3.17 (Gradient identity)**.** *Let* $a > 0$, $p \ge \frac12$, *and for* $q \in \ell^{a,p}$ *let* $G(q) = \frac14\int_0^\pi U_m(q)^4\,dx$ *(Definition 1.8). Let* $\langle u, v\rangle_0 = \sum_{j\ge1} u_j v_j$ *be the* unweighted *pairing, absolutely convergent for* $u \in \ell^{a,p+1}$, $v \in \ell^{a,p}$.

(i) $G(q) = \frac14\langle G_q(q), q\rangle_0 = \frac14\mathcal{Q}(q,q,q,q)$ *with the symmetric continuous 4-linear form* $\mathcal{Q}(q_1,q_2,q_3,q_4) = \langle T(q_1,q_2,q_3), q_4\rangle_0$, $|\mathcal{Q}| \le C_*\prod\|q_i\|_{a,p}$.
(ii) G *is real analytic on* $\ell^{a,p}$ *and* $DG(q)h = \langle G_q(q), h\rangle_0$ *for all* $h \in \ell^{a,p}$.
(iii) $G_q(q)$ *is* not *the Riesz gradient of* G *for the inner product of* $\ell^{a,p}$*; that gradient is* $(\omega_{a,p}(j)^{-2}G_q(q)_j)_j$. *In frozen weighted coordinates: for* $G^w(y) = G(\kappa_{1,1}y)$,

$$DG^w(y)h = \sum_{j\ge1} \texttt{cubicGrad m hm y j}\cdot\texttt{coeff h j}, \qquad \nabla^{\ell^2}G^w(y) = \Big(\frac{\mathcal{G}^w(y)_j}{\omega_{1,1}(j)\,\omega_{1,2}(j)}\Big)_j.$$

Proof. **Step S3.31: Quartic form.** By Lemma 2.3, $G(q) = \frac14\sum_{i,j,k,l} g_{ijkl}\hat q_i\hat q_j\hat q_k\hat q_l$ absolutely; grouping the sum over l last gives $\frac14\sum_l q_l\,T(q,q,q)_l$. The same absolute convergence, with four different arguments, defines $\mathcal{Q}$; it is symmetric because g is. Bound: $|\langle u,v\rangle_0| \le \|u\|_{0,0}\|v\|_{0,0} \le \|u\|_{a,p+1}\|v\|_{a,p}$ (Cauchy–Schwarz and Lemma 3.5), and Theorem 2.5.

Step S3.32: Derivative. $G = \frac14\mathcal{Q}\circ(q \mapsto (q,q,q,q))$ is analytic ((W1-E3)), and by symmetry $DG(q)h = \frac14\cdot 4\mathcal{Q}(q,q,q,h) = \langle T(q,q,q), h\rangle_0 = \langle G_q(q), h\rangle_0$ (Theorem 3.15(iv)).

Step S3.33: Weighted coordinates. $\langle G_q(q), h\rangle_0 = \sum_j \omega_{a,p}(j)^2\big(\omega_{a,p}(j)^{-2}G_q(q)_j\big)h_j$ identifies the Riesz gradient. For the frozen form substitute $q = \kappa_{1,1}y$, $h \mapsto \kappa_{1,1}h$ and Corollary 3.16(iv); the ℓ^2 gradient is read off from $\sum_j G_q(\kappa_{1,1}y)_j\,h_j/\omega_{1,1}(j)$. □

Source: P96-W (7) p. 278 and (8)–(9) p. 280 (where G_q is "the gradient of G" without naming the pairing).

Proposition 3.18 (The Hamiltonian field)**.** *Let* $a > 0$ *and* $p \ge \frac12$.

(i) The linear field $X_\Lambda(q,p) = (\lambda p, -\lambda q)$ *is bounded* $E_{a,p} \to E_{a,p-1}$ *with norm* $\le \sqrt{1+m}$, *and is not bounded* $E_{a,p} \to E_{a,p}$.

(ii) The nonlinear field $X_G(q,p) = (0, -G_q(q))$ is a real analytic map $E_{a,p} \to E_{a,p+1}$ (smoothing of order one), $\|X_G(q,p)\|_{E_{a,p+1}} \le C_*\|q\|_{a,p}^3$, with the Lipschitz and derivative bounds of Corollary 3.16. Its complex extension $(z,\bar z) \mapsto$ (same formula with $\mathcal{T}_{\mathbb{C}}$) is $\mathbb{C}$-analytic $E_{a,p}^{\mathbb{C}} \to E_{a,p+1}^{\mathbb{C}}$.

(iii) $X_H = X_\Lambda + X_G$ is real analytic $E_{a,p} \to E_{a,p-1}$.

(iv) (Hamiltonian structure.) $H = \Lambda + G$ is real analytic on $E_{a,p}$, and for every j, $DH(q,p)(0,e_j) = \lambda_j p_j$, $DH(q,p)(e_j,0) = \lambda_j q_j + G_q(q)_j$; that is, $X_H = (\partial_p H, -\partial_q H)$ coordinatewise, which is the system (2).

(v) (Frozen form, $a = p = 1$.) In weighted coordinates, with $D = D_\mu$, $\mu_j = \lambda_j/j$, and $S = D_{(1/j)}$, the map X^w : `WaveSpace` $\to$ `WaveSpace`,

$$X^w(y,y') = \big(Dy',\ -Dy - S^2\mathcal{G}^w(y)\big),$$

represents $X_H : E_{1,1} \to E_{1,0}$; it is real analytic on all of `WaveSpace`, and its j-th coordinates are $\omega_{1,0}(j)\big(\lambda_j$ `coeff y' j`, $-\lambda_j$ `coeff y j` $-$ `cubicGrad m hm y j`$\big)$, the right sides of the frozen conclusion (4) times $\omega_{1,0}(j) = e^j$.

Proof. **Step S3.34: Linear part.** Lemma 3.6(i)–(ii) with $r = 1$ in each component; the max norm is preserved under componentwise bounds.

Step S3.35: Nonlinear part. Corollary 3.16 with Theorem 3.15(iv) in the second component, composed with the continuous linear projection $(q,p) \mapsto q$; complex version from Theorem 3.15(ii).

Step S3.36: Sum. X_G maps into $E_{a,p+1} \subset E_{a,p-1}$ (Lemma 3.5); a sum of a bounded linear and an analytic map is analytic.

Step S3.37: Hamiltonian. $\Lambda(q,p) = \frac{1}{2}(\langle \lambda q, q\rangle_0 + \langle \lambda p, p\rangle_0)$ is the diagonal of the symmetric bilinear form $\beta(q,q') = \sum_j \lambda_j q_j q_j'$, and $|\beta(q,q')| \le \sqrt{1+m}\sum_j j|q_j||q_j'| \le \sqrt{1+m}\|q\|_{0,1/2}\|q'\|_{0,1/2} \le \sqrt{1+m}\|q\|_{a,p}\|q'\|_{a,p}$ (Lemma 3.5, $p \ge \frac{1}{2}$), so Λ is a bounded quadratic form on $E_{a,p}$. Its derivative is $\langle \lambda q, h\rangle_0 + \langle \lambda p, k\rangle_0$; add Proposition 3.17(ii) and evaluate at $(e_j,0)$, $(0,e_j)$.

Step S3.38: Frozen form. Coordinates of $X_H(\kappa_{1,1}y, \kappa_{1,1}y')$ times $\omega_{1,0}(j) = e^j$: $e^j\lambda_j y_j'/(je^j) = (\lambda_j/j)y_j'$, and e^j `cubicGrad`$_j$ $= j^{-2}\mathcal{G}^w(y)_j$ by Corollary 3.16(iv). D and S are bounded by Lemma 3.4 ($\sup \lambda_j/j = \sqrt{1+m}$, $\sup 1/j = 1$), so X^w is a polynomial map on `WaveSpace`, analytic by (W1-E3). □

Source: P96-W p. 280 ("X_G defines a real analytic map from some neighbourhood of the origin in $\ell^{a,s} \times \ell^{a,s}$ into $\ell^{a,s+1} \times \ell^{a,s+1}$ … X_Λ is unbounded of order 1"); P96-K Assumption C p. 121 ($X_P : \mathcal{P}^{a,p} \to \mathcal{P}^{a,\bar p}$, $\bar p > p$ for $d = 1$). Global analyticity is ours.

3.6. The sine system and analytic reconstruction.

Lemma 3.19 (Completeness of the sine system)**.** *(i) If $f \in L^2(0,\pi)$ and $\int_0^\pi f\phi_l\,dx = 0$ for all $l \ge 1$, then $f = 0$ almost everywhere on $(0,\pi)$. (ii) If moreover f is continuous on $[0,\pi]$, then $f = 0$ on $[0,\pi]$.*

Proof. **Step S3.39: Odd extension.** Define g on $(-\pi,\pi]$ by $g(x) = f(x)$ for $0 < x \le \pi$, $g(x) = -f(-x)$ for $-\pi < x < 0$, $g(0) = 0$, and regard it as a function on $\mathbb{R}/2\pi\mathbb{Z}$. Then $g \in L^2$, with $\|g\|^2 = 2\|f\|^2$ for the Lebesgue measure on $(-\pi,\pi]$.

Step S3.40: Fourier coefficients vanish. For $n \in \mathbb{Z}$, $\int_{-\pi}^\pi g(x)e^{-inx}dx = \int_0^\pi f(x)(e^{-inx} - e^{inx})dx = -2i\int_0^\pi f(x)\sin(nx)\,dx$ (substitute $x \mapsto -x$ on $(-\pi,0)$). This vanishes for $n = 0$, for $n \ge 1$ by hypothesis ($\sin(nx) = \sqrt{\pi/2}\,\phi_n(x)$), and for $n \le -1$ since $\sin(nx) = -\sin(|n|x)$.

Step S3.41: Hilbert basis. The Fourier coefficients of g for the normalized Haar measure are $(2\pi)^{-1}$ times the integrals above, hence all zero; by (W1-E8) (the coefficient map of a Hilbert basis is injective), $g = 0$ in L^2, so $g = 0$ a.e. and $f = 0$ a.e. on $(0, \pi)$.

Step S3.42: Continuity. If f is continuous and $f(x_0) \neq 0$ for some $x_0 \in [0, \pi]$, then $|f| > |f(x_0)|/2$ on a relatively open subinterval of $[0, \pi]$ of positive length, contradicting (i). □

Source: standard (P96-W p. 279, first paragraph, asserted); used in Step S2.10. Lean: transport of g to `AddCircle (2 * Real.pi)` and `fourierBasis`; the measure normalization is the factor $(2\pi)^{-1}$.

Lemma 3.20 (Analytic sine features)**.** *Let $a > 0$, $p \in \mathbb{R}$, $k \in \mathbb{N}$, and let $\beta : \mathbb{N}_+ \to \mathbb{R}$ satisfy $|\beta_j| \le B j^\nu$ for some $B, \nu \ge 0$. For $x \in \mathbb{C}$ with $|\operatorname{Im} x| < a$ define $\Psi(x) \in \ell^2_{\mathbb{C}}$ by*

$$\Psi(x)_j = \frac{\beta_j \, \phi_j^{(k)}(x)}{\omega_{a,p}(j)}, \qquad \phi_j^{(k)}(x) = \sqrt{2/\pi}\, j^k \sin^{(k)}(jx).$$

(i) For $0 < a' < a$, $\sum_j \sup_{|\operatorname{Im} x| \le a'} |\Psi(x)_j| \le \sqrt{2/\pi}\, B \sum_j j^{\nu+k-p} e^{-(a-a')j} < \infty$.

(ii) Ψ is holomorphic on the strip $\{|\operatorname{Im} x| < a\}$, hence $\mathbb{C}$-analytic there; $\Psi(x)$ is real for real x, and $\Psi_{\mathbb{R}} = \operatorname{re} \circ \Psi|_{\mathbb{R}} : \mathbb{R} \to \ell^2_{\mathbb{R}}$ is real analytic.

(iii) For real x and $y \in \ell^2_{\mathbb{R}}$, $\langle \Psi_{\mathbb{R}}(x), y \rangle = \sum_j \beta_j \, (\kappa_{a,p} y)_j \, \phi_j^{(k)}(x)$, absolutely convergent.

Proof. **Step S3.43: Growth on the strip.** $|\sin^{(k)}(jx)| \le \cosh(j \operatorname{Im} x) \le e^{j|\operatorname{Im} x|}$, so $|\Psi(x)_j| \le \sqrt{2/\pi} B j^{\nu+k-p} e^{-(a-a')j}$ on $|\operatorname{Im} x| \le a'$; the series converges since $a - a' > 0$.

Step S3.44: Normal convergence. The partial sums $\Psi^N(x) = \sum_{j \le N} \Psi(x)_j e_j$ are holomorphic (finite sums of entire functions times fixed vectors), and by (i) $\sup_{|\operatorname{Im} x| \le a'} \|\Psi - \Psi^N\| \le \sum_{j > N} \sup |\Psi_j| \to 0$. Every compact subset of the strip lies in some $\{|\operatorname{Im} x| \le a'\}$, so $\Psi^N \to \Psi$ locally uniformly and Ψ is holomorphic and analytic by (W1-E4).

Step S3.45: Real restriction. For real x all coordinates are real, so $\Psi(x) = \iota \Psi_{\mathbb{R}}(x)$; $\Psi_{\mathbb{R}} = \operatorname{re} \circ \Psi \circ (\mathbb{R} \hookrightarrow \mathbb{C})$ is real analytic by Lemma 3.23(ii). (iii) is `lp.inner_eq_tsum` and $y_j / \omega_{a,p}(j) = (\kappa_{a,p} y)_j$. □

Source: ours (P96-W Lemma 1, p. 278, uses complex neighbourhoods in x in the same way).

Corollary 3.21 (Analytic reconstruction)**.** *Let $Y : \mathbb{R}^n \to \ell^2_{\mathbb{R}}$ be real analytic at s_0 (Lean `AnalyticAt Real Y s0`) and $k \in \mathbb{N}$. Then $(s, x) \mapsto \partial_x^k$ `reconU m hm (Y s)`(x) and $(s, x) \mapsto \partial_x^k$ `reconV m hm (Y s)`(x) are real analytic at (s_0, x_0) for every $x_0 \in \mathbb{R}$; for $k = 0$ these are exactly the functions $(s, x) \mapsto$ `reconU m hm (Y s) x`, `reconV m hm (Y s) x`, and for $k = 1$ the x-derivative equals `derivWithin (reconU m hm (Y s)) (Set.Icc 0 Real.pi) x` for $x \in [0, \pi]$.*

Proof. **Step S3.46: Bilinear form.** Take $a = p = 1$ and $\beta_j = \lambda_j^{-1/2}$ (for U_m) or $\beta_j = \lambda_j^{1/2}$ (for V_m), so $|\beta_j| \le (1+m)^{1/4} j^{1/2}$. By Lemma 3.20(iii), the function is $(s, x) \mapsto \langle \Psi_{\mathbb{R}}^{(k)}(x), Y(s) \rangle$: for $k = 0$ this is the frozen `tsum` term by term (`coeff (Y s) j = (`$\kappa_{1,1}$ `(Y s)) j`).

Step S3.47: Analyticity. The real inner product is a continuous bilinear map; compose with Y (analytic at s_0) and $\Psi_{\mathbb{R}}^{(k)}$ (analytic at x_0) by (W1-E3) (`AnalyticAt.comp2` on the pair (s, x)).

Step S3.48: Derivatives. $\frac{d}{dx} \langle \Psi_{\mathbb{R}}^{(k)}(x), y \rangle = \langle (\Psi_{\mathbb{R}}^{(k)})'(x), y \rangle$, and $(\Psi_{\mathbb{R}}^{(k)})' = \Psi_{\mathbb{R}}^{(k+1)}$ because coordinate evaluation is continuous linear (so the derivative of Ψ has the termwise derivatives as coordinates). At $x \in [0, \pi]$ the derivative within $[0, \pi]$ equals the two-sided derivative, since the function is differentiable on $\mathbb{R}$ and $[0, \pi]$ has unique derivatives (`uniqueDiffOn_Icc`). □

Source: P96-W Lemma 1, p. 278 (real analyticity of u in t and x, argued via complex neighbourhoods); ours in this form. Consumed by W2 (frozen conclusion (5)).

3.7. **Analytic maps between weighted spaces.**

Definition 3.22 (Complexification)**.** $\iota : \ell^2_{\mathbb{R}} \to \ell^2_{\mathbb{C}}$, $(\iota y)_j = y_j$; $\mathrm{re} : \ell^2_{\mathbb{C}} \to \ell^2_{\mathbb{R}}$, $(\mathrm{re}\, z)_j = \mathrm{Re}\, z_j$; conjugation $(\bar z)_j = \overline{z_j}$; and componentwise on products, e.g. $E^{\mathbb{C}}_{a,p}$ for $E_{a,p}$, represented by $(\ell^2_{\mathbb{C}})^2$. A map f from an open subset of $\ell^2_{\mathbb{C}}$ (or a product) is *real on reals* if $f(\iota y) \in \iota(\ell^2_{\mathbb{R}})$ whenever defined.

Source: ours (P96-W p. 281: "real analytic means that H is a function of z and $\bar z$, real analytic in the real and imaginary part of z").

Lemma 3.23 (Complexification and real analyticity)**.** *(i) ι and* re *are continuous $\mathbb{R}$-linear, $\|\iota y\| = \|y\|$, $\|\mathrm{re}\, z\| \le \|z\|$, $\mathrm{re}\circ\iota = \mathrm{id}$, $\|\iota y + i\iota y'\|^2 = \|y\|^2 + \|y'\|^2$, and conjugation is a conjugate-linear isometry. All three commute with $\tau_{a,p}$, $\kappa_{a,p}$ and every real multiplier D_μ.*

(ii) If F_1, F_2 are complex Banach spaces, V a real normed space, $\ell : V \to F_1$ and $\pi : F_2 \to W$ continuous $\mathbb{R}$-linear, and f is $\mathbb{C}$-analytic at $\ell(x)$, then $\pi \circ f \circ \ell$ is $\mathbb{R}$-analytic at x.

(iii) In particular, if f is defined near ιy in $\ell^2_{\mathbb{C}}$ (or a product), $\mathbb{C}$-analytic at ιy, and real on reals near ιy, then $g = \mathrm{re}\circ f \circ \iota$ is $\mathbb{R}$-analytic at y and $\iota \circ g = f \circ \iota$ near y; its real derivative is $Dg(y)h = \mathrm{re}\, Df(\iota y)\iota h$.

Proof. (i) is coordinatewise. (ii): f is $\mathbb{R}$-analytic at $\ell(x)$ (`AnalyticAt.restrictScalars`), and composition with the $\mathbb{R}$-linear continuous maps ℓ, π preserves analyticity ((W1-E3)). (iii): (ii) with $\ell = \iota$, $\pi = \mathrm{re}$, and $\iota \circ \mathrm{re} = \mathrm{id}$ on $\iota(\ell^2_{\mathbb{R}})$; the derivative by the chain rule. □

Source: standard.

Definition 3.24 (Holomorphic maps on Banach domains)**.** Let F_1, F_2 be complex Banach spaces and $U \subset F_1$ open. A map $f : U \to F_2$ is *holomorphic* if it is complex Fréchet differentiable at every point of U (Lean `DifferentiableOn Complex f U`). We write $\|f\|_U = \sup_U \|f\|$.

Source: P96-K p. 121 ("real analytic in $D(s,r)$") and App. A p. 146; P96-W p. 281.

Lemma 3.25 (Cauchy estimate)**.** *Let f be holomorphic on the open ball $B(v,r) \subset F_1$ with $\|f\| \le M$ there. Then $\|Df(v)\| \le M/r$ in the operator norm. Consequently, if f is holomorphic on an open V with $\|f\|_V \le M$ and $B(w,\delta) \subset V$, then $\|Df(w)\| \le M/\delta$; and f is (M/δ)-Lipschitz on every convex set C with $C + B(0,\delta) \subset V$.*

Proof. **Step S3.49: Complex lines.** Fix $u \in F_1$ with $\|u\| = 1$ and $0 < r' < r$. The map $\varphi(\zeta) = f(v + \zeta u)$ is holomorphic on $|\zeta| < r$ (chain rule), in particular continuous on $|\zeta| \le r'$, with $\varphi'(0) = Df(v)u$ and $\|\varphi\| \le M$.

Step S3.50: One-variable estimate. By (W1-E4), $\|Df(v)u\| \le M/r'$; let $r' \uparrow r$ and take the supremum over u.

Step S3.51: Margins. Apply this on $B(w,\delta)$; the Lipschitz bound is the mean value inequality ((W1-E5)) on C. □

Source: P96-K Lemma A.3, p. 147, and its proof.

Lemma 3.26 (Uniform limits)**.** *Let $U \subset F_1$ be open, $f_k : U \to F_2$ holomorphic, and $f_k \to f$ uniformly on U. Then f is holomorphic on U, $Df_k \to Df$ uniformly on every ball $B(w,\delta)$ with $B(w,2\delta) \subset U$, and $\|Df(w)\| \le \|f\|_U/\delta$ there. If every f_k is real on reals, so is f.*

Proof. **Step S3.52: Cauchy in derivatives.** For $w' \in B(w,\delta)$, $B(w',\delta) \subset U$, so Lemma 3.25 applied to $f_k - f_l$ gives $\|Df_k(w') - Df_l(w')\| \le \|f_k - f_l\|_U/\delta$. Thus (Df_k) is uniformly Cauchy on $B(w,\delta)$ in the complete space of bounded $\mathbb{C}$-linear maps, with a uniform limit g.

Step S3.53: Limit. By (W1-E5) on the open ball, f is $\mathbb{C}$-differentiable there with $Df = g$. Since w is arbitrary, f is holomorphic on U; the bound on $Df(w)$ is Lemma 3.25. Real on reals: $\iota(\ell^2_{\mathbb{R}})$ is closed. □

Source: standard (used implicitly in P96-K §4, convergence of the iteration).

Lemma 3.27 (Holomorphic maps of finitely many variables are analytic)**.** *Let $U \subset \mathbb{C}^n$ be open (max norm), F a complex Banach space and $f : U \to F$ holomorphic. Then f is $\mathbb{C}$-analytic on U (Lean* `AnalyticOnNhd Complex f U`*); on every closed polydisc $\bar{P}(z_0, r) \subset U$ it is the sum of an absolutely convergent power series with coefficients bounded by $\|f\|_{\bar{P}(z_0,r)} r^{-|\alpha|}$. If $F = E^{\mathbb{C}}_{a,p}$ and f is real on reals at the real points of U, then $\mathrm{re} \circ f|_{U \cap \mathbb{R}^n}$ is real analytic into $E_{a,p}$.*

Proof. The proof of track K's Lemma `bp:p1:holomorphic-analytic` (steps `step:p1:han-margin` through `step:p1:han-real`) applies verbatim with W replaced by the complete space F: finite dimensionality of W is used there only through boundedness of a continuous map on a compact polydisc (true in any normed space), the one-variable Cauchy formula (valid for complete F, (W1-E4)), and Fubini and term-by-term integration of uniformly convergent series of continuous F-valued maps on a compact torus (Bochner integral, (W1-E6)). The real statement is Lemma 3.23(iii). □

Source: standard; track K blueprint `bp:p1:holomorphic-analytic`.

Corollary 3.28 (Real-analytic tori from complex strips)**.** *Let $s > 0$, $T_s = \{\theta \in \mathbb{C}^2 : \max_i |\mathrm{Im}\, \theta_i| < s\}$, and let $\tilde{K} : T_s \to E^{\mathbb{C}}_{1,1}$ be holomorphic, real on reals, and 1-periodic. Then $K = \mathrm{re} \circ \tilde{K}|_{\mathbb{R}^2} : \mathbb{R}^2 \to$* `WaveSpace` *satisfies* `AnalyticOnNhd Real K Set.univ`*, is $\mathbb{Z}^2$-periodic, and* `fderiv Real K θ h` *= $\mathrm{re}\, D\tilde{K}(\theta)h$ for $h \in \mathbb{R}^2$.*

Proof. Lemma 3.27 with $U = T_s$ and Lemma 3.23(iii); periodicity restricts. □

Source: ours; the analyticity clause of the frozen `IsAnalyticTorusEmbedding` (consumed by W6, W8).

Lemma 3.29 (Holomorphic flows)**.** *Let F be a complex Banach space, $V \subset F$ open, $X : V \to F$ holomorphic with $\|X\|_V \le M$, and $U \subset V$ open with $B(w, 2\delta) \subset V$ for all $w \in U$, where $0 < M < \delta$. Put $L = M/\delta < 1$.*

- *(i) For every $w \in U$ there is a unique C^1 curve $t \mapsto \phi^t(w)$, $t \in [-1, 1]$, with $\phi^0(w) = w$, $\frac{d}{dt}\phi^t(w) = X(\phi^t(w))$ and values in V; it satisfies $\|\phi^t(w) - w\| \le |t|M$. Hence $\|\phi^t - \mathrm{id}\|_U \le \|X\|_V$.*
- *(ii) For each $t \in [-1, 1]$, $w \mapsto \phi^t(w)$ is holomorphic on U and $\|D\phi^t(w)\| \le e^{L|t|}$.*
- *(iii) If $F = E^{\mathbb{C}}$ for a real space E and X is real on reals, then ϕ^t is real on reals.*
- *(iv) (Parameters.) Let $X(\cdot, \xi)$, $\xi \in \Pi$, each satisfy the hypotheses with the same V, U, M, δ. Then for $\xi, \zeta \in \Pi$ and $|t| \le 1$, with $\Delta f = f(\cdot, \xi) - f(\cdot, \zeta)$, $\|\Delta(\phi^t - \mathrm{id})\|_U \le |t|\, e^{L|t|} \|\Delta X\|_V$.*

Proof. **Step S3.54: Lipschitz region.** For $w \in U$ and $w' \in B(w, \delta)$, $B(w', \delta) \subset B(w, 2\delta) \subset V$, so $\|DX(w')\| \le L$ by Lemma 3.25, and X is L-Lipschitz on the convex ball $B(w, \delta)$.

Step S3.55: Picard iterates. Let $\phi_0(w, t) = w$ and $\phi_{k+1}(w, t) = w + \int_0^t X(\phi_k(w, \sigma))\, d\sigma$. By induction, $\|\phi_k(w, t) - w\| \le |t|M < \delta$, so the iterates stay in $B(w, \delta)$ and are continuous in (w, t), and $\|\phi_{k+1}(w, t) - \phi_k(w, t)\| \le ML^k|t|^{k+1}/(k + 1)!$ uniformly in $w \in U$.

Step S3.56: Holomorphy of the iterates. By induction, $w \mapsto \phi_k(w, t)$ is holomorphic on U with $\|D_w\phi_k(w, t)\| \le \sum_{i \le k} (L|t|)^i/i! \le e^{L|t|}$: differentiate under the integral sign ((W1-E6)) with the dominating bound $\|DX\|\, \|D_w\phi_k\| \le Le^L$ on $B(w, \delta)$.

Step S3.57: Limit. The iterates converge uniformly on $U \times [-1, 1]$ to ϕ, which satisfies the integral equation (pass to the limit under the integral), hence is a C^1 solution; uniqueness among solutions in V is (W1-E7) on $B(w, \delta)$ (a solution with values in V stays in $B(w, \delta)$ for $|t| \le 1$, since $\|\dot{\phi}\| \le M$). By Lemma 3.26, ϕ^t is holomorphic; the derivative bound passes to the limit. This proves (i), (ii).

Step S3.58: Reality. If X is real on reals, every Picard iterate of a real initial point is real, and the real subspace is closed.

Step S3.59: Parameters. Let $\psi^t = \phi^t(w,\xi) - \phi^t(w,\zeta)$; both curves lie in the convex ball $B(w,\delta)$. Then $\|\dot\psi^t\| \le \|\Delta X\|_V + L\|\psi^t\|$, $\psi^0 = 0$, and Gronwall ((W1-E7)) gives $\|\psi^t\| \le \|\Delta X\|_V\,(e^{L|t|}-1)/L \le |t|e^{L|t|}\|\Delta X\|_V$. □

Source: P96-K Lemma A.4, p. 147 (estimates (i), (iv)), and P96-K §3, (10), p. 130 (existence of the flow on a smaller domain).

SOURCE_GAP: P96-K Lemma A.4 bounds $\|X\circ\phi^s(\cdot,\xi) - X\circ\phi^s(\cdot,\zeta)\|$ by $\|DX\|_V\|\Delta\phi^s\|_U$, which needs the segment between the two points to lie in V (not assumed: V is an arbitrary open domain), and $\|DX\|_V < \infty$ (not implied by X being C^1 on V). The hypotheses above (a ball margin 2δ and $M < \delta$) supply both; in the KAM step they hold by (9)–(10) of P96-K p. 129–130.

3.8. **Support for the KAM step.**

Definition 3.30 (KAM phase space, domains, norms)**.** Let $a \ge 0$, $p \in \mathbb{R}$. The complex phase space is $\mathcal{P}^{a,p}_{\mathbb{C}} = \mathbb{C}^2 \times \mathbb{C}^2 \times \ell^{a,p}_{\mathbb{C}} \times \ell^{a,p}_{\mathbb{C}}$ with points (θ, y, u, v) (Lean: `(Fin 2 -> Complex) × (Fin 2 -> Complex) × lp ... 2 × lp ... 2` in weighted coordinates); $|\cdot|$ is the max norm on $\mathbb{C}^2$. For $s, r > 0$,

$$D(s,r) = \{(\theta,y,u,v) : \max_i |\mathrm{Im}\,\theta_i| < s,\ |y| < r^2,\ \|u\|_{a,p} + \|v\|_{a,p} < r\},$$

and for $W = (\Theta, Y, U, V) \in \mathcal{P}^{a,\bar p}_{\mathbb{C}}$,

$$|W|_{\bar p, r} = |\Theta| + \frac{|Y|}{r^2} + \frac{\|U\|_{a,\bar p}}{r} + \frac{\|V\|_{a,\bar p}}{r}.$$

For a linear map $L : \mathcal{P}^{a,p}_{\mathbb{C}} \to \mathcal{P}^{a,\bar p}_{\mathbb{C}}$, $|L|_{r,s} = \sup_{W\neq 0} |LW|_{\bar p,r}/|W|_{p,s}$. The angle θ has period one: $x = 2\pi\theta$ is P96-K's angle, a P96-K strip $|\mathrm{Im}\,x| < s_K$ is $|\mathrm{Im}\,\theta| < s_K/(2\pi)$, and the θ-component of a vector field is $(2\pi)^{-1}$ times its x-component.

Source: P96-K p. 120 ($\mathcal{P}^{a,p}$), p. 121 ((1) and $D(s,r)$), p. 129 (operator norm); period conversion ours.

Remark 3.31 (Conventions to carry into W4–W6). P96-K writes $z = (u - iv)/\sqrt{2}$, $\bar z = (u + iv)/\sqrt{2}$ (p. 124), the complex conjugate of the W0 convention $z = (q + ip)/\sqrt{2}$ (Definition 1.11). Since the normal form $\langle\omega, y\rangle + \langle\Omega, z\bar z\rangle$ and the smallness norms are invariant under $z \leftrightarrow \bar z$, this is a relabelling, not a defect; W4–W6 should keep the W0 convention and state the swap once. The domain $D(s,r)$ uses the sum $\|u\| + \|v\|$, not the max norm of `WaveSpace`; the two differ by a factor at most 2.

Lemma 3.32 (Domains)**.** *Let $0 < \sigma \le 1$, $0 < 2\sigma < s$, $r > 0$. Then $D(s,r)$ is open and convex, $D(s',r') \subset D(s,r)$ for $s' \le s$, $r' \le r$, and every $W_0 \in D(s - 2\sigma, r/2)$ satisfies $W_0 + W \in D(s-\sigma, r)$ whenever $|W|_{p,r} < \sigma/2$. Also $|W|_{\bar p,r} \le |W|_{p,r}$ for $\bar p \le p$ and $|W|_{p,r} \le |W|_{p,r'}$ for $r' \le r$.*

Proof. Openness and convexity: each defining condition is a strict sublevel set of a continuous convex function ($\theta \mapsto |\mathrm{Im}\,\theta_i|$ is a seminorm). For the margin, $|\Theta| < \sigma/2$ gives $|\mathrm{Im}(\theta+\Theta)| < s - 2\sigma + \sigma/2 < s - \sigma$; $|Y| < r^2\sigma/2 \le r^2/2$ gives $|y + Y| < r^2/4 + r^2/2 < r^2$; $\|U\| + \|V\| < r\sigma/2 \le r/2$ gives $\|u + U\| + \|v + V\| < r/2 + r/2 = r$. The norm comparisons are Lemma 3.5(i) and $1/r^2 \le 1/r'^2$, $1/r \le 1/r'$. □

Source: P96-K p. 129 ("every point in $D(s-2\sigma, r/2)$ has at least $|\cdot|_{p,r}$-distance $\sigma/2$ to the boundary of $D(s-\sigma, r)$", with $\sigma \le 1$ from (6)); checked here.

Lemma 3.33 (Hilbert-valued Fourier series on strips)**.** *Let $\mathcal{H}$ be a complex Hilbert space (in the applications $\ell^2_{\mathbb{C}}$, representing $\ell^{a,p}_{\mathbb{C}}$), $n \ge 1$, $s > 0$, $T_s = \{\theta \in \mathbb{C}^n : \max_i |\mathrm{Im}\,\theta_i| < s\}$, and $u : T_s \to \mathcal{H}$ holomorphic, 1-periodic, with $\|u\|_s = \sup_{T_s}\|u\| < \infty$. Let $\hat u_k = \int_{[0,1]^n} u(x) e^{-2\pi i k\cdot x}dx$ ($k \in \mathbb{Z}^n$).*

(i) $\sum_k \|\hat u_k\|^2 e^{4\pi|k|_1 s} \le 2^n \|u\|_s^2$.

(ii) Conversely, if $c_k \in \mathcal{H}$ satisfy $\sum_k \|c_k\| e^{2\pi|k|_1 s} < \infty$, then $v(\theta) = \sum_k c_k e^{2\pi i k\cdot\theta}$ converges absolutely and uniformly on T_s, v is holomorphic and 1-periodic on T_s, $\hat v_k = c_k$, and $\|v\|_s \le \sum_k \|c_k\| e^{2\pi|k|_1 s}$.

(iii) Every u as in (i) is the sum of its Fourier series on $T_{s'}$ for $s' < s$.

Proof. **Step S3.60: Shifted coefficients.** For $\eta \in \mathbb{R}^n$ with $\max|\eta_i| < s$, $\hat{u}_k\, e^{2\pi k\cdot\eta}$ is the k-th Fourier coefficient of the continuous periodic map $x \mapsto u(x+i\eta)$: apply the contour shift of track K's `bp:p1:fourier-basic` to the scalar holomorphic functions $\langle e, u\rangle$, $e \in \mathcal{H}$, and use that these determine vectors.

Step S3.61: Parseval. For a continuous periodic $\mathcal{H}$-valued g, $\sum_k \|\hat{g}_k\|^2 = \int_{[0,1]^n}\|g\|^2 \le \sup\|g\|^2$: expand in an orthonormal basis (e_α) of $\mathcal{H}$ (for $\ell^2_{\mathbb{C}}$, the coordinate vectors), apply scalar Parseval ((W1-E8)) to each $\langle e_\alpha, g\rangle$ and sum (Tonelli). Hence $\sum_k \|\hat{u}_k\|^2 e^{4\pi k\cdot\eta} \le \|u\|_s^2$.

Step S3.62: Orthants. For $s' < s$ and $\varepsilon \in \{\pm 1\}^n$ take $\eta = s'\varepsilon$. Since $e^{4\pi s'|k|_1} = e^{4\pi s'\varepsilon\cdot k}$ for $\varepsilon_i = \operatorname{sgn} k_i$ (any sign when $k_i = 0$), $e^{4\pi s'|k|_1} \le \sum_\varepsilon e^{4\pi s'\varepsilon\cdot k}$; summing over the 2^n choices gives (i) with s', and monotone convergence gives s.

Step S3.63: Summation. (ii): $|e^{2\pi i k\cdot\theta}| \le e^{2\pi|k|_1 s}$ on T_s; uniform convergence of holomorphic partial sums and Lemma 3.26; coefficients by term-by-term integration. (iii): by (i) and Cauchy–Schwarz, $\sum_k \|\hat{u}_k\| e^{2\pi|k|_1 s'} \le 2^{n/2}\|u\|_s (\sum_k e^{-4\pi|k|_1(s-s')})^{1/2} < \infty$; the sum v of (ii) has the same coefficients as u on $\mathbb{R}^n$, so $\langle e, u - v\rangle$ vanishes on $\mathbb{R}^n$ by scalar uniqueness (`bp:p1:fourier-basic`) and on $T_{s'}$ by the identity theorem (`bp:p1:strip-identity`). □

Source: P96-K p. 126 ("the usual L^2-bound $\sum_k \|\hat{R}_k\|^2 e^{2|k||s|} \le 2^n \|\hat{R}\|^2_{D(s)}$", period 2π; with $\theta = x/2\pi$, $e^{2|k|s_K} = e^{4\pi|k|_1 s}$ for $s = s_K/2\pi$).

Lemma 3.34 (Hilbert-matrix bound)**.** *Let A be a bounded linear operator on $\ell^2_{\mathbb{K}}$ with matrix $A_{ij} = (Ae_j)_i$ $(i, j \ge 1)$.*

- *(i) The matrix $B_{ij} = |A_{ij}|/|i-j|$ $(i \ne j)$, $B_{ii} = 0$, defines a bounded operator, $(Bv)_i = \sum_j B_{ij} v_j$ absolutely convergent, with $\|B\| \le \frac{\pi}{\sqrt{3}}\|A\|$.*
- *(ii) The diagonal (A_{ii}) defines a multiplier with norm $\le \|A\|$.*
- *(iii) If F_{ij} satisfies $|F_{ij}| \le c|A_{ij}|/|i-j|$ for $i \ne j$ and $|F_{ii}| \le c|A_{ii}|$, then F defines a bounded operator with $\|F\| \le c(1+\pi/\sqrt{3})\|A\| \le 3c\|A\|$.*
- *(iv) The same holds for $A : \ell^{a,p}_{\mathbb{K}} \to \ell^{a,\bar{p}}_{\mathbb{K}}$ with the matrix of the representative $\hat{A}$ (Definition 3.1): the ratios $|F_{ij}|/|A_{ij}|$ are unchanged by the weights.*

Proof. **Step S3.64: Row and column sums.** $\sum_j |A_{ij}|^2 = \|A^* e_i\|^2 \le \|A\|^2$ and $\sum_i |A_{ij}|^2 = \|Ae_j\|^2 \le \|A\|^2$. By Cauchy–Schwarz and $\sum_{k\in\mathbb{Z}\setminus 0} k^{-2} = \pi^2/3$ ((W1-E2)), $\sum_j B_{ij} \le \Lambda\|A\|$ and $\sum_i B_{ij} \le \Lambda\|A\|$ with $\Lambda = \pi/\sqrt{3}$.

Step S3.65: Schur test. $\|Bv\|^2 \le \sum_i \big(\sum_j B_{ij}\big)\big(\sum_j B_{ij}|v_j|^2\big) \le \Lambda\|A\| \sum_j \big(\sum_i B_{ij}\big)|v_j|^2 \le \Lambda^2\|A\|^2\|v\|^2$, via finite partial sums ((W1-E1)).

Step S3.66: Diagonal and combination. $|A_{ii}| = |\langle e_i, Ae_i\rangle| \le \|A\|$ and Lemma 3.4. For (iii), split F into off-diagonal and diagonal parts, dominated entrywise by cB and $c\,\mathrm{diag}(|A_{ii}|)$; an entrywise dominated matrix has operator norm at most that of the nonnegative dominating matrix (apply it to $|v|$). $1 + \pi/\sqrt{3} < 2.82$.

Step S3.67: Weights. $\hat{A}_{ij} = \omega_{a,\bar{p}}(i) A_{ij} / \omega_{a,p}(j)$ and likewise for F, so the hypotheses of (iii) transfer. □

Source: P96-K Lemma A.1, pp. 145–146, and its use on p. 127 (the factor $3(A_k/\alpha)$).

3.9. **Ledger.**

PLAN nodes. `bpW:w1:space` → Lemma 3.2 (with Definition 3.1 and Lemma 3.3); `embed` → Lemma 3.5 (with Lemma 3.4); `diag` → Lemma 3.6; `twosided` → Lemma 3.10 (with Definition 3.9); `young` → Lemma 3.11; `algebra` → Lemma 3.12 (relabels Lemma 2.4); `trilinear` → Lemma 3.14 and Theorem 3.15; `cubic-analytic` → Corollary 3.16; `hamfield` → Proposition 3.18; `gradient-identity` → Proposition 3.17; `sine-completeness` → Lemma 3.19.
Added nodes and consumers. W2: 3.3, 3.20, 3.21, 3.18. W3: 3.10(iii), 3.14(b), 3.23, 3.25, 3.29. W4: 3.33, 3.34, 3.8, 3.30. W5: 3.25, 3.29, 3.32. W6, W8: 3.26, 3.27, 3.28, 3.3. The algebra extension to $\frac{1}{2} < s < 1$ (Remark 3.13) is not planned.
Source gaps and conventions. P96-K Lemma A.4 (segment and bounded-derivative hypotheses, Lemma 3.29); P96-W p. 280 constant (N-1, used corrected in Lemma 3.10); P96-K's $z = (u - iv)/\sqrt{2}$ versus W0 (Remark 3.31, a relabelling); P96-W §6 printed as "Appendix A" on p. 279 (label only).

4. Milestone W3: the analytic partial Birkhoff normal form

Part 0 computed the partial quartic normal form formally (Proposition 2.11): the generating Hamiltonian F divides each nonresonant quartic monomial with at least one tangential index by its divisor, and the time-one map of X_F removes these monomials up to order six. This part makes that computation analytic. It proves (a) a quantitative lower bound for the divisors (Lemma 4.12, P96-W Lemma 4); (b) that F is a bounded quartic polynomial whose Hamiltonian field gains one weight (Theorem 4.15); (c) that the time-one map Γ of X_F is an analytic, real, symplectic, near-identity change of coordinates between balls of the weighted phase space (Theorems 4.20, 4.22, Corollary 4.25); (d) that $H \circ \Gamma = \Lambda + \bar{G} + G^{\mathrm{nor}} + K$ with $K = O(\|z\|^6)$ and a one-smoothing holomorphic field X_K (Theorem 4.23), in the form stated in §4.8 for milestones W4–W6.
Route. The flow of X_F is not constructed by an ODE existence theorem. Since X_F is a bounded cubic polynomial, its time-t map is written as an explicit power series $\Gamma_t = \sum_k t^k \varphi_k$ whose terms are continuous multilinear maps (the Lie series of the identity). The coefficient bounds follow by induction, the flow equation follows from a Leibniz identity, and analyticity is immediate from the power series. Symplecticity is the classical argument (the derivative of the pulled-back form vanishes). The remainder K is split as $(G \circ \Gamma - G) + (\Lambda \circ \Gamma - \Lambda - Z')$; the first part is handled by the gradient transport formula, the second by the homological identity, which cancels the unbounded part of X_Λ before any estimate.
Standing data. Standing assumption 1.1 ($m > 0$) and the conventions of §1 are in force. Throughout this part $a \geq 0$ and $p \geq \frac{1}{2}$ are fixed; the headline uses $a = p = 1$ (Convention 1.5). The tangential modes are $J = \{1, 2\}$ and the normal modes are $j \geq 3$. All constants are listed with their dependence on m, a, p in §4.9. The source fixes a general finite $J = \{1, \dots, n\}$; we keep n general only in Lemma 4.12.
Inputs from W1. The weighted spaces are those of Lemma 3.2 with the embedding $\ell^{a,p+1} \subset \ell^{a,p}$ of Lemma 3.5; the multiplier bound is Lemma 3.6; "holomorphic" is Definition 3.24. Facts needed here in a two-sided, complexified form are stated and proved below (Lemmas 4.3, 4.4, 4.7, 4.9; Remark 4.8).

4.1. External facts.

Each item names its Mathlib counterpart (checkout of §7 of the scoping report). "Banach" means a complete normed space over $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$.

(E1) *Multilinear maps.* Continuous multilinear maps between Banach spaces form a Banach space under the operator norm; inserting a continuous multilinear map into one slot of another, or composing with continuous linear maps, gives a continuous multilinear map of norm at most the product of the norms. The diagonal $P(w) = M(w, \dots, w)$ of a continuous d-linear M is C^∞, with $DP(w)v = \sum_{i=1}^{d} M(w, \dots, w, \underset{i}{v}, w, \dots, w)$. (Mathlib: `ContinuousMultilinearMap`, `ContinuousMultilinearMap.opNorm_le_bound`, `ContinuousMultilinearMap.hasFDerivAt`, `ContinuousMultilinearMap.compContinuousLinearMap`.)

(E2) *Power series.* If $(M_n)_{n\geq 0}$ are continuous n-linear maps with $\sum_n \|M_n\| r^n < \infty$, then $f(w) = \sum_n M_n(w,\dots,w)$ has this formal power series on the open ball of radius r, is analytic at every point of that ball, and its derivative is obtained termwise. Finite sums, compositions and continuous multilinear images of analytic maps are analytic; so is $A \mapsto A^{-1}$ on the units of a Banach algebra; and the derivative of an analytic map into a Banach space is analytic. (Mathlib: `FormalMultilinearSeries`, `HasFPowerSeriesOnBall`, `HasFPowerSeriesOnBall.analyticAt_of_mem`, `HasFPowerSeriesOnBall.fderiv`, `AnalyticAt.comp`, `analyticAt_inverse`, `AnalyticOnNhd.fderiv`, `ContinuousMultilinearMap.analyticOnNhd` and `analyticOnNhd_uncurry_of_multilinear`.)

(E3) *Termwise differentiation.* If f_k are differentiable on an open convex set U of a normed space, $\sum_k f_k(x_0)$ converges and $\sup_U \|Df_k\| \leq u_k$ with $\sum u_k < \infty$, then $\sum_k f_k$ is differentiable on U with derivative $\sum_k Df_k$. Absolutely convergent double series may be rearranged (Cauchy product). (Mathlib: `hasFDerivAt_tsum`, `hasDerivAt_tsum`, `tsum_mul_tsum_eq_tsum_sum_antidiagonal_of_summable_norm`.)

(E4) *Mean value inequality; zero derivative.* If f is differentiable on a convex set with $\|Df\| \leq M$, then $\|f(y) - f(x)\| \leq M\|y - x\|$; a differentiable function on an interval with vanishing derivative is constant. (Mathlib: `Convex.norm_image_sub_le_of_norm_hasFDerivWithin_le`, `is_const_of_deriv_eq_zero`.)

(E5) *Integrals in a parameter.* For $Y : [0,1] \times U \to V$ (V Banach, U open in a Banach space) continuous, differentiable in the second variable with $D_w Y$ continuous and bounded on $[0,1] \times U$: $w \mapsto \int_0^1 Y(t,w)\,dt$ is differentiable with derivative $\int_0^1 D_w Y(t,w)\,dt$; the fundamental theorem of calculus holds for C^1 functions $[0,1] \to V$; continuous linear maps commute with the Bochner integral; $\|\int Y\| \leq \int \|Y\|$. Over $\mathbb{K} = \mathbb{C}$ the first statement yields complex differentiability. (Mathlib: `hasFDerivAt_integral_of_dominated_of_fderiv_le`, `intervalIntegral.integral_eq_sub_of_hasDerivAt`, `ContinuousLinearMap.intervalIntegral_comp_comm`, `intervalIntegral.norm_integral_le_of_norm_le`.)

(E6) *Neumann series.* If A is a bounded operator on a Banach space with $\|A - I\| \leq \frac{1}{2}$, then A is invertible, $\|A^{-1}\| \leq 2$ and $\|A^{-1} - I\| \leq 2\|A - I\|$. (Mathlib: `NormedRing.inverse_one_sub`, `Units.oneSub`, `NormedRing.tsum_geometric_of_norm_lt_one`.)

Remark 4.1 (Flows of analytic vector fields: what Mathlib provides). Mathlib provides local existence (Picard–Lindelöf, `IsPicardLindelof`, `exists_forall_mem_closedBall_eq_hasDerivWithinAt_lipschitzOnWith`) and uniqueness (`ODE_solution_unique`) of solutions of Lipschitz ODEs in Banach spaces, and Gronwall's inequality. It does *not* provide analytic (or C^1) dependence of the flow on initial data, symplecticity of Hamiltonian flows, or the theorem that complex-differentiable maps on open subsets of a complex Banach space are analytic (only $f : \mathbb{C} \to E$ is covered: `DifferentiableOn.analyticOnNhd`). W1 constructs flows of general bounded holomorphic fields by Picard iteration and uniform limits (Lemma 3.29). This part does not use that lemma: the time-t map of the polynomial field X_F is constructed as an explicit power series (Definition 4.18), and symplecticity and invertibility are proved for it directly (Theorems 4.20, 4.22). This gives analyticity (not only holomorphy), the gain of one weight in $\Gamma_t - \mathrm{id}$, and explicit constants. Uniqueness of ODE solutions is not used.

4.2. **Coordinates, symplectic form, Hamiltonian fields.**

Definition 4.2 (Two-sided coordinate space)**.** For $a \geq 0$ and $r \in \mathbb{R}$ let $\mathfrak{W}_{a,r}$ be the space of complex families $w = (w_l)_{l \in \mathbb{Z}\setminus 0}$ with

$$\|w\|_{a,r}^2 = \sum_{l \neq 0} |l|^{2r} e^{2a|l|} |w_l|^2 < \infty.$$

Write $\mathfrak{W} = \mathfrak{W}_{a,p}$ and $\mathfrak{W}^+ = \mathfrak{W}_{a,p+1}$, with norms $\|\cdot\|$ and $\|\cdot\|_+$. For $w \in \mathfrak{W}$ write $\tilde{w} = (w_l)_{|l|\le 2}$ (tangential part) and $\hat{w} = (w_l)_{|l|\ge 3}$ (normal part), extended by zero. Put $\lambda'_l = \operatorname{sgn}(l)\lambda_{|l|}$. In Lean the carrier is `lp (fun _ : Bool × PNat => Complex) 2` in weighted coordinates (the pair $(\operatorname{sgn}, |l|)$ indexes l), so that the norm is exactly the one displayed.

Source: ours; P96-W p. 282 (the coordinates $w_{\pm j}$ and the space $\ell^{a,s}_b$ restricted to $l \neq 0$).

Lemma 4.3 (Complex coordinates and real structure)**.** *Let $E^{\mathbb{C}} = \ell^{a,p}_{\mathbb{C}} \times \ell^{a,p}_{\mathbb{C}}$ with $\|(q,p)\|_E = \max(\|q\|_{a,p}, \|p\|_{a,p})$, and define $\mathcal{Z} : E^{\mathbb{C}} \to \mathfrak{W}$ by*

$$(\mathcal{Z}x)_j = \frac{q_j + ip_j}{\sqrt{2}}, \qquad (\mathcal{Z}x)_{-j} = \frac{q_j - ip_j}{\sqrt{2}} \qquad (j \ge 1).$$

Then $\mathcal{Z}$ is a $\mathbb{C}$-linear bijection with $\|x\|_E \le \|\mathcal{Z}x\| \le \sqrt{2}\|x\|_E$; it maps $E^{\mathbb{C}}_{a,p+1}$ onto $\mathfrak{W}^+$ with the same bounds. The map $\varrho(w)_l = \overline{w_{-l}}$ is a conjugate-linear isometric involution of $\mathfrak{W}$ and of $\mathfrak{W}^+$, and $\mathcal{Z}$ maps the real space $E_{a,p}$ bijectively onto its fixed set $\mathfrak{W}_{\mathbb{R}} = \{\varrho w = w\}$. On real points, $w_j = z_j$ and $w_{-j} = \bar{z}_j$ (Definition 1.11) and $w_j w_{-j} = |z_j|^2 = I_j$.

Proof. For complex q_j, p_j the parallelogram identity gives $|q_j + ip_j|^2 + |q_j - ip_j|^2 = 2(|q_j|^2 + |p_j|^2)$, so $\|\mathcal{Z}x\|^2 = \|q\|^2 + \|p\|^2 \in [\|x\|_E^2, 2\|x\|_E^2]$. The inverse is $q_j = (w_j + w_{-j})/\sqrt{2}$, $p_j = (w_j - w_{-j})/(i\sqrt{2})$. If q, p are real then $w_{-j} = \overline{w_j}$, i.e. $\varrho w = w$; conversely $\varrho w = w$ makes both formulas real. The weights depend on $|l|$ only, so ϱ is isometric. □

Source: ours; P96-W p. 281 (definition of z) and p. 282 ($z_j = w_j$, $\bar{z}_j = w_{-j}$).

Lemma 4.4 (Symplectic form)**.** *$\Omega(w, w') = i\sum_{l\neq 0} \operatorname{sgn}(l)\, w_l w'_{-l}$ is a continuous antisymmetric bilinear form on $\mathfrak{W}$ with $|\Omega(w,w')| \le \|w\|\|w'\|$. It is nondegenerate: $\Omega(\zeta, e_{-l}) = i\operatorname{sgn}(l)\zeta_l$ for the unit vectors e_l, so $\Omega(\zeta, \cdot) = 0$ implies $\zeta = 0$. For real $x, x' \in E$, $\Omega(\mathcal{Z}x, \mathcal{Z}x') = \sum_j (q_j p'_j - p_j q'_j)$, the form $\sum_j dq_j \wedge dp_j$ of Definition 1.10. Moreover $\Omega(\varrho w, \varrho w') = \overline{\Omega(w,w')}$.*

Proof. Cauchy–Schwarz and $|l|^p e^{a|l|} \ge 1$ give the bound. Substituting the inverse formulas of Lemma 4.3, $q_j p'_j - p_j q'_j = i(w_j w'_{-j} - w_{-j} w'_j)$. The last identity follows from $\overline{\operatorname{sgn}(l)} = \operatorname{sgn}(l) = -\operatorname{sgn}(-l)$ after the substitution $l \mapsto -l$. □

Source: ours; P96-W p. 281 (symplectic structure $i\sum dz_j \wedge d\bar{z}_j$).

Definition 4.5 (Hamiltonian field; one-smoothing holomorphic class)**.** Let $U \subset \mathfrak{W}$ be open and $P : U \to \mathbb{C}$ complex differentiable. A map $X_P : U \to \mathfrak{W}$ is the *Hamiltonian field* of P if $DP(w)v = \Omega(X_P(w), v)$ for all $w \in U$, $v \in \mathfrak{W}$; by Lemma 4.4 it is unique when it exists, and $X_P(w)_l = -i\operatorname{sgn}(l)\, DP(w)e_{-l}$. P is *real* if $P(\varrho w) = \overline{P(w)}$ for $w \in U = \varrho U$; then $X_P \circ \varrho = \varrho \circ X_P$. The class $\mathcal{A}(U)$ consists of the real P that are complex differentiable on U and whose Hamiltonian field exists, takes values in $\mathfrak{W}^+$, and is complex differentiable as a map $U \to \mathfrak{W}^+$ (Mathlib: `DifferentiableOn Complex`). "Holomorphic" means complex differentiable in this sense, as in Definition 3.24.

The equations $\dot{w} = X_P(w)$ are Hamilton's equations: on real points, by Lemma 4.4, $\mathcal{Z}^{-1}X_P = (\partial_p P, -\partial_q P)$, the convention of Definition 1.10. The reality statement holds because $DP(\varrho w)\varrho v = \overline{DP(w)v}$ and Lemma 4.4 give $\Omega(X_P(\varrho w), \varrho v) = \Omega(\varrho X_P(w), \varrho v)$ for all v.
Source: ours; P96-W p. 281 (the class $A(\ell^{a,s}, \ell^{a,s+1})$ of real analytic maps from a neighbourhood of the origin; here with an explicit domain and complex differentiability, see Remark 4.24).

Definition 4.6 (Smoothing symmetric forms)**.** A continuous symmetric d-linear form $\mathfrak{P} : \mathfrak{W}^d \to \mathbb{C}$ ($d \ge 2$) is *smoothing with constant c* if there is a continuous $(d-1)$-linear map $\nabla\mathfrak{P} : \mathfrak{W}^{d-1} \to \mathfrak{W}^+$ with $\mathfrak{P}(w^1, \dots, w^{d-1}, v) = \Omega(\nabla\mathfrak{P}(w^1, \dots, w^{d-1}), v)$ and $\|\nabla\mathfrak{P}(w^1, \dots, w^{d-1})\|_+ \le c\prod\|w^i\|$. Its

diagonal $P(w)=\mathfrak{P}(w,\dots,w)$ then has $X_P(w)=d\,\nabla\mathfrak{P}(w,\dots,w)$ by (E1), and on every ball:

$$|P(w)|\le c\|w\|^d,\qquad \|X_P(w)\|_+\le dc\|w\|^{d-1},$$
$$\|X_P(y)-X_P(w)\|_+\le d(d-1)c\max(\|y\|,\|w\|)^{d-2}\|y-w\|.$$

Source: ours.

Lemma 4.7 (The linear part)**.** *$\Lambda(w)=\sum_{j\ge1}\lambda_j w_j w_{-j}$ is a continuous quadratic form on $\mathfrak{W}$ with $|\Lambda(w)|\le\sqrt{1+m}\,\|w\|^2$, and $\Lambda(\mathcal{Z}x)=\frac12\sum\lambda_j(q_j^2+p_j^2)$ for real x (Definition 1.8). Its derivative is $D\Lambda(w)v=\sum_{l\ne0}\lambda_{|l|}w_{-l}v_l$ for all $w,v\in\mathfrak{W}$. For $w\in\mathfrak{W}^+$ its Hamiltonian field is $X_\Lambda(w)_l=-i\lambda'_l w_l$, a bounded linear map $\mathfrak{W}^+\to\mathfrak{W}$ of norm $\le\sqrt{1+m}$; X_Λ does not map $\mathfrak{W}$ into itself.*

Proof. $\lambda_j\le\sqrt{1+m}\,j\le\sqrt{1+m}\,j^{2p}e^{2aj}$ since $p\ge\frac12$, and Cauchy–Schwarz gives the bound. The field is computed from Definition 4.5: $\Omega(X_\Lambda(w),v)=\sum_l\lambda_{|l|}w_lv_{-l}$. The operator bound is Lemma 3.6(i) with $r=1$. □

Source: P96-W p. 280 (X_Λ unbounded of order one) and p. 281 ($\Lambda=\sum\lambda_j|z_j|^2$).

Remark 4.8 (Requests to W1). Lemmas 4.3, 4.4, 4.7 and 4.9 are not in the W1 plan. They are proved here; if W1 adopts them, the labels here become citations. Lemma 4.3 is the pairing of Lemma 3.10(iii) composed with $(q,p)\mapsto(z,\bar z)$. Proposition 3.18 is stated for real $E_{a,p}$; this part uses its complexified, two-sided form (Lemma 4.10).

4.3. **The quartic Hamiltonian in two-sided coordinates.** For $x=(x_1,x_2,x_3,x_4)\in(\mathbb{Z}\setminus0)^4$ write $|x|=(|x_1|,\dots,|x_4|)$, $w_x=w_{x_1}w_{x_2}w_{x_3}w_{x_4}$, and

$$\mathfrak{g}_x=\frac{g_{|x|}}{\sqrt{\lambda_{|x_1|}\lambda_{|x_2|}\lambda_{|x_3|}\lambda_{|x_4|}}},\qquad \delta_x=\lambda'_{x_1}+\lambda'_{x_2}+\lambda'_{x_3}+\lambda'_{x_4},$$

with g from Lemma 2.2. Call x *paired* if the multiset $\{x_t:x_t>0\}$ equals $\{-x_t:x_t<0\}$, i.e. x is a permutation of $(r,-r,s,-s)$; paired x have $\delta_x=0$. Let $\mathcal{T}=\{x:\min_t|x_t|\le2\}$ (at least one tangential index).

Lemma 4.9 (Majorant for the cubic sums)**.** *For nonnegative $u,u',u''\in\ell^{a,p}$ put $\mathcal{M}(u,u',u'')_l=\lambda_l^{-1/2}\sum_{i,j,k\ge1}|g_{ijkl}|\frac{u_iu'_ju''_k}{\sqrt{\lambda_i\lambda_j\lambda_k}}$. The sums converge and*

$$\|\mathcal{M}(u,u',u'')\|_{a,p+1}\le C_*\|u\|_{a,p}\|u'\|_{a,p}\|u''\|_{a,p},\qquad C_*=\frac{K_{p+1/2}^2}{\pi},$$

with K_s from Lemma 2.4. C_* does not depend on m; for $a=p=1$, $C_*=8(1+2\zeta(3))/\pi$.

Proof. Steps S2.6–S2.8 of the proof of Theorem 2.5 bound exactly the absolute series $\sum|g_{ijkl}||\hat q_i\hat q'_j\hat q''_k|$; apply them with $q=u$, $q'=u'$, $q''=u''$. □

Source: P96-W (13)–(14), p. 284 (constant "c"); the constant is that of Theorem 2.5.

Lemma 4.10 (Quartic forms)**.** *For $\mathcal{S}\subset(\mathbb{Z}\setminus0)^4$ invariant under permutations of the four entries and under $x\mapsto-x$, and coefficients (c_x) with $c_x=c_{\sigma x}=\overline{c_{-x}}$ and $|c_x|\le|\mathfrak{g}_x|$, the form*

$$\mathfrak{P}(w^1,w^2,w^3,w^4)=\frac{1}{16}\sum_{x\in\mathcal{S}}c_x\,w^1_{x_1}w^2_{x_2}w^3_{x_3}w^4_{x_4}$$

converges absolutely, is continuous, symmetric, and smoothing with constant $C_*/4$, with $\nabla\mathfrak{P}(w^1,w^2,w^3)_l=-\frac{i\,\mathrm{sgn}(l)}{16}\sum_{y:(y,-l)\in\mathcal{S}}c_{(y,-l)}w^1_{y_1}w^2_{y_2}w^3_{y_3}$; its diagonal P is real. In particular, with $c=\mathfrak{g}$ restricted to the sets $(\mathbb{Z}\setminus0)^4$, $\mathcal{T}$, $\mathcal{T}\cap\{\text{paired}\}$, $\mathcal{T}^c$, we obtain the forms $\mathfrak{G}$, $\mathfrak{G}^{\tan}$, $\bar{\mathfrak{G}}$, $\mathfrak{G}^{\mathrm{nor}}$ and their diagonals $G,G^{\tan},\bar G,G^{\mathrm{nor}}$; $G=G^{\tan}+G^{\mathrm{nor}}$; and

$$\|X_P(w)\|_+\le C_*\|w\|^3,\qquad |P(w)|\le\tfrac{C_*}{4}\|w\|^4\qquad(P\in\{G,G^{\tan},\bar G,G^{\mathrm{nor}}\}).$$

Proof. **Step S4.1: Majorize.** Given w^1, w^2, w^3 let $u_i^t = |w_i^t| + |w_{-i}^t|$ $(i \geq 1)$, so $\|u^t\|_{a,p} \leq \sqrt{2}\|w^t\|$. Summing over the signs of $y \in (\mathbb{Z} \setminus 0)^3$, $\sum_y |\mathfrak{g}_{(y,-l)}||w^1_{y_1} w^2_{y_2} w^3_{y_3}| \leq \mathcal{M}(u^1, u^2, u^3)_{|l|}$.

Step S4.2: Bound the gradient. Hence, by Lemma 4.9, $\|\nabla\mathfrak{P}\|_+^2 \leq 2 \cdot 16^{-2} \|\mathcal{M}(u^1, u^2, u^3)\|_{a,p+1}^2$, so $\|\nabla\mathfrak{P}\|_+ \leq \frac{\sqrt{2}}{16} C_* (\sqrt{2})^3 \prod \|w^t\| = \frac{C_*}{4} \prod \|w^t\|$.

Step S4.3: Identify. By Lemma 4.4, $\Omega(\nabla\mathfrak{P}(w^1, w^2, w^3), v) = i \sum_l \operatorname{sgn}(l) \nabla\mathfrak{P}(\cdot)_l v_{-l} = \frac{1}{16} \sum_l \sum_y c_{(y,-l)} w^1_{y_1} w^2_{y_2} w^3_{y_3} v_{-l}$, which is $\mathfrak{P}(w^1, w^2, w^3, v)$ after $l \mapsto -l$; the four-fold series converges absolutely by Step S4.1 and Cauchy–Schwarz against v. Symmetry follows from the permutation invariance of $\mathcal{S}$ and c. Reality: $P(\varrho w) = \frac{1}{16} \sum c_x \overline{w_{-x}} = \frac{1}{16} \sum_x c_{-x} \overline{w_x} = \overline{P(w)}$. The bounds are Definition 4.6 with $d = 4$, $c = C_*/4$. $\mathfrak{g}$ is invariant under permutations and $x \mapsto -x$ and real; the four sets are invariant; $(\mathbb{Z} \setminus 0)^4 = \mathcal{T} \sqcup \mathcal{T}^c$. □

Source: P96-W p. 282 (expansion of H in the w-coordinates) and (13)–(14), p. 284. The prefactor $\frac{1}{16}$ is the corrected one (E-PDE-W-1, Remark 2.12).

Lemma 4.11 (Identification with Part 0)**.** *For real $x = (q, p) \in E_{a,p}$: $G(\mathcal{Z}x) = G(q)$ of Definition 1.8 when $a > 0$ (for $a = 0$ this is the definition of G on $E_{0,p}$); $\mathcal{Z}^{-1} X_G(\mathcal{Z}x) = (0, -T(q,q,q))$ with T from Theorem 2.5, which is $(0, -G_q(q))$ for $a > 0$; $G^{\mathrm{nor}}(w) = G^{\mathrm{nor}}(\hat{w})$ and $X_{G^{\mathrm{nor}}}(w)$ has vanishing tangential components, so $|G^{\mathrm{nor}}(w)| \leq \frac{C_*}{4}\|\hat{w}\|^4$ and $\|X_{G^{\mathrm{nor}}}(w)\|_+ \leq C_* \|\hat{w}\|^3$; and for all $w \in \mathfrak{W}$, with $I_j := w_j w_{-j}$,*

$$\bar{G}(w) = \tfrac{1}{2} \textstyle\sum_{i,j \in J} A_{ij} I_i I_j + \sum_{j \geq 3,\, i \in J} B_{ji} I_i I_j,$$

with the corrected A, B of Theorem 2.10. Finally $x \in \mathcal{T}$ with $\mathfrak{g}_x \neq 0$ has $\delta_x = 0$ if and only if x is paired.

Proof. **Step S4.4: The quartic.** Expanding $\prod_t (z_{i_t} + \bar{z}_{i_t})$ in (5) into the 16 sign choices gives $G(\mathcal{Z}x) = \frac{1}{16} \sum_x \mathfrak{g}_x w_x$; absolute convergence is Lemma 4.10, and $G(q)$ agrees with (5) by Lemma 2.3.

Step S4.5: The field. On real points $\mathcal{Z}^{-1} X_G = (\partial_p G, -\partial_q G)$ (paragraph after Definition 4.5), $\partial_p G = 0$, and $\partial_{q_l} G = T(q,q,q)_l$ by the formula of Lemma 2.3 (equivalently Proposition 3.17).

Step S4.6: Normal part. Every $x \in \mathcal{T}^c$ has all $|x_t| \geq 3$.

Step S4.7: Count the paired terms. For integers $1 \leq r < s$ with $r \in J$: the ordered x that are permutations of $(r, -r, s, -s)$ are 24 in number, with $\mathfrak{g}_x = g_{rrss}/(\lambda_r \lambda_s) = 1/(\pi \lambda_r \lambda_s)$ and $w_x = I_r I_s$, contributing $\frac{24}{16\pi\lambda_r\lambda_s} I_r I_s = \frac{3}{2\pi\lambda_r\lambda_s} I_r I_s$. For $r = s \in J$ there are 6 orderings, $\mathfrak{g}_x = \frac{3}{2\pi\lambda_r^2}$, contributing $\frac{9}{16\pi\lambda_r^2} I_r^2 = \frac{1}{2} A_{rr} I_r^2$. These are the coefficients of Theorem 2.10.

Step S4.8: Resonance. $\mathfrak{g}_x \neq 0$ and $\delta_x = 0$ force pairing by Lemma 2.7 (applied to the indices $|x|$ with signs $\operatorname{sgn} x_t$). □

Source: P96-W pp. 282–284; Theorem 2.10 for the coefficients.

4.4. Quantitative small divisors.

Lemma 4.12 (Divisor lower bound; P96-W Lemma 4)**.** *Let $x \in (\mathbb{Z} \setminus 0)^4$ satisfy the selection rule $\varepsilon_1|x_1| + \cdots + \varepsilon_4|x_4| = 0$ for some $\varepsilon \in \{\pm 1\}^4$, and let x be not paired. With $n = \min_t |x_t|$,*

$$|\delta_x| \geq \frac{m}{27\,(n^2 + m)^{3/2}}.$$

Consequently, if $x \in \mathcal{T}$, $\mathfrak{g}_x \neq 0$ and x is not paired, then $|\delta_x| \geq \delta_*(m) := \dfrac{m}{27(4+m)^{3/2}}$.

Proof. Let $a \le b \le c \le d$ be $|x_1|, \dots, |x_4|$ sorted, so $n = a$, and write $f(s) = \sqrt{s^2 + m}$, $\lambda_k = f(k)$. Recall $f'(s) = s/f(s)$ is increasing and $f''(s) = m/f(s)^3$ is positive and decreasing on $s \ge 0$. Put $\beta = m/(27\lambda_a^3)$ and note $\beta \le \frac{1}{27\lambda_a}$ because $m \le \lambda_a^2$.

Step S4.9: Selection rule. We show $d = a+b+c$ or $d = b+c-a$. Solving the relation for d gives $d = \pm a \pm b \pm c$. Since $1 \le a \le b \le c \le d$: $d = a - b + c \le c$ forces $a = b$, $d = c$, hence $d = b + c - a$; $d = a + b - c \le b$ forces $a = b = c = d$, hence $d = b + c - a$; every choice with at least two minus signs gives $d \le c - a < c$ or $d \le 0$, impossible.

Step S4.10: Signs. $|\delta_x|$ is unchanged by $x \mapsto -x$, so assume at most two of the x_t are negative. Pairing and δ_x depend only on the signed multiset, so it suffices to bound, for each sign pattern on (a, b, c, d), the corresponding divisor.

Step S4.11: No minus sign. $\delta = \lambda_a + \lambda_b + \lambda_c + \lambda_d \ge 4 > \beta$.

Step S4.12: One minus sign. Each such divisor is $\ge \lambda_a + \lambda_b + \lambda_c - \lambda_d$. Since $f(k) - k = m/(f(k) + k)$,

$$\lambda_a + \lambda_b + \lambda_c - \lambda_d = (a + b + c - d) + m\Big(\tfrac{1}{\lambda_a + a} + \tfrac{1}{\lambda_b + b} + \tfrac{1}{\lambda_c + c} - \tfrac{1}{\lambda_d + d}\Big) \ge \frac{m}{2\lambda_a} \ge \beta,$$

using $a + b + c \ge d$ (Step S4.9), $\lambda_c + c \le \lambda_d + d$ and $\lambda_a \ge 1$.

Step S4.13: Two minus signs, outer pairs. $\delta_3 = \lambda_c + \lambda_d - \lambda_a - \lambda_b$ vanishes identically iff $a = b = c = d$ (then x is paired); otherwise $c > a$ or $d > b$, and since $\lambda_{k+1} - \lambda_k \ge f'(k) \ge f'(a) = a/\lambda_a$ for $k \ge a$, $\delta_3 \ge a/\lambda_a \ge \beta$. Likewise $\delta_2 = (\lambda_b - \lambda_a) + (\lambda_d - \lambda_c)$ is paired iff $a = b$ and $c = d$, and otherwise $\delta_2 \ge a/\lambda_a \ge \beta$.

Step S4.14: Two minus signs, inner pair. $\delta_1 = \lambda_a + \lambda_d - \lambda_b - \lambda_c$ is paired iff $a = b$ and $c = d$. Otherwise:

(i) if $d = b + c - a$, put $t = b - a = d - c \ge 1$ (if $t = 0$ the tuple is paired). As f' increases and $c \ge b$, $\lambda_d - \lambda_c = \int_c^{c+t} f' \ge \int_b^{b+t} f'$, so $\delta_1 \ge \int_a^{a+t} \big(f'(s+t) - f'(s)\big)\,ds \ge \int_a^{a+1} \int_s^{s+1} f''(r)\,dr\,ds \ge f''(a+2)$, and $f''(a+2) = m/((a+2)^2 + m)^{3/2} \ge m/(27\lambda_a^3)$ because $(a+2)^2 \le 9a^2$;

(ii) if $d = a+b+c$, then $\lambda_d - \lambda_c \ge \lambda_{d-(c-a)} - \lambda_a = \lambda_{2a+b} - \lambda_a$ (shift both endpoints down by $c-a$), so $\delta_1 \ge \lambda_{b+2a} - \lambda_b \ge 2af'(b) \ge 2/\sqrt{1+m} \ge \beta$, the last since $m/\lambda_a^3 \le m/(\sqrt{1+m}\,\lambda_a^2) \le 1/\sqrt{1+m}$.

Step S4.15: Conclusion. Patterns with two minus signs are exactly $\pm\delta_1, \pm\delta_2, \pm\delta_3$, and pairing of x coincides with the identical vanishing in each case. The corollary: $\mathfrak{g}_x \ne 0$ implies the selection rule (Lemma 2.2), $n \le 2$ on $\mathcal{T}$, and $n \mapsto m/(n^2 + m)^{3/2}$ is decreasing. □

Source: P96-W Lemma 4, p. 283; proof pp. 285–286 ("some absolute constant c"; here $c = \frac{1}{27}$).
SOURCE_GAP: Two steps of the printed proof are incorrect, although the lemma is true. (1) P96-W p. 285, Case 1: "$\delta'(m) = \frac{1}{2}(\frac{1}{\lambda_i} + \frac{1}{\lambda_j} + \frac{1}{\lambda_k} - \frac{1}{\lambda_l}) \ge \frac{1}{\lambda_i}$" is false: for $(i, j, k, l) = (1, 10, 10, 19)$ (selection rule $1-10-10+19 = 0$) and $m = 1$ the left side is ≈ 0.427 while $1/\lambda_1 \approx 0.707$. The correct bound is $\ge \frac{1}{2\lambda_i}$, which halves the constant (Step S4.12 avoids calculus). (2) P96-W p. 286, Case 2: "it suffices to study $\delta = \delta_{+--+}$" because $\delta_{-+-+}, \delta_{--++} \ge \delta_{+--+}$. When $i = j < k = l$ the divisor δ_{+--+} is resonant (identically 0) while $\delta_{--++} = 2(\lambda_k - \lambda_i)$ is not, so the reduction gives only $\delta_{--++} \ge 0$; the missing bound is Step S4.13. Also on p. 286, "$\lambda_{j+2i} - \lambda_j \ge 2(\lambda_{j+1} - \lambda_j) \ge 2if'(j)$" should read $\ge 2i(\lambda_{j+1} - \lambda_j)$; the final bound is unaffected.

Remark 4.13. The bound is attained in order: $\lambda_{n+1} - 2\lambda_n + \lambda_{n-1} \sim m/n^3$ (P96-W p. 286). For the headline, $\delta_*(m) \sim m/216$ as $m \to 0$ and $\delta_*(m) \sim 1/(27\sqrt{m})$ as $m \to \infty$; no uniformity in m is claimed (Standing assumption 1.1).

4.5. **The generating Hamiltonian.**

Definition 4.14 (Generating Hamiltonian)**.** For $x \in \mathcal{T}$ not paired put $f_x = \mathfrak{g}_x/\delta_x$, and $f_x = 0$ otherwise. (In Lean, division by 0 returns 0; Lemma 4.12 guarantees $\delta_x \neq 0$ wherever $\mathfrak{g}_x \neq 0$ and f_x is used.) Let

$$\mathfrak{F}(w^1,\dots,w^4) = \frac{i}{16}\sum_x f_x\, w^1_{x_1} w^2_{x_2} w^3_{x_3} w^4_{x_4}, \qquad F(w) = \mathfrak{F}(w,w,w,w),$$

and $S = 4\nabla\mathfrak{F} : \mathfrak{W}^3 \to \mathfrak{W}^+$, so that $X_F(w) = S(w,w,w)$.

Source: P96-W (12), p. 282, with the corrected prefactor $\frac{1}{16}$ and the time convention of Definition 1.11 (the printed $iF_{ijkl} = G_{ijkl}/\delta$ becomes $F = \frac{i}{16}\sum \mathfrak{g}/\delta\, w^4$); the same F as in Proposition 2.11.

Theorem 4.15 (F is bounded and one-smoothing)**.** *$\mathfrak{F}$ is a continuous symmetric 4-linear form, smoothing with constant $C_*/(4\delta_*)$, and F is real. Hence S is a continuous symmetric trilinear map $\mathfrak{W}^3 \to \mathfrak{W}^+$ with*

$$\|S(w^1,w^2,w^3)\|_+ \le C_F\|w^1\|\|w^2\|\|w^3\|, \qquad C_F = C_F(m) = \frac{C_*}{\delta_*(m)} = \frac{27\,C_*\,(4+m)^{3/2}}{m},$$

$\Omega(S(w^1,w^2,w^3),v) = 4\mathfrak{F}(w^1,w^2,w^3,v)$ is symmetric in all four arguments, $S(\varrho w^1,\varrho w^2,\varrho w^3) = \varrho S(w^1,w^2,w^3)$, and $|F(w)| \le \frac{C_F}{4}\|w\|^4$. X_F is a cubic polynomial, analytic on $\mathfrak{W}$ with values in $\mathfrak{W}^+$.

Proof. **Step S4.16: Coefficients.** f is invariant under permutations; $f_{-x} = \mathfrak{g}_x/(-\delta_x) = -f_x$; and $|f_x| \le |\mathfrak{g}_x|/\delta_*$ by Lemma 4.12. So $c_x := i\delta_* f_x$ satisfies the hypotheses of Lemma 4.10 ($c_{-x} = i\delta_* f_{-x} = -i\delta_* f_x$, so $\overline{c_{-x}} = i\delta_* f_x = c_x$), and $\mathfrak{F} = \delta_*^{-1}\mathfrak{P}_c$.

Step S4.17: Conclusions. Lemma 4.10 gives smoothing with constant $C_*/(4\delta_*)$ and reality; $S = 4\nabla\mathfrak{F}$ gives C_F. The reality of S follows from the reality of $\mathfrak{F}$ as in Definition 4.5, applied to the multilinear form. Analyticity: (E2). □

Source: P96-W (13)–(14), p. 284.

Proposition 4.16 (Homological identity)**.** *Let $Z' = \bar G - G^{\mathrm{tan}}$, the diagonal of $\mathfrak{z} = \bar{\mathfrak{G}} - \mathfrak{G}^{\mathrm{tan}}$ (a form as in Lemma 4.10, smoothing with constant $C_*/4$, so $\|X_{Z'}(w)\|_+ \le C_*\|w\|^3$). For every $w \in \mathfrak{W}$,*

$$D\Lambda(w)\,X_F(w) = Z'(w), \qquad \textit{equivalently} \qquad G + D\Lambda\cdot X_F = \bar G + G^{\mathrm{nor}}.$$

Proof. **Step S4.18: Field coordinates.** By Definition 4.14 and Lemma 4.10, $X_F(w)_l = 4\nabla\mathfrak{F}(w,w,w)_l = \frac{\mathrm{sgn}(l)}{4}\sum_y f_{(y,-l)}w_y$.

Step S4.19: Euler identity. By Lemma 4.7, $D\Lambda(w)X_F(w) = \sum_l \lambda_{|l|}w_{-l}X_F(w)_l$. Substituting $l = -r$ and $\lambda_{|r|}\,\mathrm{sgn}(-r) = -\lambda'_r$, this equals $-\frac14\sum_r \lambda'_r w_r \sum_y f_{(y,r)}w_y = -\frac{1}{16}\sum_x \delta_x f_x w_x$, by symmetrizing over the position of r in x (f symmetric). All series converge absolutely: $\lambda_{|l|} \le \sqrt{1+m}|l|^{p+1}e^{a|l|}$ and $\|X_F(w)\|_+ < \infty$ with Cauchy–Schwarz.

Step S4.20: Divide. $\delta_x f_x = \mathfrak{g}_x$ for $x \in \mathcal{T}$ not paired (both vanish if $\mathfrak{g}_x = 0$; Lemma 4.12 otherwise) and $\delta_x f_x = 0$ else. So the sum is $-\frac{1}{16}\sum_{\mathcal{T}\setminus\mathrm{paired}} \mathfrak{g}_x w_x = \bar G - G^{\mathrm{tan}}$ by Lemma 4.10. The second form uses $G = G^{\mathrm{tan}} + G^{\mathrm{nor}}$. □

Source: P96-W p. 283 ("$G + \{\Lambda, F\} = \bar G + \hat G$"); the identity of Proposition 2.11 with $\{\Lambda,F\} = D\Lambda\cdot X_F$, now on all of $\mathfrak{W}$.

Remark 4.17. The identity holds on $\mathfrak{W}$ although X_Λ is defined only on $\mathfrak{W}^+$: it is stated with $D\Lambda(w)$, which is a bounded functional on $\mathfrak{W}$ (Lemma 4.7). This is what lets the remainder estimate below avoid the unbounded field.

4.6. **The Lie series of the flow.**

Definition 4.18 (Lie coefficients)**.** Define continuous $(2k+1)$-linear maps $\Phi_k : \mathfrak{W}^{2k+1} \to \mathfrak{W}$ by $\Phi_0(w) = w$ and

$$\Phi_{k+1}(w^1, \dots, w^{2k+3}) = \frac{1}{k+1} \sum_{i=1}^{2k+1} \Phi_k\big(w^1, \dots, w^{i-1}, S(w^i, w^{i+1}, w^{i+2}), w^{i+3}, \dots, w^{2k+3}\big),$$

and let $\varphi_k(w) = \Phi_k(w, \dots, w)$. For a C^1 map Q from an open subset of $\mathfrak{W}$ to a Banach space, let $(LQ)(w) = DQ(w)\, X_F(w)$.

Source: ours (the Lie series $\Gamma_t = \sum_k t^k L^k(\mathrm{id})/k!$; P96-W p. 283 uses the time-one map of X_F without constructing it).

Lemma 4.19 (Coefficient bounds and the Leibniz identity)**.** *For all $k \geq 0$:*

(i) $\varphi_{k+1} = \frac{1}{k+1} L\varphi_k$*, hence* $\varphi_k = L^k(\mathrm{id})/k!$*;*

(ii) $\|\Phi_k\| \leq (2C_F)^k$*; for* $k \geq 1$*,* Φ_k *takes values in* $\mathfrak{W}^+$ *and* $\|\Phi_k\|_{\mathfrak{W}^{2k+1} \to \mathfrak{W}^+} \leq (2C_F)^k$*;*

(iii) $\Phi_k(\varrho w^1, \dots) = \varrho \Phi_k(w^1, \dots)$*;*

(iv) $(k+1)\,\varphi_{k+1} = \sum_{a+b+c=k} S(\varphi_a, \varphi_b, \varphi_c)$ *and*

$$(k+1)\, D\varphi_{k+1}(w)v = \sum_{a+b+c=k} 3\, S\big(D\varphi_a(w)v, \varphi_b(w), \varphi_c(w)\big).$$

Proof. **Step S4.21: Recursion.** (i) is (E1): $D\varphi_k(w)v = \sum_i \Phi_k(w, \dots, \underset{i}{v}, \dots, w)$ with $v = S(w, w, w)$.

Step S4.22: Norms. By (E1), $\|\Phi_{k+1}\| \leq \frac{2k+1}{k+1} C_F \|\Phi_k\| \leq 2C_F \|\Phi_k\|$ (as $\|\cdot\| \leq \|\cdot\|_+$). For values in $\mathfrak{W}^+$: $\Phi_1 = S$, and for $k \geq 1$ each summand is Φ_k (into $\mathfrak{W}^+$) applied to arguments in $\mathfrak{W}$, so the same recursion holds for the $\mathfrak{W}^+$-valued norms.

Step S4.23: Reality. Induction from Theorem 4.15.

Step S4.24: Leibniz rule. For a continuous multilinear B and C^∞ maps $Q_1, \dots, Q_r$, the product and chain rules give $L\, B(Q_1, \dots, Q_r) = \sum_i B(Q_1, \dots, LQ_i, \dots, Q_r)$, and by induction on n, $L^n B(Q_1, \dots, Q_r) = \sum_{n_1 + \dots + n_r = n} \frac{n!}{n_1! \cdots n_r!} B(L^{n_1} Q_1, \dots, L^{n_r} Q_r)$.

Step S4.25: The identity. $L\,\mathrm{id} = X_F = S(\mathrm{id}, \mathrm{id}, \mathrm{id})$, so by (i) and Step S4.24, $(k+1)!\,\varphi_{k+1} = L^k(S(\mathrm{id}, \mathrm{id}, \mathrm{id})) = \sum_{a+b+c=k} \frac{k!}{a!b!c!} S(a!\varphi_a, b!\varphi_b, c!\varphi_c)$. Differentiating and using the symmetry of S gives the second identity. □

Source: ours.

Theorem 4.20 (The flow of X_F)**.** *Let* $\rho_0 = \rho_0(m) = (40C_F)^{-1/2}$*,* $B_\rho = \{w \in \mathfrak{W} : \|w\| < \rho\}$*, and for real* $|t| < 2$ *and* $w \in B_{\rho_0}$

$$\Gamma_t(w) = \sum_{k \geq 0} t^k \varphi_k(w).$$

Then, with $r = 2|t|C_F\|w\|^2 \leq \frac{1}{10}$*:*

(i) (Analyticity.) *The series converges absolutely;* $w \mapsto \Gamma_t(w)$ *has the power series* $(t^k \Phi_k)$ *on* B_{ρ_0} *and is analytic there;* $\Gamma_t - \mathrm{id}$ *is analytic* $B_{\rho_0} \to \mathfrak{W}^+$*;* $(t, w) \mapsto \Gamma_t(w)$ *is* C^1 *on* $(-2, 2) \times B_{\rho_0}$*.*

(ii) (Flow and transport equations.) $\Gamma_0 = \mathrm{id}$*,* $\partial_t \Gamma_t(w) = X_F(\Gamma_t(w))$ *and* $\partial_t \Gamma_t(w) = D\Gamma_t(w) X_F(w)$*.*

(iii) (Estimates.) $\|\Gamma_t(w)\| \leq \frac{10}{9}\|w\|$*,* $\|\Gamma_t(w) - w\|_+ \leq 3|t|C_F\|w\|^3$*, and* $\|D\Gamma_t(w) - I\|_{\mathfrak{W} \to \mathfrak{W}^+} \leq 10|t|C_F\|w\|^2 \leq \frac{1}{2}$*. Consequently* $D\Gamma_t(w)$ *maps* $\mathfrak{W}^+$ *into itself and is invertible on* $\mathfrak{W}$ *and on* $\mathfrak{W}^+$*, with inverses of norm* ≤ 2 *and* $\|D\Gamma_t(w)^{-1} - I\|_{\mathfrak{W} \to \mathfrak{W}^+} \leq 20|t|C_F\|w\|^2$*.*

(iv) (Reality.) $\Gamma_t(\varrho w) = \varrho\Gamma_t(w)$.
(v) (Variational equation.) $\partial_t D\Gamma_t(w)v = DX_F(\Gamma_t(w))\, D\Gamma_t(w)v$, *where* $DX_F(y)\xi = 3S(y,y,\xi)$.
(vi) (Inverse.) *With* $\rho_1 = \rho_0/2$*: for* $|t| \le 1$ *and* $w \in B_{\rho_1}$, $\Gamma_{-t}(\Gamma_t(w)) = w$. *Hence* $\Gamma = \Gamma_1$ *is injective on* B_{ρ_1}, $\Gamma(B_{\rho_1}) \subset B_{10\rho_1/9}$, *and* $\Gamma(B_{\rho_1}) \supset B_{9\rho_1/10}$.

Proof. **Step S4.26: Convergence and analyticity.** By Lemma 4.19(ii), $|t|^k\|\varphi_k(w)\| \le r^k\|w\|$ and $r \le 4C_F\rho_0^2 = \frac{1}{10}$; the power series $(t^k\Phi_k)$ has $\sum_k |t|^k\|\Phi_k\|\rho^{2k+1} < \infty$ for $\rho < \rho_0$, so (E2) applies, in $\mathfrak{W}$ and (for $k \ge 1$) in $\mathfrak{W}^+$. The partial derivatives of $t^k\varphi_k(w)$ in t and w are bounded by $k|t|^{k-1}(2C_F)^k\rho^{2k+1}$ and $(2k+1)|t|^k(2C_F)^k\rho^{2k}$ on $|t| < 2$, $\|w\| < \rho < \rho_0$; these are summable, so (E3) gives joint C^1 with termwise derivatives.

Step S4.27: Flow equation. Termwise, $\partial_t\Gamma_t = \sum_k (k+1)t^k\varphi_{k+1} = \sum_k t^k \sum_{a+b+c=k} S(\varphi_a,\varphi_b,\varphi_c)$ (Lemma 4.19(iv)), which is $S(\Gamma_t,\Gamma_t,\Gamma_t)$ by the Cauchy product of absolutely convergent series by (E3) and continuity of S.

Step S4.28: Transport equation. Termwise, $\partial_t\Gamma_t = \sum_k t^k L\varphi_k = \big(\sum_k t^k D\varphi_k\big)X_F = D\Gamma_t\, X_F$ by Lemma 4.19(i) and Step S4.26.

Step S4.29: Estimates. $\sum_k r^k \le \frac{10}{9}$ and $\sum_{k\ge1} r^k \le \frac{10}{9}r$ give the first two bounds ($\frac{10}{9}\cdot 2 \le 3$). For the third, $\|D\varphi_k(w)\|_{\mathfrak{W}\to\mathfrak{W}^+} \le (2k+1)(2C_F)^k\|w\|^{2k}$ and $\sum_{k\ge1}(2k+1)r^k = \frac{2r}{(1-r)^2} + \frac{r}{1-r} \le 5r$ for $r \le \frac14$. Since $\|\cdot\| \le \|\cdot\|_+$, the same operator bounds hold on $\mathfrak{W}$ and on $\mathfrak{W}^+$, and (E6) gives the inverses; $D\Gamma_t^{-1} - I = -D\Gamma_t^{-1}(D\Gamma_t - I)$ with $\|D\Gamma_t^{-1}\|_{\mathfrak{W}^+\to\mathfrak{W}^+} \le 2$.

Step S4.30: Reality. Lemma 4.19(iii), termwise; t is real.

Step S4.31: Variational equation. Termwise, by Lemma 4.19(iv) and the Cauchy product,

$$\partial_t D\Gamma_t v = \sum_k (k+1)t^k D\varphi_{k+1}v = \sum_k t^k \sum_{a+b+c=k} 3S(D\varphi_a v,\varphi_b,\varphi_c) = 3S(D\Gamma_t v,\Gamma_t,\Gamma_t).$$

Step S4.32: Inverse. For $w \in B_{\rho_1}$ and s between 0 and t, $\|\Gamma_s(w)\| \le \frac{10}{9}\|w\| < \rho_0$, so $\psi(s) = \Gamma_{-s}(\Gamma_s(w))$ is defined and C^1 (chain rule, Step S4.26). Its derivative is $-(\partial_\tau\Gamma_\tau)|_{\tau=-s}(\Gamma_s w) + D\Gamma_{-s}(\Gamma_s w)\,\partial_s\Gamma_s(w) = -D\Gamma_{-s}(\Gamma_s w)X_F(\Gamma_s w) + D\Gamma_{-s}(\Gamma_s w)X_F(\Gamma_s w) = 0$, by the transport equation at the first term and the flow equation at the second. By (E4), $\psi(t) = \psi(0) = w$. For the image: if $\|y\| < \frac{9}{10}\rho_1$ then $w = \Gamma_{-1}(y) \in B_{\rho_1}$ and $\Gamma(w) = y$ (the same argument with $t = -1$). □

Source: P96-W p. 285 ("in a sufficiently small neighbourhood of the origin the time-1-map is well defined and gives rise to a real analytic, symplectic change of coordinates Γ with the estimates $\|\Gamma - \mathrm{id}\|_{a,s+1} = O(\|w\|_{a,s}^3)$, $\|D\Gamma - I\|_{a,s+1,s}^{\mathrm{op}} = O(\|w\|_{a,s}^2)$"), asserted there without proof; constructed and proved here. Lean: every `tsum` above is absolutely convergent by Step S4.26.

Remark 4.21. By uniqueness of solutions of Lipschitz ODEs (`ODE_solution_unique`), $t \mapsto \Gamma_t(w)$ is the solution of $\dot y = X_F(y)$, $y(0) = w$; this identification is not used.

Theorem 4.22 (Symplecticity and gradient transport)**.** *For* $|t| < 2$, $w \in B_{\rho_0}$ *and* $\xi,\eta \in \mathfrak{W}$*:*

$$\Omega\big(D\Gamma_t(w)\xi, D\Gamma_t(w)\eta\big) = \Omega(\xi,\eta).$$

Consequently, if P *is complex differentiable near* $\Gamma_t(B_\rho)$ *(*$\rho \le \rho_0$*) with Hamiltonian field* X_P*, then* $P\circ\Gamma_t$ *has Hamiltonian field*

$$X_{P\circ\Gamma_t}(w) = D\Gamma_t(w)^{-1}X_P(\Gamma_t(w)) \qquad (w \in B_\rho),$$

which takes values in $\mathfrak{W}^+$ *when* X_P *does.*

Proof. **Step S4.33: Skewness of DX_F.** $\Omega(DX_F(y)\xi,\eta) = 3\Omega(S(y,y,\xi),\eta) = 12\mathfrak{F}(y,y,\xi,\eta)$ is symmetric in (ξ,η) (Theorem 4.15); with antisymmetry of Ω, $\Omega(DX_F(y)\xi,\eta)+\Omega(\xi,DX_F(y)\eta)=0$.

Step S4.34: Constant form. $g(t) = \Omega(D\Gamma_t\xi, D\Gamma_t\eta)$ is differentiable with $g'(t) = \Omega(DX_F(\Gamma_t)D\Gamma_t\xi, D\Gamma_t\eta) + \Omega(D\Gamma_t\xi, DX_F(\Gamma_t)D\Gamma_t\eta) = 0$ by Theorem 4.20(v), bilinearity and continuity of Ω, and Step S4.33. As $D\Gamma_0 = I$, $g \equiv \Omega(\xi,\eta)$ by (E4).

Step S4.35: Gradient transport.

$$D(P\circ\Gamma_t)(w)v = \Omega(X_P(\Gamma_t w), D\Gamma_t v) = \Omega\big(D\Gamma_t\, D\Gamma_t^{-1}X_P(\Gamma_t w), D\Gamma_t v\big) = \Omega(D\Gamma_t^{-1}X_P(\Gamma_t w), v),$$

using invertibility on $\mathfrak{W}$ (Theorem 4.20(iii)) and Step S4.34. Invertibility on $\mathfrak{W}^+$ gives the last claim. □

Source: P96-W p. 285 ("symplectic"; "$\Gamma^* X_H = D\Gamma^{-1}X_H\circ\Gamma = X_{H\circ\Gamma}$"), asserted there.

4.7. **The normal form.** Let $H = \Lambda + G$ on $\mathfrak{W}$ (finite by Lemmas 4.7, 4.10); on real points it is the Hamiltonian of Definition 1.8. Its Hamiltonian field $X_H = X_\Lambda + X_G$ is defined on $\mathfrak{W}^+$, with values in $\mathfrak{W}$.

Theorem 4.23 (Analytic partial Birkhoff normal form; P96-W Main Proposition for $J=\{1,2\}$)**.** *Let $\Gamma = \Gamma_1$ and $\rho_1 = \rho_1(m) = \frac12(40C_F(m))^{-1/2}$. Define on B_{ρ_1}*

$$K = H\circ\Gamma - \Lambda - \bar G - G^{\mathrm{nor}}.$$

Then:

- *(i) K is analytic on B_{ρ_1} and real, and $|K(w)| \le C_K\|w\|^6$ with $C_K = 10\,C_*C_F$;*
- *(ii) $K \in \mathcal{A}(B_{\rho_1})$ (Definition 4.5): its Hamiltonian field X_K is a holomorphic map $B_{\rho_1}\to\mathfrak{W}^+$, real, with $\|X_K(w)\|_+ \le C_K'\|w\|^5$, $C_K' = 66\,C_*C_F$;*
- *(iii) $\bar G$ and G^{nor} are as in Lemma 4.11: $\bar G = \frac12\langle AI, I\rangle + \langle BI, Z\rangle$ with the corrected coefficients, G^{nor} depends only on $\hat w$ and $|G^{\mathrm{nor}}| \le \frac{C_*}{4}\|\hat w\|^4$; both are in $\mathcal{A}(\mathfrak{W})$ with cubic fields bounded by $C_*\|w\|^3$ (resp. $C_*\|\hat w\|^3$);*
- *(iv)* (Conjugacy.) *For $w\in B_{\rho_1}\cap\mathfrak{W}^+$, $\Gamma(w)\in\mathfrak{W}^+$ and*

$$D\Gamma(w)\big[X_\Lambda(w) + X_{\bar G}(w) + X_{G^{\mathrm{nor}}}(w) + X_K(w)\big] = X_H(\Gamma(w)).$$

Proof. Write $Z' = \bar G - G^{\tan}$ (Proposition 4.16). Since $Z' + G - \bar G - G^{\mathrm{nor}} = 0$,

$$K = K_1 + K_2, \qquad K_1 = G\circ\Gamma - G, \qquad K_2 = \Lambda\circ\Gamma - \Lambda - Z'.$$

Step S4.36: Analyticity and reality. Λ, G, $\bar G$, G^{nor} are continuous polynomials and Γ is analytic on B_{ρ_1} (Theorem 4.20(i)); compose by (E2). Reality: each polynomial is real and $\Gamma\circ\varrho = \varrho\circ\Gamma$.

Step S4.37: The bound for K_1. On the segment from w to $\Gamma(w)$ every point y has $\|y\| \le \frac{10}{9}\|w\|$, and $|DG(y)v| = |\Omega(X_G(y),v)| \le C_*\|y\|^3\|v\|$. By (E4) and Theorem 4.20(iii), $|K_1(w)| \le C_*(\frac{10}{9})^3\|w\|^3\cdot 3C_F\|w\|^3 \le 5C_*C_F\|w\|^6$.

Step S4.38: The bound for K_2. Let $h(t) = \Lambda(\Gamma_t w) - \Lambda(w) - tZ'(w)$ for $t\in[0,1]$. Then $h(0)=0$ and, by the flow equation and Proposition 4.16 at $y=\Gamma_t w$, $h'(t) = D\Lambda(\Gamma_t w)X_F(\Gamma_t w) - Z'(w) = Z'(\Gamma_t w) - Z'(w)$. As in Step S4.37 (with $\|X_{Z'}(y)\|_+ \le C_*\|y\|^3$), $|h'(t)| \le 5C_*C_F\,t\|w\|^6$, so $|K_2(w)| = |h(1)| \le 5C_*C_F\|w\|^6$. Together with Step S4.37, (i) holds.

Step S4.39: The field of K_1. By Theorem 4.22, $X_{K_1}(w) = D\Gamma(w)^{-1}X_G(\Gamma w) - X_G(w) = D\Gamma^{-1}\big[X_G(\Gamma w) - X_G(w)\big] + (D\Gamma^{-1} - I)X_G(w)$. By Definition 4.6 (with $d=4$, $c = C_*/4$) and Theorem 4.20(iii), $\|X_{K_1}(w)\|_+ \le 2\cdot 3C_*(\frac{10}{9})^2\|w\|^2\cdot 3C_F\|w\|^3 + 20C_F\|w\|^2\cdot C_*\|w\|^3 \le 44C_*C_F\|w\|^5$. It is analytic into $\mathfrak{W}^+$: $w\mapsto D(\Gamma-\mathrm{id})(w)\in L(\mathfrak{W},\mathfrak{W}^+)$ is analytic by (E2), restriction $L(\mathfrak{W},\mathfrak{W}^+)\to L(\mathfrak{W}^+)$ is bounded linear, inversion is analytic on the units of $L(\mathfrak{W}^+)$, and $X_G\circ\Gamma : B_{\rho_1}\to\mathfrak{W}^+$ is analytic.

Step S4.40: The field of K_2. By the fundamental theorem of calculus (E5) and Step S4.38, $K_2(w)=\int_0^1\big(Z'(\Gamma_t w)-Z'(w)\big)dt$. The integrand is C^1 in (t,w) and, by Theorem 4.22,

$$D_w\big(Z'(\Gamma_t w)-Z'(w)\big)v=\Omega(Y(t,w),v),\qquad Y(t,w)=D\Gamma_t(w)^{-1}X_{Z'}(\Gamma_t w)-X_{Z'}(w)\in\mathfrak{W}^+.$$

As in Step S4.39, $\|Y(t,w)\|_+\le 44C_*C_F\,t\|w\|^5$, and Y is continuous in (t,w) and analytic in w with D_wY bounded on $[0,1]\times B_\rho$ for each $\rho<\rho_1$ (from the power series of Theorem 4.20). By (E5), $DK_2(w)v=\int_0^1\Omega(Y(t,w),v)dt=\Omega\big(\int_0^1 Y(t,w)dt,v\big)$, so $X_{K_2}(w)=\int_0^1 Y(t,w)\,dt$; it is complex differentiable $B_{\rho_1}\to\mathfrak{W}^+$ by (E5), and $\|X_{K_2}(w)\|_+\le 22C_*C_F\|w\|^5$.

Step S4.41: Conclusion of (ii). $X_K=X_{K_1}+X_{K_2}$, with $44+22=66$; reality from Definition 4.5 and (i).

Step S4.42: (iii). Lemmas 4.10 and 4.11.

Step S4.43: Conjugacy. Let $w\in B_{\rho_1}\cap\mathfrak{W}^+$. Then $\Gamma(w)=w+(\Gamma(w)-w)\in\mathfrak{W}^+$, so $X_H(\Gamma w)\in\mathfrak{W}$ is defined, and $DH(y)u=\Omega(X_H(y),u)$ for $y\in\mathfrak{W}^+$, $u\in\mathfrak{W}$ (Lemma 4.7). By the chain rule and Theorem 4.22, $D(H\circ\Gamma)(w)v=\Omega(X_H(\Gamma w),D\Gamma v)=\Omega(D\Gamma^{-1}X_H(\Gamma w),v)$. On the other hand $D(H\circ\Gamma)(w)v=D(\Lambda+\bar G+G^{\mathrm{nor}}+K)(w)v=\Omega(X_\Lambda(w)+X_{\bar G}(w)+X_{G^{\mathrm{nor}}}(w)+X_K(w),v)$. Nondegeneracy (Lemma 4.4) gives the identity after applying $D\Gamma(w)$. □

Source: P96-W Main Proposition, pp. 281–282; proof pp. 282–285, (12)–(14). Deviations: $J=\{1,2\}$; fixed m (the source's uniformity on compact m-intervals and analytic dependence on m are not needed by W-NLW2 and are omitted); the corrected coefficients (E-PDE-W-1); time convention of Definition 1.11.
SOURCE_GAP: P96-W p. 285 derives $X_K\in A(\ell^{a,s},\ell^{a,s+1})$ from "$\Gamma^*X_H=X_{H\circ\Gamma}\in A(\ell^{a,s},\ell^{a,s+1})$" and "$[X_F,X_H]=X_{\{H,F\}}\in A(\ell^{a,s},\ell^{a,s+1})$". For $H=\Lambda+G$ neither field is in that class (X_Λ is unbounded of order one, p. 280), and the cancellation of the unbounded terms in X_K is not shown. Steps S4.38 and S4.40 supply it: the unbounded part enters only through $D\Lambda\cdot X_F=Z'$ (Proposition 4.16), a bounded quartic. The source also deduces analyticity from coordinatewise analyticity and local boundedness ([9, Appendix A] of P96-W); here it comes from explicit power series and parametric integrals.

Remark 4.24 (Regularity class). K and Γ are analytic (convergent power series). For X_K the proof gives complex differentiability (`DifferentiableOn Complex`) into $\mathfrak{W}^+$, which is what Cauchy estimates along complex lines need (apply the one-variable Cauchy formula to $\zeta\mapsto X_K(w+\zeta v)$, Mathlib `DifferentiableOn.analyticOnNhd` for $\mathbb{C}\to E$). Analyticity of X_K would follow from the classical fact that complex-differentiable maps on open subsets of complex Banach spaces are analytic, which Mathlib does not provide (Remark 4.1); it is not used.

Corollary 4.25 (Real change of coordinates). *Let $r_1=\rho_1/\sqrt{2}$ and $B_r^E=\{x\in E_{a,p}:\|x\|_E<r\}$ (real). The map $\Gamma^{\mathbb{R}}=\mathcal{Z}^{-1}\circ\Gamma\circ\mathcal{Z}:B_{r_1}^E\to E_{a,p}$ is well defined (real-valued) and:*

- *(i) it is real analytic: it has on $B_{r_1}^E$ the real power series whose $(2k+1)$-st term is $\mathcal{Z}^{-1}\Phi_k(\mathcal{Z}\cdot,\dots,\mathcal{Z}\cdot)$ restricted to $E_{a,p}^{2k+1}$ (Mathlib `AnalyticOnNhd Real`);*
- *(ii) $\|\Gamma^{\mathbb{R}}(x)-x\|_{E_{a,p+1}}\le 6\sqrt{2}\,C_F\|x\|_E^3$ and $\Gamma^{\mathbb{R}}(B_{r_1}^E)\subset B_{2r_1}^E$;*
- *(iii) it is symplectic for $\sum dq_j\wedge dp_j$, injective, with inverse $\mathcal{Z}^{-1}\Gamma_{-1}\mathcal{Z}$ on its image, and $\Gamma^{\mathbb{R}}(B_{r_1}^E)\supset B_{r_1/2}^E$;*
- *(iv) for real $x\in B_{r_1}^E$: $H(\Gamma^{\mathbb{R}}x)=\Lambda(x)+\bar G(x)+G^{\mathrm{nor}}(x)+K(\mathcal{Z}x)$, where on real points $\bar G=\frac12\langle AI,I\rangle+\langle BI,Z\rangle$ with $I_j=(q_j^2+p_j^2)/2$.*

Proof. $\|\mathcal{Z}x\|\le\sqrt{2}\|x\|_E<\rho_1$, and Γ preserves $\mathfrak{W}_{\mathbb{R}}$ (Theorem 4.20(iv)), so $\Gamma^{\mathbb{R}}$ is real-valued (Lemma 4.3). (i): each Φ_k commutes with ϱ (Lemma 4.19(iii)), so its restriction to real arguments is real; the real multilinear maps have norm $\le 2^{(2k+1)/2}\|\Phi_k\|$ ($\|\mathcal{Z}\|\le\sqrt{2}$, $\|\mathcal{Z}^{-1}\|\le 1$), so the radius is at least $\rho_0/\sqrt{2}$; apply (E2) over $\mathbb{R}$. (ii): Theorem 4.20(iii) and $\|\mathcal{Z}x\|\le\sqrt{2}\|x\|_E$ ($3\cdot 2^{3/2}\le 6\sqrt{2}$; $\frac{10}{9}\sqrt{2}<2$). (iii): Lemma 4.4, Theorem 4.22, Theorem 4.20(vi): for $y\in B_{r_1/2}^E$,

$\|\mathcal{Z}y\| < \rho_1/2$, so $w = \Gamma_{-1}(\mathcal{Z}y)$ is real with $\|\mathcal{Z}^{-1}w\|_E \le \|w\| \le \frac{5}{9}\rho_1 < r_1$ and $\Gamma(w) = \mathcal{Z}y$. (iv): Theorem 4.23 and Lemma 4.11. □

Source: P96-W p. 281 ("real analytic, symplectic change of coordinates Γ in some neighbourhood of the origin in $\ell^{a,s}$") and p. 290 (use of Γ in real coordinates).

Corollary 4.26 (Invariant tori are transported)**.** *Let $\nu \in \mathbb{R}^2$ and $\Psi : \mathbb{R}^2 \to B_{\rho_1} \cap \mathfrak{W}^+$ be differentiable as a map into $\mathfrak{W}$ with $D\Psi(\theta)\nu = X_{H'}(\Psi(\theta))$ for all θ, where $X_{H'} = X_\Lambda + X_{\bar G} + X_{G^{\mathrm{nor}}} + X_K$. Then $\Gamma \circ \Psi$ takes values in $\mathfrak{W}^+$ and $D(\Gamma \circ \Psi)(\theta)\nu = X_H(\Gamma \circ \Psi(\theta))$. If Ψ is real ($\varrho\Psi = \Psi$), then $(Q, P) = \mathcal{Z}^{-1}\Gamma \circ \Psi$ satisfies, for every $j \ge 1$, $D_\nu Q_j = \lambda_j P_j$ and $D_\nu P_j = -\lambda_j Q_j - T(Q, Q, Q)_j$, which is the coefficient system (2) ($T = G_q$ for $a > 0$).*

Proof. Chain rule (Γ is differentiable on $\mathfrak{W}$) and Theorem 4.23(iv). Coordinates are continuous linear functionals, and on real points $\mathcal{Z}^{-1}X_H = (\lambda p, -\lambda q - T(q, q, q))$ by Lemmas 4.7 and 4.11. □

Source: P96-W p. 290 ("These tori carry the quasi-periodic motions $\gamma = \Gamma \circ \Psi(\dots)$").

4.8. **Interface to milestones W4–W6.**

Remark 4.27 (The exact form that P96-K Theorem A receives). After Theorem 4.23, the Hamiltonian is $H' = N_0 + \bar G + G^{\mathrm{nor}} + K$, $N_0 = \Lambda$, on the complex ball $B_{\rho_1} \subset \mathfrak{W}$, with:

- *Phase space and weights.* $\mathfrak{W} = \mathfrak{W}_{a,p}$ (complexification of $E_{a,p}$ via $\mathcal{Z}$; $a = p = 1$ for the headline). The variables w_l and w_{-l} are *independent* complex variables on B_{ρ_1}; reality is $\varrho w = w$. This is needed because the KAM domain complexifies the angles, so that $w_{-j} \ne \overline{w_j}$ there. In P96-K's notation (p. 120) the normal coordinates are $u_j = q_{j+2}$, $v_j = p_{j+2}$ ($j \ge 1$; KP96 p. 25, Step 1, with $q_j^{\mathrm{KP}} = \sqrt{2}z_j$), in $\ell^{a,p}$ with weights $(j+2)^p e^{a(j+2)}$; these are equivalent to P96-K's $j^p e^{aj}$ within the factor $3^p e^{2a}$, and P96-K's sum norm $\|u\| + \|v\|$ is within a factor 2 of the max norm. W4 must carry these two conversions.
- *Smoothing.* P96-K Assumption C with $\bar p = p + 1$ (so $\bar p - p = 1$, $\Delta = \min(\bar p - p, 1) = 1$): the fields of $\bar G$, G^{nor}, K map into $\mathfrak{W}^+ = \mathfrak{W}_{a,p+1}$; in particular their normal components $(\partial_{p_j} R, -\partial_{q_j} R)_{j \ge 3}$ lie in $\ell^{a,p+1}$, which is P96-K's $(P_v, -P_u) \in \ell^{a,\bar p}$.
- *Regularity class.* $\bar G$, G^{nor}: real quartic polynomials (analytic on $\mathfrak{W}$); $K \in \mathcal{A}(B_{\rho_1})$: analytic, real, with holomorphic real field $X_K : B_{\rho_1} \to \mathfrak{W}^+$ (Definition 4.5, Remark 4.24). No dependence on parameters: the parameter ξ of P96-K enters later, through the action-angle substitution (KP96 p. 25, Step 1), and its Lipschitz seminorms are then obtained by Cauchy estimates in ξ (KP96 p. 26, Step 3).
- *Size.* On B_{ρ_1}: $|K| \le C_K\|w\|^6$, $\|X_K\|_+ \le C_K'\|w\|^5$, $|G^{\mathrm{nor}}| \le \frac{C_*}{4}\|\hat w\|^4$, $\|X_{G^{\mathrm{nor}}}\|_+ \le C_*\|\hat w\|^3$. This is the hypothesis $|R| = O(\|\hat z\|^4) + O(\|z\|^g)$, $g = 6$, of the Cantor Manifold Theorem (P96-W p. 289) with $R = G^{\mathrm{nor}} + K$, strengthened by the field bounds that KP96 p. 26, Step 3 uses ("the hypotheses for R_u, R_v"). The exponent condition $g = 6 > 4 + (4 - \Delta)/\kappa = \frac{11}{2}$ uses $\kappa = 2$ (Remark 2.14).
- *Normal form.* $\bar G = \frac{1}{2}\langle AI, I\rangle + \langle BI, Z\rangle$ with $I_j = w_j w_{-j}$, the corrected A, B (Theorem 2.10); nondegeneracy is Theorem 2.13.
- *Return trip.* Tori of H' in $B_{\rho_1} \cap \mathfrak{W}^+$ are carried to tori of H by Corollary 4.26; the KAM embeddings take values in the $\bar p$-space (KP96 p. 28, Step 5), so $\Psi \in \mathfrak{W}^+$ holds. Analyticity in θ of $\Gamma^{\mathbb{R}} \circ \Psi^{\mathbb{R}}$ follows from Corollary 4.25(i) and analyticity of $\Psi^{\mathbb{R}}$; injectivity of $\Gamma^{\mathbb{R}}$ transports the family injectivity (6) of W-NLW2; the closeness $\|\Gamma^{\mathbb{R}}(x) - x\| \le 6\sqrt{2}C_F\|x\|^3$ is $o(r)$ in (3).

Source: P96-K Assumption C and the norm (1), p. 121; P96-W Cantor Manifold Theorem, assumption C, pp. 288–289, and p. 290; KP96 §7, Steps 1–5, pp. 25–28.

4.9. **Constants and their dependence on m.** They are chosen in this order; each depends only on the listed data.

Constant	Value	Depends on
C_*	$K_{p+1/2}^2/\pi$; $= 8(1+2\zeta(3))/\pi \approx 8.67$ for $a=p=1$	p only (not m, not a)
$\delta_*(m)$	$m/(27(4+m)^{3/2})$	m
$C_F(m)$	$C_*/\delta_* = 27C_*(4+m)^{3/2}/m$	m, p
$\rho_0(m)$, $\rho_1(m)$	$(40C_F)^{-1/2}$, $\frac{1}{2}(40C_F)^{-1/2}$	m, p
$C_K(m)$, $C_K'(m)$	$10C_*C_F$, $66C_*C_F$	m, p
Λ bound	$\sqrt{1+m}$	m

Asymptotically $C_F \sim 216C_*/m$ as $m \to 0$ and $C_F \sim 27C_*\sqrt{m}$ as $m \to \infty$; nothing is uniform in m, as Standing assumption 1.1 allows. Lean totality: the divisions in f_x are guarded by Lemma 4.12; $\sqrt{\lambda_j} > 0$; every `tsum` is absolutely convergent (Lemmas 4.9, 4.10, Theorem 4.20); the integral in Step S4.40 has a continuous integrand on $[0,1]$; derivatives are stated as `HasFDerivAt` at points of differentiability.

4.10. **Summary of status.**

- *Proved here*: Lemmas 4.3, 4.4, 4.7, 4.9, 4.10, 4.11, 4.12, 4.19; Theorems 4.15, 4.20, 4.22, 4.23; Proposition 4.16; Corollaries 4.25, 4.26.
- *External*: (E1)–(E6), all in Mathlib.
- *From W1*: Lemmas 3.2, 3.5, 3.6, Definition 3.24, Proposition 3.17 (the last also covered by Lemma 2.3).
- *Source defects*: P96-W pp. 285–286 (two steps of the proof of Lemma 4 and a slip); P96-W p. 285 (regularity of X_K not shown); P96-W pp. 282, 284 factor 16 (E-PDE-W-1, already recorded).

5. The proposed headline W-NLW2

Status: PROPOSED (not frozen, not approved; for independent audit and owner review).

This section states the headline proposed in the scoping report (`docs/explorations/pde-kam-and-leapfrogging.md`, §2.3), adopted for the source/design phase by owner decision D30, with every definition written out in the conventions of §1. It is a proposal for later independent statement audit and owner approval; nothing here is frozen. W-NLW2 is *the D30 first headline (a declared weakening of P96-W's Main Theorem for $J = \{1,2\}$)*; it is not the full specialized Main Theorem. The weakenings are listed in Remark 5.7.

5.1. **Definitions.** Fix $m > 0$, the weights $a = p = 1$ (Convention 1.5), the phase space $E = E_{1,1}$ with norm $\|(q,p)\|_E = \max(\|q\|_{1,1}, \|p\|_{1,1})$, the reconstruction U_m, V_m (Definition 1.7) and the cubic gradient G_q (Definition 1.9). Let $P = (0,\infty)^2$, $r(I) = \sqrt{I_1 + I_2}$ for $I \in P$, and $B_h = \{I \in \mathbb{R}^2 : \max_i |I_i| < h\}$.

Definition 5.1 (Linear torus)**.** For $I \in P$, $K_I^0 : \mathbb{R}^2 \to E$ is the map of Convention 1.13.

Definition 5.2 (Analytic torus embedding)**.** A map $K : \mathbb{R}^2 \to E$ is an *analytic torus embedding* if it is $\mathbb{Z}^2$-periodic, real analytic (each point has a neighbourhood on which K is the restriction of a holomorphic map from a complex neighbourhood in $\mathbb{C}^2$ to the complexification $E_{\mathbb{C}}$), injective modulo $\mathbb{Z}^2$ ($K(\theta) = K(\theta') \Rightarrow \theta - \theta' \in \mathbb{Z}^2$), and has injective derivative $DK(\theta) : \mathbb{R}^2 \to E$ at every θ.

Definition 5.3 (Derivative along a frequency)**.** For $\nu \in \mathbb{R}^2$ and differentiable K, $D_\nu K(\theta) = DK(\theta)\nu$.

Definition 5.4 (Diophantine)**.** $\nu \in \mathbb{R}^2$ is (γ,τ)-Diophantine if $|k \cdot \nu| \geq \gamma |k|_1^{-\tau}$ for all $k \in \mathbb{Z}^2 \setminus \{0\}$, $|k|_1 = |k_1| + |k_2|$. (Same as the frozen track-K definition `KAM.IsDiophantine`.)

5.2. **Statement.**

Theorem 5.5 (W-NLW2, proposed)**.** *For every $m > 0$ there exist $r_0 > 0$, $C > 0$, $\tau > 1$, a function $\eta : (0, r_0) \to [0, \infty)$ with $\eta(r) \to 0$ as $r \downarrow 0$, a Borel set $\Gamma \subset \{I \in P : r(I) < r_0\}$, a continuous map $\nu : \Gamma \to \mathbb{R}^2$, and maps $K_I = (Q_I, P_I) : \mathbb{R}^2 \to E$ $(I \in \Gamma)$ such that $(I, \theta) \mapsto K_I(\theta)$ is continuous on $\Gamma \times \mathbb{R}^2$ and:*

(1) (Density one in actions.) $\lim_{h \downarrow 0} \dfrac{\mathrm{Leb}_2(\Gamma \cap B_h)}{\mathrm{Leb}_2(P \cap B_h)} = 1.$

(2) (Two independent frequencies.) *For every $I \in \Gamma$ there is $\gamma_I > 0$ such that $\nu(I)$ is (γ_I, τ)-Diophantine, and $\nu(I) \to (\lambda_1, \lambda_2)/(2\pi)$ as $I \to 0$ in Γ.*

(3) (Small embedded tori.) *Each K_I is an analytic torus embedding, and for all $I \in \Gamma$*

$$\sup_\theta \|K_I(\theta) - K_I^0(\theta)\|_E \le \eta(r(I))\, r(I), \qquad \sup_\theta \|K_I(\theta)\|_E \le C\, r(I).$$

(4) (Exact invariance.) *For every $I \in \Gamma$, $\theta \in \mathbb{R}^2$ and $j \ge 1$,*

$$D_{\nu(I)} Q_{I,j}(\theta) = \lambda_j P_{I,j}(\theta), \qquad D_{\nu(I)} P_{I,j}(\theta) = -\lambda_j Q_{I,j}(\theta) - G_q(Q_I(\theta))_j.$$

(5) (PDE solutions.) *For every $I \in \Gamma$ and $\theta_0 \in \mathbb{R}^2$, the function $u(t,x) = U_m\big(Q_I(\theta_0 + t\nu(I))\big)(x)$ is real analytic in a neighbourhood of every point of $\mathbb{R} \times [0, \pi]$, is a classical solution of (1) on $\mathbb{R}$, satisfies $u_t(t, \cdot) = V_m\big(P_I(\theta_0 + t\nu(I))\big)$, and*

$$\sup_{t \in \mathbb{R},\, x \in [0,\pi]} \big(|u| + |u_x| + |u_t|\big) \le C\, r(I).$$

(6) (Family injectivity.) *For all $I, I' \in \Gamma$ and $\theta, \theta' \in \mathbb{R}^2$, if $K_I(\theta) = K_{I'}(\theta')$ then $I = I'$ and $\theta - \theta' \in \mathbb{Z}^2$. Thus distinct parameters give disjoint tori, and each torus is embedded (with (3)).*

Remark 5.6 (Reading of the statement).

- *Quantifier order.* $r_0, C, \tau, \eta, \Gamma, \nu, K$ are chosen after m and before I and θ_0; γ_I may degenerate as $I \to 0$. No uniformity in m is claimed.
- *Invariance* in (4) is the coefficient system (2) along the linear flow on the torus (period-one angles, Convention 1.13); by Proposition 2.6(i) it yields (5) once the required time regularity is checked (W2). The cubic term is the physical force by (4).
- *Normalization independence.* The statement uses only the fixed real Hamiltonian, the standard actions $I_j = (q_j^2 + p_j^2)/2$ (through K_I^0) and period-one angles; it contains none of the printed P96-W constants. The frequency asymptotics in (2) are only the leading term; the twist A of Theorem 2.10 enters the proof, not the statement.
- *Family injectivity* (6) is proved in W8 from the source's family construction: the tori are images of the parameter tori under a single map that is injective on the union of the parameter tori (P96-W Main Theorem pp. 272–273, "embedding" of the family; Cantor Manifold Theorem and Remark 1 pp. 288–289).
- *Deviations from the source* (to be audited): pure cubic nonlinearity; $J = \{1, 2\}$; $a = p = 1$; period-one angles; Borel density-one set instead of a Cantor set (isolated points are allowed, cf. KP96 p. 24, Remark 2); linear stability, Whitney regularity, and the Lipschitz extension of P96-W Remark 2 p. 289 are omitted; the closeness to K_I^0 is a little-$o(r)$ bound rather than the $O(r^\sigma)$, $\sigma > 1$, of the radial map of the Cantor Manifold Theorem (P96-W p. 289), because the linearizing parametrization of Remark 1 there has a worse estimate (KP96 §7, Step 7).

Remark 5.7 (Declared weakenings). Relative to P96-W's Main Theorem specialized to $J = \{1, 2\}$, W-NLW2 declares the following weakenings (orchestrator ruling W-A2, recorded for owner review). *Deferred together* (D30 Q3): linear stability of the tori; Whitney regularity of the family; and the Lipschitz regularity of the family in Cartesian coordinates, i.e. the source's Lipschitz Cantor manifold (P96-W pp. 272–273, 288–289); W-NLW2 requires only continuity of $(I, \theta) \mapsto K_I(\theta)$ on

$\Gamma \times \mathbb{R}^2$. *Retained*: each torus is an analytic embedding (3), and the family is injective (6), so the union of the tori is a disjoint union of embedded tori; what is not asserted is that the induced map on $\Gamma \times \mathbb{T}^2$ is a Lipschitz (bi-Lipschitz-parametrized) embedding. *Also declared*: Borel density-one parameter set instead of a Cantor set; $o(r)$ closeness instead of $O(r^\sigma)$; pure cubic nonlinearity, $a = p = 1$, period-one angles (Remark 5.6).

Source: P96-W Main Theorem pp. 272–273 (condition (4) holds for $J = \{1, 2\}$), Lemma 1 p. 278, Cantor Manifold Theorem and Remark 1 p. 289, assembly pp. 290–291; KP96 Theorem 2 p. 19 and §7 pp. 25–31; P96-K Theorems A, D.